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Exploring Jet Substructure in Public LHC Data

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Declaração

Declaro que o presente documento é um trabalho original da minha autoria e que cumpre todos os requisitos do Código de Conduta e Boas Práticas da Universidade de Lisboa.

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RESUMO

As correlações de spin colineares, induzidas pela polarização linear dos gluões intermédios na cascata partónica, produzem uma modulação azimutal entre ramificações sucessivas nos jatos. Este trabalho estuda-as com dados do CMS Open Data, de colisões protão-protão a $\sqrt{s} = 13$ TeV registadas em 2016, e a respetiva simulação Pythia 8. Construiu-se uma cadeia de análise reprodutível para observar a modulação do ângulo $\Delta\psi_{12}$ entre os planos das duas ramificações mais duras na desaglomeração de Lund de jatos com $p_{\mathrm{T}} > 700$ GeV e a modulação azimutal do correlador de energia de três pontos. Como os canais $g \to q\bar{q}$ e $g \to gg$ contribuem com sinais opostos, desenvolveu-se uma seleção em duas etapas com redes neuronais para enriquecer as amostras em ramificações $g \to q\bar{q}$, identificadas na simulação com o algoritmo de neutralização de sabor intercalada. Sem seleção, obteve-se $\langle \cos 2\Delta\psi_{12} \rangle = -0.078 \pm 0.003$ nos dados, em acordo com a simulação. Na simulação, a seleção intensifica a modulação de -0.12 para -0.27 quando a pureza sobe de 0.18 para cerca de metade, e os dados acompanham-na dentro da sua precisão. A decomposição por canal mostra, contudo, que todas as classes contêm uma componente importante sem origem no spin e que a seleção desloca a modulação de cada classe, pelo que a modulação selecionada não pode ser atribuída apenas ao canal $g \to q\bar{q}$. O correlador de energia apresenta uma pequena modulação positiva, quase insensível ao canal. Estes resultados são consistentes com a recente observação do CMS, mas estabelecer as correlações de spin requer uma previsão sem elas, processada pela mesma seleção.

Palavras-chave: Subestrutura de jatos, Correlações de spin colineares, Polarização do glúão, Plano de Lund, CMS Open Data, Aprendizagem automática

ABSTRACT

Collinear spin correlations, induced by the linear polarisation of intermediate gluons in the parton shower, produce an azimuthal modulation between successive splittings inside jets. This work studies them using CMS Open Data datasets, created from proton-proton collisions at $\sqrt{s} = 13$ TeV recorded in 2016 and the corresponding Pythia 8 simulation. A reproducible pipeline was built in order to observe the modulation of the angle $\Delta\psi_{12}$ between the planes of the two hardest splittings in the Lund declustering of jets with $p_{\mathrm{T}} > 700$ GeV, and the azimuthal modulation of the three-point energy correlator. Given that the $g \to q\bar{q}$ and $g \to gg$ channels contribute with opposite signs, a two-stage neural-network selection was developed to enrich the samples in $g \to q\bar{q}$ splittings, which were labelled in simulation using the interleaved flavour neutralisation algorithm. Before any selection was done, the value $\langle \cos 2\Delta\psi_{12} \rangle = -0.078 \pm 0.003$ was obtained in data, in agreement with the simulation. The implemented selection strengthens the modulation from -0.12 to -0.27 in simulation as the purity rises from 0.18 to about one half, and the data follow the simulation wherever their precision allows. The decomposition by channel shows, however, that every class carries a large non-spin component and that the selection shifts the modulation of each class, so the selected modulation cannot be interpreted as that of the $g \to q\bar{q}$ channel alone. The energy correlator shows a small positive modulation, nearly insensitive to the channel. These results are consistent with the recent CMS observation, but establishing the spin correlations requires a prediction without them, processed through the same selection.

Keywords: Jet substructure, Collinear spin correlations, Gluon polarisation, Lund jet plane, CMS Open Data, Machine learning

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LIST OF ABBREVIATIONS

AK8	Jets clustered with the anti-k_{T} algorithm and $R = 0.8$
ALICE	A Large Ion Collider Experiment
AOD	Analysis Object Data
ATLAS	A Toroidal LHC ApparatuS
C/A	Cambridge/Aachen algorithm
CERN	European Organization for Nuclear Research
CMS	Compact Muon Solenoid
CMSSW	CMS Software
CP5	Pythia 8 tune derived by CMS
DGLAP	Dokshitzer-Gribov-Lipatov-Altarelli-Parisi
DNN	Deep neural network
ECAL	Electromagnetic calorimeter
EEEC	Three-point energy correlator
FWLite	CMS Framework Lite
HCAL	Hadronic calorimeter
HLT	High-Level Trigger
IFN	Interleaved flavour neutralisation
IRC	Infrared and collinear
LHC	Large Hadron Collider
LHCb	Large Hadron Collider beauty
LIP	Laboratório de Instrumentação e Física Experimental de Partículas
MiniAOD	Reduced version of the AOD format, retaining the particle-flow candidates
NanoAOD	Further reduced version of the AOD format, without constituent information
PR	Precision-recall
PUPPI	Pileup per particle identification
QCD	Quantum Chromodynamics
ReLU	Rectified linear unit
ROC	Receiver operating characteristic

1 INTRODUCTION

1.1 Context and motivation

The Large Hadron Collider (LHC) at CERN produces the highest energy proton-proton collisions ever achieved in a laboratory and, with them, an enormous sample of strongly interacting final states. The majority of these collisions are governed by Quantum Chromodynamics (QCD), the gauge theory of the strong interaction. The comprehension and study of these collisions are not only a prerequisite for suppressing backgrounds in searches for new phenomena, but also allow the structure of the hadronic final state to be examined and compared to theory directly, at scales where the coupling is weak enough for perturbation theory to be applied and yet the dynamics still remain rich.

The most prominent objects in these final states are jets. Jets are collimated sprays of hadrons that emerge from the fragmentation of high-energy quarks and gluons. A jet is not a fundamental object but instead a definition. A clustering algorithm specifies which detector inputs are grouped together, and a recombination scheme specifies how momentum is assigned to the result. This defines a jet, from which an interesting and important question then arises. What does the radiation inside the jets look like? This domain is called jet substructure, which over the past decade has grown from a collection of tagging tools into a precision probe of QCD dynamics in its own right.

In order to organise and look at the internal structure of a jet, one can use the Lund jet plane [1]. The Lund jet plane is a two-dimensional map of a jet's branching history on which soft, collinear, and non-perturbative regions of QCD radiation occupy distinct territories. This map records where the jet's internal emissions occur, but it sets aside an intriguing piece of information, the orientation of each splitting relative to the one before. This orientation is not random. Through the linear polarisation of intermediate gluons, successive splittings are correlated in their relative azimuthal angle, producing a characteristic modulation in the distribution of that angle. This is a genuine quantum interference effect which survives only because the intermediate gluon is described by an amplitude rather than a probability, and it is absent from any picture that treats successive emissions as independent classical branchings.

These collinear spin correlations are well understood theoretically and are implemented in modern parton shower algorithms, yet establishing them in collision data has proved difficult. They were observed only recently by the CMS Collaboration, in jets enriched in $g \to q\bar{q}$ splittings [2]. The main difficulty is that the effect carries opposite signs in different splitting channels. The $g \to q\bar{q}$ and $g \to gg$ channels contribute with opposite sign, so a measurement that sums all channels sees the two contributions partially cancel. This interference is not absent from such an inclusive measurement. It is diluted, and the apparent size reflects the accidental flavour composition of the sample rather than the strength of the underlying physics. The measured distributions also receive contributions unrelated to the spin, which a measurement must be able to separate from the effect.

This is the obstacle the present work addresses. A meaningful measurement of the effect requires knowing which splitting produced each emission. However, the flavour of an individual splitting inside a jet is not directly observable and must be inferred. Working entirely on public CERN Open Data [3], more precisely with CMS Open Data [4], this dissertation builds a reproducible pipeline from the released samples through Lund declustering to the azimuthal spin observable, and introduces a flavour selection

that enriches the selected samples in the $g \to q\bar{q}$ channel. The resulting modulation is reported as a function of the threshold of that selection, with the purity of each selected sample determined in simulation. The behaviour of each channel under the selection is also studied in simulation, which shows to what extent the modulation of a selected sample reflects that of the $g \to q\bar{q}$ channel.

The Lund declustering angle $\Delta\psi_{12}$ is not the only way to reach this effect. The same $\cos 2\psi$ modulation also imprints itself on the flow of energy within a jet, and can be accessed through a three-point energy correlator, which probes the relative azimuthal orientation of energy deposited by triplets of particles without reference to any clustering history. The two observables are complementary. One reads the effect from the reconstructed branching sequence, the other from the energy flow directly, and each carries a different exposure to the choices and imperfections of the measurement. This work develops both, treating the Lund observable as the primary measurement and the energy correlator as a complementary probe of the same underlying physics.

The fact that this analysis rests on public data is not incidental. The CERN Open Data Portal has made it possible for work outside the experimental collaborations to produce meaningful results and to carry out genuinely new measurements. The present work is intended as a contribution in that spirit. It builds an end-to-end pipeline from the CMS collision data and simulation to the final measurements, establishing along the way a workflow for accessing, handling and measuring with CMS Open Data, whose code is publicly available [5].

1.2 Objectives

The main aim of this dissertation is to study collinear spin correlations in QCD jets using public LHC data and to establish the flavour selection that such a measurement requires in order to be meaningful. For the purpose of achieving that goal, the work is organised around a sequence of objectives. The first is to build a reproducible pipeline over CERN Open Data samples capable of extracting the necessary variables to begin a jet substructure analysis. This means that the data samples must contain jet constituent information, thus, the data format chosen must retain that level of detail, ruling out the more compact samples in favour of one that preserves per-constituent quantities.

With that pipeline in place, the second objective is to construct the primary and secondary Lund jet planes and to verify that they reproduce the known features of QCD radiation, since this agreement is what establishes that the declustering is sound before any finer observable is built on top of it. Following that, the third objective is to implement the azimuthal observable $\Delta\psi_{12}$ between the hardest primary and secondary branchings, with the grooming condition $z > 0.1$, and to study its inclusive distribution before attempting to sharpen it. The same objective includes the azimuthal modulation of the three-point energy correlator, implemented as a complementary probe that does not rely on the declustering.

The fourth objective is the central one, and the point at which this work departs from a purely inclusive analysis. Because the spin-interference signal carries opposite signs in the $g \to q\bar{q}$ and $g \to gg$ channels, an inclusive measurement dilutes it. Enriching the sample in a single channel is therefore not a refinement but a precondition for observing the effect of that channel. This objective is thus to develop a flavour selection capable of enriching the $g \to q\bar{q}$ channel, and to quantify both how well it performs and how pure a sample it can reach in principle. The fifth objective builds directly on it. It is to measure $\langle \cos 2\Delta\psi_{12} \rangle$ as a function of the flavour selection, in data and in simulation, and to determine to what extent the modulation of the selected samples reflects that of the $g \to q\bar{q}$ channel.

Finally, the work aims to identify the sources of systematic uncertainty affecting this measurement and the dominant limitations of the approach, so that the extensions it opens up can be set out on solid ground.

1.3 Thesis outline

Chapter 2 develops the theoretical material on which the measurement rests, such as the elements of QCD relevant to jet formation, the definition and reconstruction of jets and the algorithms used to build and decluster them, the Lund jet plane, and the origin of collinear spin correlations together with the two observables through which this work accesses them. Chapter 3 turns to the data and the analysis, and covers the CERN Open Data resources and the CMS detector, the data format and dataset used, and the pipeline that runs from the released samples through Lund declustering to the azimuthal observable, the flavour selection, and the treatment of systematic uncertainties. Chapter 4 presents the validation of the pipeline and the measurements of both observables, and discusses the results in the light of the expectations set out in Chapter 2 and of the CMS measurement. Finally, Chapter 5 returns to the objectives of Section 1.2, states what has been established and with what confidence, and outlines the extensions that the present framework makes possible. Appendix A collects the numerical values behind the figures of Chapter 4.

2 THEORETICAL BACKGROUND

2.1 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the quantum field theory of the strong interaction, a non-Abelian gauge theory built on the colour symmetry group $SU(3)$ [6]. Its fundamental degrees of freedom are quarks, which carry colour charge, and gluons, the gauge bosons that mediate the interaction. The non-Abelian character of the theory has a decisive consequence absent from Quantum Electrodynamics. The gluons themselves carry colour, and therefore couple directly to one another. To a large extent, the dynamics of the strong interaction are the dynamics of this gluon self-coupling, and the two properties of QCD that govern everything in this chapter both follow from it.

2.1.1 Asymptotic freedom and confinement

The first property is asymptotic freedom. The strong coupling α_s decreases as the momentum transfer of an interaction increases, so that at the high scales probed in LHC collisions the quarks and gluons interact weakly and perturbation theory applies [7, 8]. The running of the coupling is captured at leading order by

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + \alpha_s(\mu^2)\,\frac{33 - 2n_f}{12\pi}\,\ln\frac{Q^2}{\mu^2}}, \tag{2.1}$$

where Q is the scale of the process, μ a reference scale, and n_f the number of active quark flavours. The positive sign of the coefficient $33 - 2n_f$, which holds for any physical number of flavours, is what makes the coupling fall at large Q. It is a direct imprint of the gluon self-interaction, and its discovery established that a perturbative treatment of hard processes is justified. It is convenient to trade the reference scale for a single dimensionful constant Λ_{QCD}, the scale at which the perturbative coupling formally diverges, of an order of a few hundred MeV. In terms of it, the coupling is small for $Q \gg \Lambda_{\text{QCD}}$ and grows strong as Q approaches it.

This growth is the second property called confinement. At low momentum transfer, α_s becomes large and colour-charged particles cannot be isolated. A quark or gluon produced in a hard scattering does not propagate freely to a detector, because as it separates from other colour charges, the energy stored in the colour field between them grows with distance until it becomes energetically favourable to create a new quark-antiquark pair from the vacuum. The original parton is thereby dressed into a spray of colour-neutral hadrons. This is why the strongly interacting final state reaches the detector not as partons but as the collimated sprays introduced in Chapter 1, and why any comparison between a perturbative calculation and a measurement must bridge the divide between the partonic and hadronic descriptions.

2.1.2 Soft and collinear radiation

The bridge between the hard parton and the observed hadrons is built from successive emissions, and the pattern of those emissions is far from uniform. When a parton radiates a gluon, the emission probability

at first order in the coupling takes the schematic form

$$d\mathcal{P} \;\propto\; \frac{2\alpha_s C}{\pi}\,\frac{d\theta}{\theta}\,\frac{dE}{E}, \tag{2.2}$$

where E and θ are the energy and emission angle of the radiated gluon, and C is a colour factor. The two factors carry the characteristic structure of QCD radiation. The factor $d\theta/\theta$ diverges as $\theta \to 0$ meaning that emissions are enhanced when the gluon is radiated close to the direction of its emitter. This is known as the collinear divergence. The factor dE/E diverges as $E \to 0$ meaning that emissions are enhanced when the gluon is soft. This is known as the soft divergence. Together they mean that a hard parton radiates predominantly softly and at small angles, and that the number of such emissions is, in a precise logarithmic sense, large.

The collinear structure is described more completely by the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) splitting functions [6], which give the probability that a parton of one type splits into two, as a function of the momentum fraction z carried by one of the products. For the three splittings relevant to this work they read, at leading order,

$$P_{q\to qg}(z) = C_F\,\frac{1+z^2}{1-z}, \tag{2.3}$$

$$P_{g\to gg}(z) = C_A\,\left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z)\right], \tag{2.4}$$

$$P_{g\to q\bar{q}}(z) = T_R\,\left[z^2 + (1-z)^2\right], \tag{2.5}$$

with colour factors $C_F = 4/3$, $C_A = 3$ and $T_R = 1/2$. The distinction between the gluon splitting channels of equations (2.4) and (2.5), one producing a pair of gluons and the other a quark-antiquark pair, will prove central. The two channels differ not only in these rates but in the spin structure of their emissions, and it is that difference which the spin observables of Section 2.4 are designed to probe.

These divergences are not a pathology. They are integrable, and in any physical, infrared-safe observable the divergent real emissions cancel against corresponding virtual corrections, leaving a finite result. This cancellation is the requirement that shapes the very definition of a jet in Section 2.2. Their logarithmic structure, however, is physical and measurable, and it is exactly the structure that the Lund jet plane of Section 2.3 is built to expose. By mapping each emission onto logarithmic axes of angle and transverse momentum, it turns the double logarithmic enhancement of equation (2.2) into an approximately uniform density, against which finer correlations stand out.

2.1.3 Parton showers and hadronization

Because a single hard parton typically radiates many times before hadronizing, the emissions can be organised into a sequence ordered from the hardest, widest-angle splittings down to the soft and collinear ones. This sequence is what is known as a parton shower. Modern Monte Carlo event generators implement this sequence probabilistically through the splitting functions of equations (2.3)-(2.5), evolving each parton down from the hard scale of the collision to a low scale of order 1 GeV, at which the perturbative description breaks down, and confinement takes over. Below that scale, the partons are converted into hadrons by phenomenological hadronization models, tuned to data rather than derived from first principles.

This picture, a hard, perturbatively calculable core, dressed by a shower of increasingly soft and collinear emissions and finally hadronized, underlies the whole of the analysis that follows. The observables studied in this work are built to read the perturbative shower through the hadrons that reach the detector, and for this reason the measurement is deliberately confined to a hard kinematic regime, where the perturbative component dominates, and non-perturbative corrections are comparatively suppressed.

2.2 Jets

As settled in Section 2.1, a coloured parton produced in a hard scattering does not reach the detector directly because it radiates and hadronizes before arriving as a collimated spray of hadrons. A jet is then an object, which by grouping those hadrons and combining their momentum back together, inverts the process approximately and estimates the momentum of the original parton that produced them. Therefore, a jet is a reconstructed physical object, built from a construction algorithm, with properties that depend on this specific algorithm used to build it. This section sets out the requirement which any jet definition must satisfy, in order to be usable in perturbative QCD, and three algorithms, used in this work and built on that requirement.

2.2.1 Jet algorithms and infrared-collinear safety

A jet is built using two components. A jet algorithm establishes which inputs, from detector-level energy deposits, reconstructed particles, or, in a calculation, partons, are grouped together into a jet. A recombination scheme picks up those grouped inputs and specifies how their momenta are combined in order to compute the momentum of the jet. Across this work, the inputs are the particle-flow candidates, described in Chapter 3, and momenta are combined in the E-scheme, in which four-momenta are simply added. This scheme is Lorentz invariant and returns jets with non-zero mass.

Though, not every conceivable grouping prescription is admissible. The necessary requirement, and the one that eliminates most of them is infrared and collinear (IRC) safety. In order for an observable to be infrared safe, its value must remain unchanged by the addition of an arbitrarily soft emission. For it to be collinear safe, this value must also remain unchanged if one of its inputs is replaced by two collinear inputs sharing its momentum. So, a jet algorithm is IRC safe when the set of jets it returns is stable under both of these operations. A soft particle must not create or destroy a jet and a collinear splitting must not change how particles are assigned to jets.

This requirement is not optional because of what follows directly from Section 2.1. The emission probability of equation (2.2) diverges in both the soft and the collinear limits. In any IRC-safe observable these divergent real emissions cancel against equally divergent virtual corrections, leaving a finite result. Opposingly, in an IRC-unsafe observable, that cancellation is ruined and a fixed-order perturbative prediction returns infinity rather than a finite number. An IRC-unsafe jet definition is therefore not merely inconvenient but it also cannot be compared with perturbative QCD at all. Since the purpose of measuring jets is to confront the theory with data, IRC safety is a precondition and not a refinement.

In the historical development of jet algorithms this criterion has been conclusive. The first definitions that appeared were cone algorithms, which grouped particles falling within a fixed angular radius of a candidate axis. These algorithms were simple and seed-based and were not IRC safe because a soft particle could seed a new candidate cone, or a collinear splitting could change which particles acted

as seeds, and either of these flaws could alter the resulting jets. A lot of effort was put in order to fix them and eventually producing seedless variants that restored collinear safety but made them very slow. Alternatively, another family of algorithms, the sequential recombination algorithms, are IRC safe by construction and this work uses them.

Since the same mechanism reappears when flavour is assigned to jets in Section 2.2.4, it is worth making concrete how such a problem arises. Consider a seed-based cone algorithm applied to an event with two hard particles whose separation is slightly greater than the cone radius, so that each seeds its own jet. By adding a single soft particle between them, it can seed a third cone that overlaps both, merging the two hard particles into one jet where before there were only two, which, again, is an infrared-unsafe change, since an arbitrarily soft emission has altered the hard jets. A collinear splitting produces the collinear-unsafe failure. Replacing one hard seed by two collinear particles, each carrying half its transverse momentum, may leave neither above the seed threshold, so that a jet which existed before the splitting is absent after it. Therefore, in both cases, the divergent emissions of equation (2.2) feed directly into the jets themselves, and the perturbative prediction diverges with them.

Going back to the family introduced before, the sequential algorithms avoid both failures by never relying on seeds or on fixed geometric cones. In these algorithms, jets are built by repeatedly merging inputs according to a distance measure, so that soft or collinear additions change only the order in which already nearby inputs are combined and never the final assignment of hard momentum jets. This structural difference is the reason why this entire family is IRC safe and it is why it is recommended for any measurement to be compared with theory over the cone algorithms.

A sequential recombination algorithm is defined by a distance measure between each pair of inputs and between each input and the beam. The generalised-k_T family [9] writes these as

$$d_{ij} = \min\left(p_{\mathsf{T},i}^{2p},\, p_{\mathsf{T},j}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \tag{2.6}$$

$$d_{iB} = p_{\mathsf{T},i}^{2p}, \tag{2.7}$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ is the squared angular separation of the inputs in rapidity y and azimuth ϕ, R is the jet radius parameter, and the integer p selects the algorithm. The algorithm, at each step, finds the smallest of all the distances. If it is a d_{ij}, the inputs i and j are recombined into a single input. If it is a d_{iB}, the input i is declared a final jet and removed. This procedure repeats until no inputs remain. The returned jets are IRC safe for any integer value of p, but the value of p changes drastically how the clustering proceeds. The common choices $p = -1$, $p = 0$ and $p = +1$, give, respectively, the anti-k_T, Cambridge/Aachen and k_T algorithms, of which the first two are the tools this work relies on and are described in the following subsections.

In all three of those algorithms, the parameter R appears and sets the angular reach of the clustering through the ratio $\Delta R_{ij}^2/R^2$ in equation (2.6). A smaller R yields narrower jets that resolve nearby structure but also capture less of a single parton radiation. A larger R collects more of that radiation at the price of admitting more contamination from unrelated soft activity. The most appropriate choice depends on what we intend to use the jets for, and the two algorithms this work relies on are used with different values of R for correspondingly different goals. This point is, again, taken up in Sections 2.2.2 and 2.2.3.

2.2.2 The anti-k_T algorithm

The choice $p = -1$ in equations (2.6) and (2.7) defines the anti-k_T algorithm [10]. With this value the inter-particle distance becomes

$$d_{ij} = \min\left(\frac{1}{p_{\mathsf{T},i}^2}, \frac{1}{p_{\mathsf{T},j}^2}\right)\frac{\Delta R_{ij}^2}{R^2}, \tag{2.8}$$

so that the distance scales with the inverse squared transverse momenta of the inputs. This makes the smallest distances, and thus the first recombinations, commit to the hardest particles. The clustering is therefore driven by the hard inputs. So, a hard particle accumulates the softer inputs around it before soft inputs cluster among themselves.

The name ”anti” of this algorithm reflects this inversion. In the k_T algorithm, the one with the choice of $p = 1$, the distance scales with the positive powers of transverse momentum, making the soft inputs cluster first and the jet being assembled from the bottom, following the inverse of the physical shower order. The anti-k_T algorithm does the reverse of it. When taking $p = -1$, the soft-first ordering of k_T turns into a hard-first one. Effectively, it is the k_T algorithm with the role of soft and hard interchanged, and the ”anti” part shows exactly that inversion. The two sit at opposite ends of the generalised family, with Cambridge/Aachen, free of any transverse momentum dependence, between them.

There is a geometric consequence because of this ordering. A hard particle with no other hard particle within an angular distance $2R$ gathers all softer inputs within a distance R of it, producing a jet that is exactly circular in the (y, ϕ) plane. When two hard particles are closer than $2R$, the border between their jets is set by their relative transverse momenta with the harder one retaining the larger share, but the jets remain regular in shape. Due to soft radiation being last in the clustering, it cannot deform these borders. They are spread across the hard jets without changing their shape.

The region of the (y, ϕ) plane from which a jet gathers its inputs is a useful way to describe this stability. This area can be probed operationally by adding a large number of infinitesimal soft particles to an event and recording which jet each of these particles is assigned to. For the majority of the algorithms, the resulting areas are irregular and fluctuate from jet to jet, because soft particles impact the clustering. However, for the anti-k_T algorithm, where soft particles are clustered last and cannot reshape the hard jets, the areas are close to the ideal value πR^2 and fluctuate little. A stable and expected area is what makes the subtraction of pile-up and underlying-event contamination tractable, since that contamination scales with the area over which each jet collects it.

This regularity is what made the anti-k_T algorithm the standard jet definition at the LHC. Since the jet border is stable against soft radiation, the area over which a jet collects contamination from pile-up and the underlying event is well defined and almost constant, and the corresponding energy corrections are simple to calibrate. Therefore, the anti-k_T algorithm combines the IRC safety guaranteed for the whole generalised-k_T family with a jet form that is simple to treat experimentally. This combination thus provides regular borders, which the other two generalised-k_T algorithms do not.

The importance of the anti-k_T is also a matter of history. When the LHC started taking data, an IRC safe jet definition was needed, so that measurements could be compared with the increasingly precise perturbative calculations that became available. The algorithm was also required to be fast enough to run along the very large event samples which the LHC would produce. The anti-k_T algorithm, together with its efficient implementation in the FastJet package [11], met both needs, and it was adopted as the

baseline jet definition by the LHC experiments, a role it has held since.

It is because of this that the anti-k_T algorithm was used in this work to define the jets, with radius parameter $R = 0.8$. Jets constructed in this way are conventionally denoted as AK8. However, the algorithm does not provide a physically meaningful account of how the jet was assembled. This is because its sequence is ordered by decreasing hardness, so that the hardest particles are clustered first. A clustering sequence follows the radiation process only if it undoes the latest splittings first. Since a parton shower evolves from the hardest, widest-angle splittings down to the soft and collinear ones, this requires an ordering by increasing hardness, as in the k_T algorithm, or by increasing angle, as in the C/A algorithm. The intermediate recombinations of the anti-k_T algorithm therefore bear no correspondence to the sequence of physical splittings in a parton shower. The clustering history of an anti-k_T jet is, in this sense, an artefact of the algorithm rather than a record of the underlying radiation. Recovering that history, which is what the substructure observables of this work require, calls for a different member of the family, applied to the constituents of the jet once it has been defined. That algorithm is the Cambridge/Aachen algorithm and it is the subject of the next subsection.

2.2.3 The Cambridge/Aachen algorithm

From equation 2.6 and equation 2.7, if one chooses $p = 0$, the Cambridge/Aachen (C/A) algorithm is defined [12]-[13]. With this choice, the transverse momentum factors drop out of the distance measure entirely, leaving the equation to a purely angular form:

$$d_{ij} = \frac{\Delta R_{ij}^2}{R^2}. \tag{2.9}$$

The clustering process with this algorithm proceeds by repeatedly merging the pair of inputs closest in angle, independently of how hard or soft they are. Where the anti-k_T algorithm is driven by the hardest inputs, the C/A algorithm orders them by angular separation alone. This means that the first merges are the most collinear and the last the widest.

This type of angular ordering is what makes the C/A algorithm the optimal tool choice for evaluating jet substructure. As discussed in Section 2.1, the parton shower is ordered in angle, with the widest angle splittings happening earliest in the physical branching sequence. Therefore, the C/A algorithm mirrors, in reverse, the order in which the shower developed. By undoing the clustering process one step at a time, an approximation of the branching sequence is retraced from the widest splitting to the narrowest splitting, and each step exposes a single emission, characterised by the angular separation ΔR and the relative transverse momentum k_t of the two branches it separates. This agreement between the C/A history and the physical shower is what the Lund jet plane, discussed in Section 2.3, is built to exploit, and it is the reason jet substructure observables are extracted from the C/A history rather than the anti-k_T one.

The declustering process can be asserted explicitly. Starting from the single C/A object that contains the whole jet, one undoes the last recombination, splitting it into two subjets a and b, labelled so that $p_{t,a} > p_{t,b}$, meaning a is the harder branch and b the softer branch. This softer branch is taken as the emission and then, the declustering process, follows the harder branch a, repeating the previous step of splitting it. Iterating along the hard branch until it can no longer be declustered traces out the primary sequence of the jet. This is an ordered list of emissions, from the widest and earliest to the most collinear and latest.

Every splitting in this sequence is characterised by a small set of kinematic variables. The angular separation of the two branches ΔR_{ab}, and their relative transverse momentum $k_t = p_{t,b}\,\Delta R_{ab}$, together locate the emission on the two scales that matter. This characterises the splitting by how hard and how collinear it is. A third variable, the momentum fraction $z = p_{t,b}/(p_{t,a} + p_{t,b})$, computes how the momentum was distributed between the emission and the emitter. These are the quantities from which the Lund jet plane of section 2.3 is constructed, and computing them at every declustering step is the immediate goal of the C/A history.

By following only the harder branch, at each step, one captures the primary emissions, although it discards the structure within the softer branches. In order to recover that structure, it is necessary to decluster it recursively. By choosing a primary emission, the softer branch b is itself declustered in the same way, exposing the emission within it and generating a secondary sequence of emissions associated with that primary splitting. The primary sequence records how the leading parton radiated and the secondary sequences record how those radiated partons radiated in turn. In this work, both primary and secondary emissions' observables are needed in order to compare the orientation of a primary splitting with that of the secondary splitting it seeds.

In practice, the anti-k_T and C/A algorithms are applied in succession and complement each other. A jet is first defined with the anti-k_T algorithm, since it fixes its constituents and its regular boundary. Those constituents are then extracted and reclustered with the C/A algorithm, in order to obtain an angularly ordered history that can expose the emissions within a jet. In this work, the radius parameter of the C/A reclustering step was set to $R = 2$, larger than the anti-k_T value of $R = 0.8$, to ensure that all the constituents of the original jet are gathered into a single C/A object and that the declustering history is complete with no emission lost at the edge. Thankfully, the kinematics of the jet are unchanged by performing this operation because the same constituents with the same momentum are involved, however they now become organised into a sequence from which the physical splittings can be read.

The division of labour between the two algorithms is thus deliberate. The anti-k_T algorithm supplies a jet with a stable, regular boundary, well suited to the energy calibration and pile-up subtraction discussed in Section 2.2.2. The C/A algorithm applied to that jet's constituents supplies a history that mirrors the physical showering sequence with angularly ordered branching history which substructure requires. Neither algorithm alone would suffice. The anti-k_T algorithm gives a clean jet but an unphysical history, while the C/A algorithm gives a physical history but an irregular, contamination-prone boundary. Using each where it is strong is what makes the substructure analysis of this work both experimentally robust and physically interpretable.

2.2.4 Flavoured jets and the IFN algorithm

In the previous subsections, the algorithms cluster inputs on the basis of their kinematics alone, which is sufficient for most purposes. However, for the core measurement of this work, kinematics alone are not enough. As foreseen in Chapter 1, the goal is to explore the spin correlations of gluon splitting, and that requires distinguishing the channels of equations (2.4), (2.5) and (2.3), which produce, respectively, a pair of gluons, a quark or antiquark with a gluon and a quark-antiquark pair. The channels differ not only in their rates but also their spin interference signal, which carries different signs and magnitudes. Evaluating this effect at its real strength thus requires separating channels, and, in addition, that requires flavour assignment to the splittings inside a jet. This raises the obstacle of how one can assign flavour to a jet, or even to a branch inside one, in a way that remains theoretically well defined and experimentally robust.

The most obvious procedure is to sum the flavour of a jet's constituents, by counting a quark as $+1$ and an antiquark as -1 for each quark flavour separately, so that the net flavour of a jet is that of the parton which initiated it. While this procedure is simple, it is not IRC safe and the root of the problem is precisely the $g \to q\bar{q}$ splitting. A gluon carries no net flavour, but it can split into a quark-antiquark pair that is soft or wide in angle or even both. This quark and antiquark can then be assigned to different jets, or even to different branches of the same jet, making each acquire a net flavour that the gluon never possessed. Because the splitting can be arbitrarily soft, the misassignment occurs with a probability that diverges in the same manner as the emission spectrum of equation (2.2), and no virtual correction is assigned to cancel it. This is the same mechanism for kinematic observables, identified in Section 2.2.1 but now for the flavour. Here, this simple procedure is ruined by exactly the soft and collinear radiation that QCD produces most abundantly, and a fixed order calculation of it fails to converge. Thus, flavour assigned sloppily is not a physical quantity and it cannot be compared with theory.

Restoring IRC safety to jet flavour has been pursued for two decades. The first safe prescriptions achieved IRC safety by making the clustering distance itself depend on the flavour of the objects being combined [14], which succeeded in suppressing the divergence, but at the cost of producing jets that no longer had the kinematics of the fundamental algorithm. This means that a flavour aware jet and its equivalent anti-k_T or C/A jet were no longer the same object. For a measurement that requires both kinematics of a jet and the flavour of its structure, that cost is too high. In this work, the approach adopted was using the Interleaved Flavour Neutralisation (IFN) [15], which was designed exactly to avoid that issue. IFN leaves the kinematics of the clustering entirely untouched and treats flavour as a separate layer that runs alongside the normal clustering sequence. The jets, which are returned by this algorithm, are identical to the C/A jets of Section 2.2.3. This includes constituents and momenta. What IFN gives in addition is a flavour label for each, computed by a procedure that is stable against soft and collinear emissions.

The principle behind this new procedure is neutralisation. While the clustering occurs, IFN tracks down the flavour carried by every object in the sequence, and, at each stage, asks whether a flavoured object can be matched to a nearby object of opposite flavour, which is the partner who would have originated the common $g \to q\bar{q}$ splitting. Whenever it finds such a pair, the two flavours are neutralised against each other, so that the non-genuine flavour is cancelled at the moment it would otherwise be passed up the clustering tree and inherited by a jet. This makes the quark and antiquark born of the same soft or collinear gluon splitting reunite in flavour, even though their kinematics remain wherever the C/A algorithm placed them. The matching is ruled by a flavour distance that weighs how soft and how collinear a candidate pair is,

$$u_{ik} = \mathsf{max}(p_{\mathsf{T},i}, p_{\mathsf{T},k})^{\alpha}\ \mathsf{min}(p_{\mathsf{T},i}, p_{\mathsf{T},k})^{2-\alpha}\ \Omega_{ik}^2, \tag{2.10}$$

with the angular measure

$$\Omega_{ik}^2 = 2\left[\frac{\mathsf{cosh}(\omega\,\Delta y_{ik}) - 1}{\omega^2} - (\mathsf{cos}\,\Delta\phi_{ik} - 1)\right], \tag{2.11}$$

where Δy_{ik} and $\Delta\phi_{ik}$ are the rapidity and azimuthal separations of i and k. At small angles $\Omega_{ik}^2 \to \Delta R_{ik}^2$, and u_{ik} takes the form of the flavour-k_T distance of [14]. When i is about to be clustered with j, its flavour is neutralised against a flavoured object k of opposite flavour if $u_{ik} < u_{ij}$. In this way it is exactly the soft and collinear configurations that are neutralised, while genuinely distinct hard flavours are left in place. IRC safety requires $\alpha + \omega > 2$ [15], and this work uses $\alpha = 2$ and $\omega = 3 - \alpha = 1$.

What IFN brings to this work is that it does not merely give a prescription that labels the final jet. It exposes the flavour at every step of the clustering and not only at the end. This is the feature that the present

analysis required. The clustering of a C/A history produces at each step a splitting of one object into two branches. IFN attaches to those branches a flavour label that is IRC safe and, therefore, meaningful at fixed order in perturbation theory. The abstract distinction between the discussed channels in equations (2.4), (2.5) and (2.3) is thereby made concrete, and it can be read off at each splitting in the history of a jet. It is on this per splitting flavour that the channel selection at the heart of the measurement is built. The algorithm is distributed as a plugin to the FastJet package [11], and the configuration adopted here, together with the way its per-splitting labels are combined into the selection itself, is set out in Chapter 3.

2.3 The Lund jet plane

As described in Section 2.2.3, the declustering of the C/A history associates with every jet an ordered sequence of emissions, each carrying an angular separation ΔR, a relative transverse momentum k_t and a momentum fraction z. This sequence, on its own, is just a list of numbers. Jets have different numbers of emissions and the raw variables reach many orders of magnitude. However, the Lund jet plane [1] can turn this list into something clear by mapping each emission to a single point in a two-dimensional plane whose coordinates are chosen so that the structure of QCD radiation is exposed in its simple form.

2.3.1 Coordinates and the uniform density

The procedure to construct the Lund jet plane is direct. For each jet, the primary declustering of the C/A history, of Section 2.2.3, yields a list of emissions. Every emission in that list is mapped to a single point on the plane using its angular separation ΔR and relative transverse momentum k_t as coordinates. When repeating this for every emission in a sample and overlaying the resulting points, it produces a two-dimensional distribution, which is then normalised to the number of jets inside the sample, giving the average density of emission per unit area. It is this averaged distribution, rather than the pattern of any single jet, that is referred to as the Lund jet plane and that carries the statistical information exploited below.

Each primary emission is placed on the plane at the coordinates:

$$\left(\ln \frac{1}{\Delta R},\, \ln k_{\mathsf{T}}\right), \tag{2.12}$$

the horizontal axis measures the inverse angle of the splitting and the vertical axis its hardness, both logarithmically. The choice of logarithmic axes is not cosmetic. Recall from Section 2.1 that the emission probability of a gluon emission carries the double structure $d\mathcal{P} \propto \alpha_s\, d\theta/\theta\; dE/E$, which is one factor for the collinear enhancement and one for the soft enhancement. When written in terms of Lund coordinates of equation (2.12), each of these factors becomes flat. The measure $d\theta/\theta$ is simply $d\ln(1/\Delta R)$, and the energy factor maps onto $d\ln k_{\mathsf{T}}$. The doubly divergent emission spectrum of the raw variables therefore becomes, on the Lund plane, an approximately uniform density of emissions per unit area,

$$\rho \;=\; \frac{1}{N_{\text{jet}}}\,\frac{dn_{\text{emissions}}}{d\ln(1/\Delta R)\; d\ln k_{\mathsf{T}}} \;\approx\; \text{const} \times \frac{\alpha_s}{\pi}, \tag{2.13}$$

up to the slow running of α_s and the effects discussed below. It is this flatness that makes the plane so useful against a uniform background. Any physical effect beyond the leading soft-collinear approximation

therefore appears as a localised enhancement or suppression, rather than being obscured by the rapidly falling raw spectrum. This uniform density is precisely what is observed in practice. Figure 2.1 shows the average primary Lund plane obtained from a simulated sample of QCD jets. Through the perturbative majority of the plane the density is approximately constant, as anticipated by equation (2.13), falling away only towards the boundaries of the kinematically accessible region.

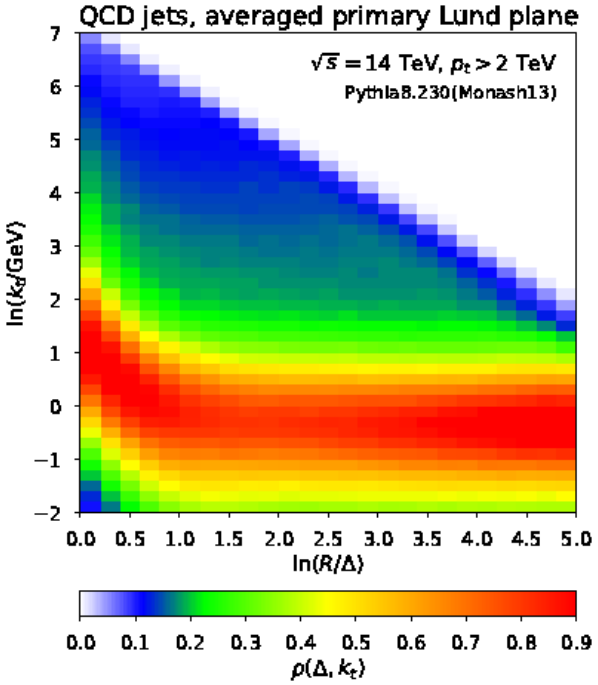


Figure 2.1 – The average primary Lund plane density, ρ, for jets clustered with the C/A algorithm and $R = 1$ having $p_t > 2$ TeV and $|y| < 2.5$, in a simulated QCD dijet sample [1].

2.3.2 Regions of the plane

The reachable area of the plane is delimited, and its interior can be divided into different regions with distinct physical source, sketched in Figure 2.2. The upper boundary is set by the kinematics of the jet. No emission can be harder than the jet which contains it, so it caps $\ln k_\mathrm{T}$ from above. Along the right of the plane, at large $\ln(1/\Delta R)$, lie the collinear emissions, those radiated at small angle to their emitter. Towards the bottom, at small $\ln k_\mathrm{T}$, lie the soft emissions. The perturbative description of Section 2.1 holds across the majority of this region, where k_T is large enough for α_s to remain small, making the density there be the near-constant value of equation (2.13).

The exception is the region in the lower-right corner, at both small angle and low transverse momentum. In that region, k_T falls to the scale of a few hundred MeV, α_s grows large, and the perturbative description breaks down. This region is known as the non-perturbative region and it is populated by hadronisation and by the soft, wide-angle contamination present inside the event. This way, one strong feature of the Lund jet plane is that these non-perturbative effects are confined to a restricted corner of the plane, clearly separated from the perturbative part and not mixed with the whole observable. A measurement can therefore restrict, by eye or by a specific cut, to the region where theory is applicable.

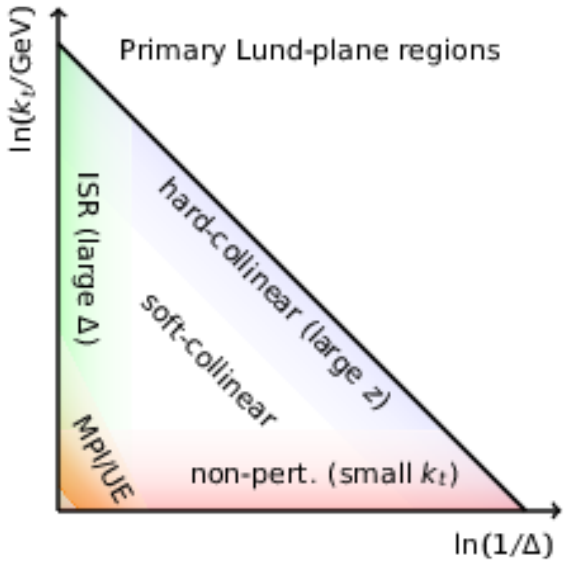


Figure 2.2 – Schematic representation of the different regions of the Lund plane [1].

2.3.3 Grooming and the primary and secondary planes

This clear separation motivates grooming methods. Grooming is the intentional removal of soft, wide-angle emissions that dominate non-perturbative regions of the Lund jet plane. The grooming method used in this work is SoftDrop [16], which walks through the declustering and, at each step, keeps the harder branch unless the softer one satisfies the following:

$$z > z_{\text{cut}} \left(\frac{\Delta R}{R} \right)^{\beta}, \tag{2.14}$$

in which case both branches are kept and the declustering procedure continues. The parameter z_{cut} specifies how soft an emission must be to be discarded, and β manages the angular reach of the grooming. The choice $\beta = 0$, adopted here for the SoftDrop in this work, makes the condition purely one of momentum fraction, $z > z_{\text{cut}}$, independent of angle. In this condition, SoftDrop becomes a modified Mass-Drop Tagger [17]. Applying this grooming method becomes very visible on the Lund jet plane because it removes the emissions in the soft, low k_{T} band, clearing the non-perturbative corner and leaving a plane populated only by the harder splittings.

In the present work, two versions of the Lund jet plane are built and used as follows. The primary Lund plane is built with the emissions along the hard branch of the jet and describes how the leading parton radiated. The secondary Lund plane is built by taking the chosen primary emission and declustering its softer branch in turn, using the same method as the primary Lund plane, as described in Section 2.2.3. The resulting points describe how that radiated parton itself went on to radiate. The two planes together deliver complementary information and both are needed for the spin observables of Section 2.4, which compare the orientation of a primary splitting with that of the secondary splitting it seeds. Figure 2.3 illustrates these two levels of structure for two example jets. The middle row shows the full Lund diagram, in which every emission, primary and secondary, is represented, while the bottom row shows the primary Lund plane alone, retaining a single point for each primary splitting.

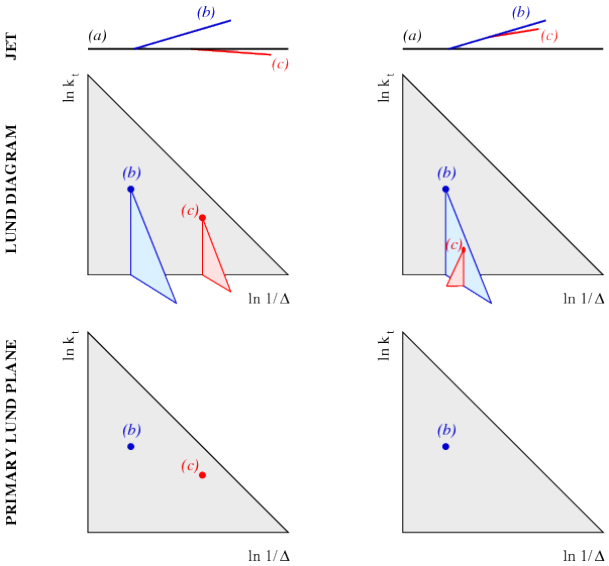


Figure 2.3 – Different representations for two jets. Top: the particles inside the jet. Middle: the full Lund diagram. Bottom: the primary Lund plane [1].

2.4 Collinear spin correlations

In the previous sections, the tools assembled describe a parton shower as a sequence of independent splittings, each blind to those before it. This description is incomplete since intermediate partons in a shower carry spin and their spin is correlated with the kinematics of the splitting that produces them. Furthermore, when that parton branches, that correlation is transmitted to the azimuthal orientation of its own decay products. So, successive splittings are therefore not independent, meaning that the plane of one branching is correlated with the plane of the next. These are the collinear spin correlations which this work presents, and they are the subject of this section.

This correlation effect is a quantum interference phenomenon, with no counterpart in a classical picture of sequential radiation, and it shows in a distinctive and robust way. This phenomenon manifests as a modulation of the azimuthal angle between successive splitting planes with a π periodicity instead of 2π, varying directly as a function of $\cos 2\psi$. This section develops that statement in four steps. Section 2.4.1 shows the physical mechanism and the origin of the $\cos 2\psi$ modulation. Section 2.4.2 shows how the size and sign of the modulation depend on the splitting channel, the property that makes flavour identification essential to the measurement. Sections 2.4.3 and 2.4.4 then introduce the two observables through which this work accesses the effect. These observables are the Lund declustering angle $\Delta\psi_{12}$, computed from the branching history of Section 2.3, and the three-point energy correlator, which reaches the same physics through the flow of energy within the jet.

2.4.1 Mechanism and the $\cos 2\psi$ modulation

The origin of the correlation lies in the spin of the intermediate gluon [18]. Consider a gluon produced in a first splitting that subsequently undergoes a second splitting of its own. This corresponds to the setup of the primary and secondary Lund planes discussed in Section 2.3. Since the gluon is a spin-one particle, a hard, nearly on-shell gluon possesses only two physical polarisation states, both transverse to its direction of motion. In the collinear limit, these states are most naturally described not as circular polarisations, but as two orthogonal linear polarisations, each representing an oscillation of the field along

a fixed axis in the plane transverse to the gluon momentum.

The polarisation of the intermediate gluon is not random. The first splitting, which produced it, selects a preferred transverse axis. Therefore, the gluon emerges linearly polarized with its polarisation correlated with the plane of that first branching. This is the main and essential point. A gluon radiated in a splitting carries a memory of the orientation of that splitting, engraved in the direction of its linear polarisation.

Afterward, that memory is read out by the second splitting. The probability that a linearly polarised gluon branches into a given final state depends on the azimuthal orientation of the second splitting plane with respect to the polarisation axis. Writing ψ for the angle between the first and the second splitting planes, the differential rate acquires a dependence on ψ through this alignment, and the two splittings become correlated in their relative azimuth.

The period of that correlation is fixed by the nature of the polarisation. A linear polarisation is specified by an axis, not by a direction along it. Rotating the polarisation by π returns the same physical state, since an oscillation along an axis is unchanged by reversing its sense. The azimuthal dependence must therefore be invariant under $\psi \to \psi + \pi$, which forbids a $\cos\psi$ term and admits $\cos 2\psi$ as its leading harmonic. This is the distinctive signature of the effect. The result has to have a modulation of period π rather than 2π, reflecting the headless, axial character of linear polarisation. Collinear spin correlations announce themselves not as any azimuthal dependence, but specifically as a $\cos 2\psi$ dependence.

At second order in the strong coupling and in the collinear limit, the cross section differential in ψ therefore takes the form

$$\frac{d\sigma}{d\psi} \propto a_0 \left(1 + \frac{a_2}{a_0} \cos 2\psi\right), \tag{2.15}$$

where a_0 is the spin-averaged rate, or, in other words, the result one would obtain by ignoring the polarisation of the intermediate gluon, and a_2 is the coefficient of the interference term [18]. The ratio a_2/a_0 measures the strength of the correlation and vanishes if the intermediate gluon carries no polarisation memory, and its magnitude sets the depth of the $\cos 2\psi$ modulation. Its sign determines the phase, whether the second splitting preferentially aligns with the first splitting plane, giving maxima at $\psi = 0, \pi$, or preferentially lies perpendicular to it, giving maxima at $\psi = \pm\pi/2$. The measurement of these spin correlations reduces, in essence, to the measurement of this single ratio, and the whole of the analysis that follows is directed towards extracting it cleanly.

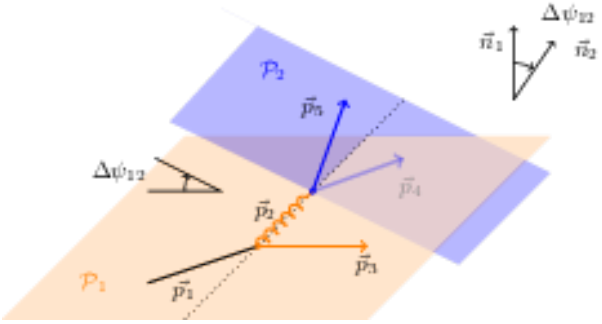


Figure 2.4 – Azimuthal angles are defined between successive splitting planes for the $1 \to 23$ splitting, $\mathcal{P}_1 \supset \{\vec{p}_2, \vec{p}_3\}$, and the $2 \to 45$ splitting, $\mathcal{P}_2 \supset \{\vec{p}_4, \vec{p}_5\}$. The figure also depicts the vectors normal to the two planes, $\vec{n}_1$ and $\vec{n}_2$ [18].

2.4.2 Channel dependence of a_2/a_0

The process described in Section 2.4.1 was made for a generic intermediate gluon, however, the strength and sign of the resulting modulation are not universal. These parameters depend on the nature of the second splitting, or, in other words, on the channel through which the polarised gluon decays. The two channels available are those of equations (2.4) and (2.5). The gluon can split into a pair of gluons, $g \to gg$, or into a quark-antiquark pair, $g \to q\bar{q}$. Each of these channels reads out the polarisation of the parent gluon, but each one of them does so with opposite sign and different strength [18].

The reason behind this difference comes from the spin of the produced particles. The angular distribution of a splitting depends on how the daughters' spins couple to the polarisation axis of the parent, and this coupling differs between a final state of two vector gluons and one of two spin-half fermions [18]. For a gluon fully linearly polarised along a fixed transverse direction, the splitting functions of equations (2.4) and (2.5) acquire a dependence on the azimuthal angle ϕ between the plane of the splitting and the polarisation axis [18],

$$P_{g\to gg}(z,\phi) = C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z) + z(1-z)\cos 2\phi\right], \tag{2.16}$$

$$P_{g\to q\bar{q}}(z,\phi) = T_R \left[z^2 + (1-z)^2 - 2z(1-z)\cos 2\phi\right], \tag{2.17}$$

which reduce to equations (2.4) and (2.5) when averaged over ϕ. The $\cos 2\phi$ term enters the two channels with opposite signs. The gluon produced in the first splitting is only partially polarised. It emerges linearly polarised in the plane of that splitting, with a degree of polarisation that depends on the momentum fraction z_1 it carries,

$$A_{q\to qg}(z_1) = \frac{2(1-z_1)}{1+(1-z_1)^2}, \qquad A_{g\to gg}(z_1) = \frac{(1-z_1)^2}{\left[1-z_1(1-z_1)\right]^2}. \tag{2.18}$$

At this order the ratio of equation (2.15) factorises into one factor from each splitting [18],

$$\frac{a_2}{a_0} = A(z_1)\, B(z_2), \tag{2.19}$$

where $B(z_2)$ is the ratio of the $\cos 2\phi$ term to the ϕ-independent term of equations (2.16) and (2.17),

$$B_{g\to gg}(z_2) = \frac{z_2^2(1-z_2)^2}{\left[1-z_2(1-z_2)\right]^2}, \qquad B_{g\to q\bar{q}}(z_2) = -\frac{2z_2(1-z_2)}{1-2z_2(1-z_2)}. \tag{2.20}$$

For symmetric splittings, $z_2 = 1/2$, $B_{g\to q\bar{q}}$ reaches -1, while $B_{g\to gg}$ never exceeds $1/9$.

This sign difference is the main goal which this work is built to overcome. If one does a measurement without distinguishing the two channels of the second splitting, the contributions of both sum together, leaving a result which almost cancels itself. The endured signal measured in this case is not the intrinsic strength of the correlation but a residue, because its magnitude and even its sign are set by the balance between the $g \to gg$ and $g \to q\bar{q}$ channels present in the sample. However, this interference is not absent. It is just diluted and this dilution reflects the flavour composition of the selection rather than any property of QCD.

Thus, it is required to isolate a single channel in order to make a reliable measurement of the effect. This work shows the selection procedure of the $g \to q\bar{q}$ splittings, and measures the parameter a_2/a_0 on that subsample alone, replacing the diluted inclusive residue with the coefficient of a single channel. This

is precisely what the infrared-safe and collinear-safe flavour labelling of Section 2.2.4 makes possible. IFN tags the flavour, exposed during the declustering, of each splitting, allowing the $g \to q\bar{q}$ channel to be enriched and the $\cos 2\psi$ modulation to be measured as a function of the purity of that enrichment. The channel dependence described here is therefore not just a feature of the physics but the reason the flavour selection developed in this work is required, and it is the procedure that connects the spin observable of the next section to the flavoured-jet algorithm of Section 2.2.4.

2.4.3 The Lund observable $\Delta\psi_{12}$

The effect described in the previous sections can be studied using a concrete observable, given by the Lund declustering of Section 2.3, in order to identify the two splittings whose planes are to be compared. This construction was proposed in [18], and the observable is the azimuthal angle $\Delta\psi_{12}$ between the hardest primary and secondary branchings of a jet.

The two splittings are selected as follows. When declustering the primary Lund sequence of a jet, one selects the splitting which passes the SoftDrop condition of equation (2.14) and has the largest relative transverse momentum k_T among all the primary sequence. This splitting is then called the primary splitting. Then, declustering the soft branch of this selected primary splitting, we build the secondary sequence generated by it. Thus, with the same procedure as in the primary sequence, one selects the splitting which passes the SoftDrop condition of equation (2.14) and has the largest relative transverse momentum k_T among all the secondary sequence. This splitting is called the secondary splitting. Each of the two chosen splittings defines a plane in momentum space. These are the plane $\mathcal{P}_1$, spanned by the branches of the first, and the plane $\mathcal{P}_2$ spanned by the branches of the second. With this, $\Delta\psi_{12}$ is the azimuthal angle between them, illustrated in Figure 2.4.

By construction, $\Delta\psi_{12}$ is the realisation for jets of the abstract angle ψ of equation (2.15). Its distribution therefore inherits the $\cos 2\psi$ modulation derived in Section 2.4.1,

$$\frac{1}{\sigma}\frac{d\sigma}{d\Delta\psi_{12}} \;\propto\; 1 + \frac{a_2}{a_0}\cos 2\Delta\psi_{12}, \tag{2.21}$$

and the coefficient a_2/a_0 extracted from it is the quantity that carries the spin-correlation physics. Two features of the definition are essential to what follows. First, because the second splitting is drawn from the secondary sequence seeded by the first, $\Delta\psi_{12}$ compares precisely the two successive branchings between which the polarisation memory of Section 2.4.1 is transmitted. Second, when that second splitting is the decay of an intermediate gluon, its channel $g \to gg$ or $g \to q\bar{q}$ can be identified by the flavour labelling of Section 2.2.4. Splittings of the form $q \to qg$, in which no intermediate gluon is present, carry no correlation and act only as a background to be removed. It is this flavour resolution that makes $\Delta\psi_{12}$ not merely a measurement of the inclusive, diluted modulation, but a channel-resolved probe of the effect. By restricting to secondary splittings tagged as $g \to q\bar{q}$, and thereby separating them both from $g \to gg$ splittings and from the correlation-free $q \to qg$ background, the sign cancellation of Section 2.4.2 is undone, and a_2/a_0 can be measured on a single channel.

2.4.4 The three-point energy correlator

Although the Lund observable described in the previous section gives a result for the collinear spin correlations of Section 2.4.1, it is not the only route to do so. The $\cos 2\psi$ modulation can be accessed through the energy flow within a jet and without any reference to any clustering history, using the three-point energy correlator (EEEC) [19]. Where $\Delta\psi_{12}$ reconstructs the two splitting planes by declustering, the energy correlator reads the effect directly from the angular and energy distribution of the jet's constituents, and the two therefore provide complementary handles on the same underlying physics.

The EEEC measures the energy-weighted distribution of triplets of particles as a function of their relative angles. Concretely, it is defined as the energy-weighted sum over all triplets of particles in the jet,

$$\frac{1}{\sigma}\frac{d^3\Sigma}{d\psi\, d\theta_S\, d\theta_L} = \sum_{i,j,k} \frac{p_{\mathrm{T},i}\, p_{\mathrm{T},j}\, p_{\mathrm{T},k}}{p_{\mathrm{T,jet}}^3}\, \delta(\psi - \psi_{ijk})\, \delta(\theta_S - \theta_{S,ijk})\, \delta(\theta_L - \theta_{L,ijk}), \tag{2.22}$$

where the sum runs over all triplets (i, j, k) of the jet's constituents, each weighted by the product of their transverse momenta normalised to that of the jet. Here θ_S and θ_L are the smallest and largest opening angles of the triplet, and ψ is the azimuthal angle between the plane spanned by the two closest particles and that spanned by the third. Its power for the present purpose is that this last variable retains sensitivity to the azimuthal orientation discussed in Section 2.4.1. This means that the correlator's dependence on ψ carries the same $\cos 2\psi$ modulation, governed by the same interference between gluon polarisation states. Being built from energy weights and angles alone, the correlator is infrared and collinear safe by construction and requires no declustering, no grooming and no jet-flavour assignment.

This makes the energy correlator an independent cross-check on the physics extracted through $\Delta\psi_{12}$. It reaches the same $\cos 2\psi$ effect by a route that shares none of the reconstruction choices on which the Lund observable depends, so that agreement between the two would establish that the measured modulation is a property of the radiation rather than an artefact of any one analysis chain. Note that the coefficient a_2/a_0 extracted from the energy correlator is not expected to coincide numerically with that from $\Delta\psi_{12}$, since the two observables define the azimuthal angle through different objects and carry different sensitivities to the kinematic selection. What the two share is the sign and the qualitative behaviour dictated by the underlying gluon-polarisation interference, and it is at that level that they cross-check one another. The correlator is developed in this work as a complementary probe alongside the primary $\Delta\psi_{12}$ measurement, and its implementation and results are presented in Chapter 4.

3 DATA AND ANALYSIS

The preceding chapter established the observables through which collinear spin correlations can be accessed. Following that, this chapter describes how they are measured in practice and presents the data on which this analysis is built, the software environment in which it runs, and the sequence of steps that leads from the raw available events to the azimuthal distributions of Chapter 4. The goal of this chapter is to reveal all the steps necessary to make this analysis, securing reproducibility. That said, enough detail is given, and the code used is available publicly in [5], so that the analysis can be repeated in full from public inputs alone.

The chapter is organised as follows. Section 3.1 introduces the CERN Open Data initiative and the principles that make an analysis of this kind possible outside the experimental collaborations. Section 3.2 describes the relevant features of the CMS detector, and Section 3.3 the data format and the specific dataset used. Section 3.4 presents the analysis pipeline, from the extraction of jet constituents to the computation of the spin observables. Section 3.5 describes the flavour selection that isolates the $g \to q\bar{q}$ channel, and Section 3.6 assesses the systematic uncertainties affecting the measurement.

3.1 CERN Open Data

The data analysed in this work were obtained through the CERN Open Data Portal [3], the platform through which the LHC experiments release reconstructed collision data and the associated simulation to the public. Launched in 2014, the portal was the first initiative of its kind in high-energy physics, and it has since grown into a major repository holding data from the ALICE, ATLAS, CMS and LHCb collaborations. The first releases came from the CMS experiment, and it is CMS data that this work uses.

The CERN Open Data Portal does not provide detector output but data at the same level of reconstruction used in collaborations' own published analysis. These datasets include physics objects such as jets, tracks and reconstructed particles, with the calibration already applied and run quality information required to use them correctly. Alongside the collision data, the portal also releases the corresponding Monte Carlo simulations, the software environment necessary to analyse the datasets and documentation describing their format and content. All these features of the portal make it possible for any user to use their data for their own analysis and interpret the results.

The importance of this portal in this work is twofold. First, as made clear in this chapter's introduction, it makes the measurement reproducible. Every input from the collision and respective simulated datasets is publicly available, so that the analysis can be repeated in full by anyone. Second, the portal specifies, for each dataset, the steps required to use them correctly. For each dataset used in this work, CMS provides a list of available and validated runs, with the proper calibrations and standard energy corrections and residual effects, applied by them.

This type of analysis are feasible outside the collaborations has been demonstrated by a growing body of work built on CMS Open Data. Studies using released CMS samples have addressed problems ranging from the discrimination of quark and gluon initiated jets [20] to the measurement of N-point energy correlators inside jets [21], the latter closely related to the observables of the present work. These analyses establish both that public data retain the information needed for substructure measurements

and that results of genuine physics interest can be extracted from them. The present work is situated in this line. It applies the same public data to the measurement of collinear spin correlations, an effect not previously accessed in this way.

The only small downside of working with public data is the choice of software environment. The released CMS data are stored in formats tied to the experiment's reconstruction software, and reading them requires a compatible environment. CMS provides this in the form of a dedicated container image, inside docker [22], that reproduces the software stack in which the data were generated. In this work that environment was used to read the released files and extract the jet and constituent information required by the analysis, which was then carried out in a lighter, independent environment. The details of this arrangement are given in Section 3.3 and Section 3.4.

3.2 The CMS detector

The collision data used and analysed in this work was recorded by the Compact Muon Solenoid (CMS). This giant detector, alongside with ATLAS, is a general-purpose detector, at the LHC [23]. It is a cylindrical apparatus with 21.6 m in length and 14.6 m of diameter, built in concentric layers around the beam pipe with the characteristic feature of having a superconducting solenoid, with 6 m in internal diameter, capable of producing a uniform 3.8 T magnetic field, that bends the trajectories of charged particles allowing for their momenta to be measured. Its layers are capable of capturing relevant features of this analysis and they will be described below in the order of a particle traversing them.

The CMS coordinate system has the origin centered at the nominal collision point inside the detector. Its y-axis points vertically upwards and the x-axis points radially inwards towards the center of the LHC. This makes the z-axis point along the beam axis towards Jura mountains from LHC Point 5 [23]. The azimuthal angle ϕ is measured from the x-axis in the x-y plane and the radial coordinate in this plane is denoted by r. The polar angle θ is measured from the z axis, although it is conventional to use the pseudorapidity $\eta = -\ln\tan(\theta/2)$, because differences in η are invariant under boosts along the beam. The transverse momentum is denoted as p_T and it is the component of momentum in the plane perpendicular to the beam. These are the variables which jets and their constituents are described in this work.

Starting from the layer closest to the interaction point, the inner-tracker is an all-silicon system of pixel and strip detectors covering $|\eta| < 2.5$. This layer is placed inside the solenoid's field and it reconstructs the trajectories of charged particles, measures their momenta and detects the primary interaction vertex. Since substructure observables used in this work are built from the trajectories and momenta of a jet's charged constituents, this tracker is the subdetector which these measurements directly rely on.

There are two calorimeters surrounding the inner-tracker. The electromagnetic calorimeter (ECAL) uses lead tungstate scintillating crystal to measure the energy of electrons and photons. Surrounding it, the hadron calorimeter (HCAL), a brass/scintillator device, measures the energy of the hadrons. Together, these two calorimeters cover the central region and, with more calorimeters forward extending the coverage to $|\eta| < 5$, give CMS a near complete solid-angle coverage needed to measure the flow of energy in an event. The muon system outside the solenoid, plays no important role for this analysis.

When CMS reconstructs jets, it does not rely on any one of these subdetectors in isolation. Instead, it uses its own particle-flow algorithm, tuned specifically for the CMS detector [24], which combines signals from all the subdetectors, including tracks from the inner-tracker and energy deposits from the calorimeters, to reconstruct a list of individual particles in each event, the particle-flow candidates.

It is these candidates that are the inputs to the jets' clustering algorithms of Section 2.2.1, and their charged component measured by the tracker that most directly determines the directions from which the substructure observables are built. This particle-flow algorithm is therefore the point at which the detector described here connects to the analysis that follows, and the data format in which its output is released is the subject of the next section. The following figure 3.1, gives a solid perspective of the CMS detector.

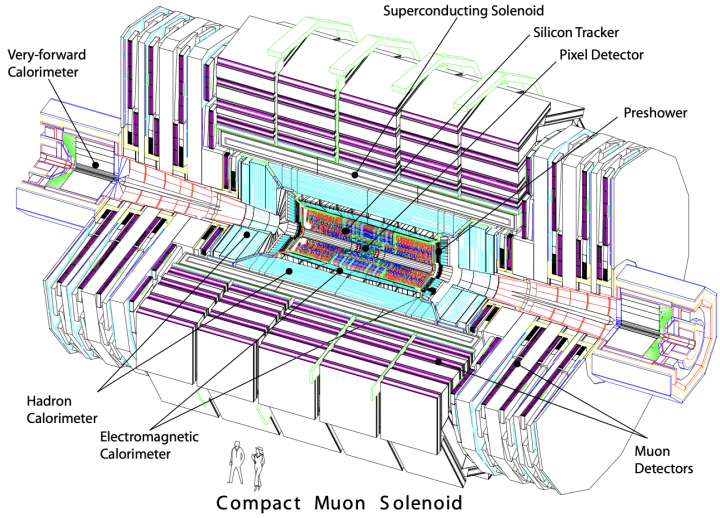


Figure 3.1 – A perspective view of the CMS detector [23].

3.3 Data format and datasets

The released reconstructed data, by CMS, in the CERN Open data, comes in a hierarchy of formats. Each format sacrifices completeness for size making the choice of format the first decision required for an analysis using their datasets, since it fixes what information is available and what variables are at hand. The three formats relevant here are summarised in Table 3.1.

Table 3.1 – The three CMS data-tier formats, in decreasing order of size and detail. Sizes are approximate and quoted relative to the baseline AOD.

Format	Content	Size
AOD	High-level physics objects plus substantial lower-level reconstruction detail.	baseline
MiniAOD	Analysis-level physics objects similar to AOD but slimmed: many collections are reduced, extra information is packed into lighter objects, and less commonly used low-level detail is removed.	$\sim 1/10$ of AOD
NanoAOD	Compact ntuple-style content: a limited set of per-object observables and precomputed IDs; little to no low-level detector detail (e.g., Particle Flow candidates dropped).	$\sim 1/20$ of MiniAOD

For this work, the requirement is having access to the constituents of a jet. The substructure observables of Chapter 2 are built by declustering a jet into its constituents, so this analysis requires the individual particle-flow candidates that build the jet and not just a summary of the jet's kinematics. This immediately invalidates the NanoAOD data format since the particle-flow candidates are not stored there. The AOD format stores the particle-flow candidates, however it also stores other low-level reconstruction details which this analysis does not need making it occupy a large volume of space. This leaves the MiniAOD

data format, which occupies the middle ground of space and still stores particle-flow candidates required for substructure while discarding the low-level information that is not needed. Therefore, it is for this reason that the MiniAOD data format was the choice made for the format to be used throughout this work.

Between all the MiniAOD datasets, this work analysed the JetHT primary dataset from the 2016 data-taking period, in MiniAOD format [25]. Its characteristics are collected in Table 3.2. The data are proton–proton collisions at a centre-of-mass energy of $\sqrt{s} = 13$ TeV, recorded during run period G of 2016 and released as part of the Ultra-Legacy reprocessing, the final and most complete calibration of the Run 2 data.

Table 3.2 – Characteristics of the JetHT primary dataset [25].

Property	Value
Dataset	`/JetHT/Run2016G-UL2016_MiniAODv2-v2/MINIAOD`
Collision	proton–proton, $\sqrt{s} = 13$ TeV
Run period	2016G (runs 278820–280385)
Events	1.21×10^8
Size	4.1 TiB (1236 files)

The JetHT dataset is well suited for this analysis since it is enriched in jet events. CMS records events online that pass a two-level trigger system, composed by a hardware Level-1 trigger followed by a software High Level Trigger (HLT) [26]. The events of this dataset are then those accepted by a set of HLT paths, which require either a large scalar sum of jet transverse momenta or the presence of one or two energetic jets. As a result the sample is dominated by the high-transverse-momentum, high-multiplicity final states in which jet substructure is most informative. Since the dataset is served by triggers based on large-radius jets, events are required to pass one of the `HLT_AK8PFJet` paths. The analysis is restricted to jets with transverse momentum above 700 GeV, where the unprescaled paths of highest threshold, `HLT_AK8PFJet450` and `HLT_AK8PFJet500`, are almost fully efficient. In this region any jet passing the selection also satisfies these paths, so the trigger requirement introduces no turn-on bias, and the sample sits in the hard, perturbative regime motivated in Section 2.1.

Furthermore, as required for all analysis of this dataset, the data is restricted to a list of runs certified as good physics, provided with the dataset release. So, events not in these certified runs do not enter this analysis. In addition, jets in this released sample do not require any energy calibrations to be applied to them, since they already carry the standard jet energy corrections applied by CMS in the Ultra-Legacy reprocessing.

Alongside the JetHT collision dataset, this analysis uses a simulated sample of QCD multijet events. This is because the flavour of a splitting inside a jet cannot be accessed in data, so it is not possible to determine the secondary splitting channel. What the simulation provides is the set of generated partons, carrying flavour, which are clustered in the same way as the reconstructed particles in data. First the partons are clustered into jets with the anti-k_T algorithm of section 2.2.2 and then, through the Cambridge/Aachen algorithm of Section 2.2.3 and the IFN algorithm of Section 2.2.4, into a branching history in which each splitting acquires a flavour. It is from this process that the channel of each splitting, $g \to q\bar{q}$, $g \to gg$ or $q \to qg$, is labelled. This flavour labelling process and how it is further used in collision data is discussed with more detail in Section 3.5. The simulated samples are QCD multijet datasets generated with Pythia 8 [27] using the CP5 tune [28], passed through the full simulation of the CMS detector and released in the same format as the collision data (MiniAODSIM). They are produced in intervals of the partonic transverse momentum $\hat{p}_\mathrm{T}$. The interval $800 < \hat{p}_\mathrm{T} < 1000$ GeV [29] is used to train the flavour selection, and the independent interval $600 < \hat{p}_\mathrm{T} < 800$ GeV [30] to compute its purity.

From these two datasets, there are some quantities extracted from each event, which are listed in Table 3.3. From each jet in collision datasets, their constituents are stored, since they are needed for the declustering of Chapter 2 and to build the Lund jet plane. The electric charge of each constituent is extracted in addition to its kinematics, as it is used later in the flavour selection of Section 3.5. In the simulated datasets, the same jet and constituent quantities are extracted, together with the final-state generated partons of the event, each with its identity. These partons carry the flavour information from which the splitting flavour is determined, as described in Section 3.5.

Table 3.3 – Quantities extracted from each event. The jet and constituent quantities are stored for both the collision data and the simulation. The generated partons are available in the simulation only.

Level	Quantity	Description
Jet	`jet_pt`, `jet_eta`, `jet_phi`, `jet_mass`	Transverse momentum, pseudorapidity, azimuth and mass of each jet
Constituent	`const_pt`, `const_eta`, `const_phi`, `const_mass`	Kinematics of each jet constituent
	`const_charge`	Electric charge of each constituent
Parton	`gen_pt`, `gen_eta`, `gen_phi`, `gen_mass`	Kinematics of each final-state parton (simulation only)
	`gen_pdgId`	Identity of each parton (simulation only)

The jets themselves are taken from the `slimmedJetsAK8` collection, which are the anti-k_T jets with $R = 0.8$ stored in MiniAOD, in a way that the jet definition of Section 2.2.2 is applied by CMS, making it not necessary to repeat this step in this analysis. Beyond the requirement of the transverse momentum described earlier in this section, jets are required to lie within $|\eta| < 1.7$, keeping them in the central, well-instrumented region of the detector. Events are further required to contain at least two such jets separated in azimuth by more than 2 radians, selecting the back to back dijet topology in which the two hard jets recoil against one another. In addition, jet's constituents are required to have $p_\mathrm{T} > 1$GeV in order to be recorded along with the jet. This selection follows closely that of the CMS measurement on which the present analysis is based [2], applied here to the earlier public dataset. The complete set of requirements is summarised in Table 3.4.

Table 3.4 – Event and jet selection. The trigger and certified-run requirements apply to the collision data only.

Requirement	Value
Trigger	`HLT_AK8PFJet` (data only)
Certified runs	validated-run list (data only)
Jet collection	`slimmedJetsAK8`, anti-k_T $R = 0.8$
Jet p_T	> 700 GeV
Jet $\|\eta\|$	< 1.7
Dijet topology	at least two jets with $\Delta\phi > 2$ rad
Constituent p_T	> 1 GeV

3.4 Analysis pipeline

The analysis made in this work, proceeds in a sequence of steps that start from the collision and simulated datasets, described in Section 3.3, and end with the final azimuthal distributions. These steps are split into two parts, separated by the software environment in which each one runs. The first part is the extraction of the jet's properties and constituent information from the datasets. This step, as indicated by CMS, must be done inside the CMS Open Data container or the CMS Virtual Machine , since reading the MiniAOD

format requires the experiment's own reconstruction software. In this work, this step was done using the CMS Open Data docker container (CMSSW) with, for each dataset, the recommended version of it. The second step covers declustering of the jets and the computation of the observables. It is done in a separate environment built on ROOT [31], FastJet [11] and Python. This separation makes the user able to use the CMSSW software just for a single step which is completely mandatory and allows the rest of the analysis to run in a modern and portable environment.

The stages of the pipeline are illustrated in Figure 3.2 and described in the following subsections. Section 3.4.1 describes the extraction of the jet ntuples from the collision and simulated samples. Section 3.4.2 describes the reclustering and declustering of each jet and the construction of the Lund observables and of $\Delta\psi_{12}$. Section 3.4.3 describes how the same procedure is applied to the generated partons in simulation in order to label the channel of each splitting. Section 3.4.4 describes the computation of the three-point energy correlator. The outputs of these stages feed the flavour selection of Section 3.5 and the measurement presented in Chapter 4. Since the datasets are extremely large, with a large number of jets and its variables, every stage has parallelisation integration. For the most demanding step of this analysis, which was the extraction of the variables, the code was run on LIP's computing cluster, which made a huge difference to the time required to complete this step. The code used is publicly available [5], so that each stage can be reproduced from the released data alone.

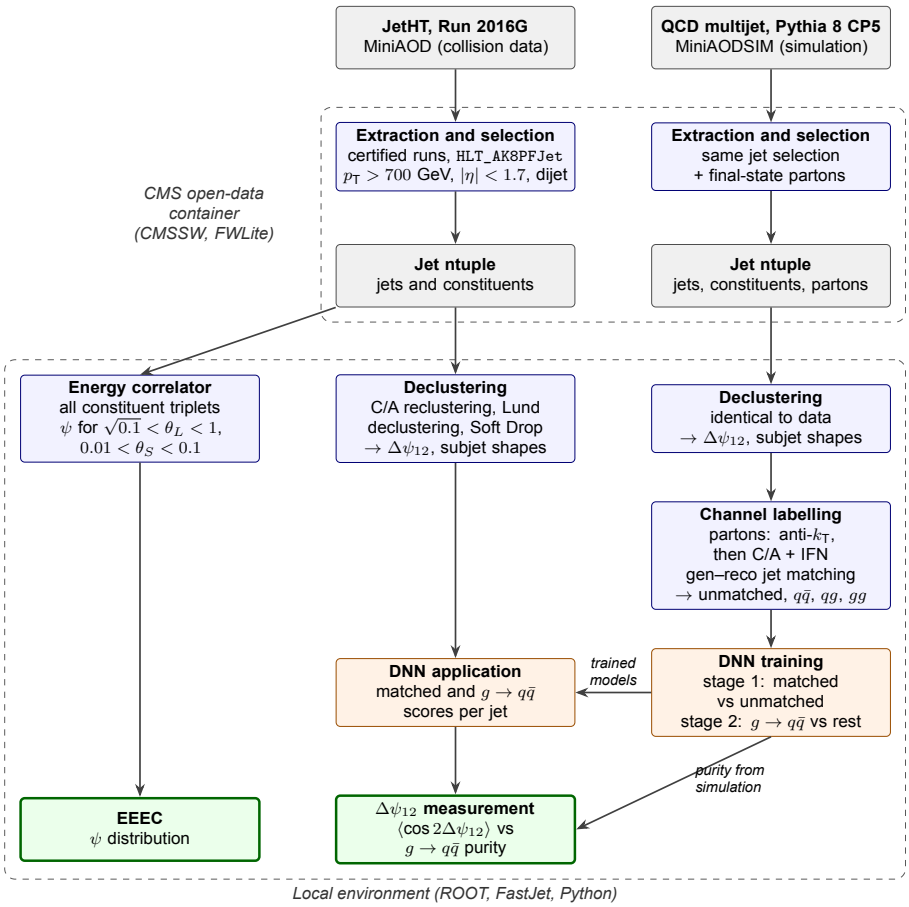


Figure 3.2 – Schematic of the analysis pipeline. The extraction stage runs inside the CMS environment and produces the jet ntuples. The declustering, the generator-level labelling and the energy-correlator computation run outside it, and feed the flavour selection and the final measurement.

3.4.1 Extraction

The first stage of this work's pipeline is to convert the released MiniAOD and MiniAODSIM datasets into compact ROOT ntuples with all the extracted information necessary for the rest of the analysis. This stage runs inside the CMS open data container, where the collections present inside the datasets are accessed through FWLite [32], a lightweight Python interface to the CMS event data model. The script developed by this work is capable of analysing datasets without downloading them. In order to achieve that, the dataset's files are listed in the file indices provided with each record on the CERN Open Data Portal and can be read remotely from the portal's storage. This completely avoids downloading/transferring the several terabytes of the full dataset but sacrifices a little bit of the execution time of the script. Since there are multiple files and all different from each other, this script is parallelised as an array of jobs on a computing cluster. Each one of these jobs processes a portion of all the dataset's files, creating for each file its own output file. After all the jobs are completed, a script was made to merge all the output files into a single one with all the extracted variables of the dataset.

For the collision data, each event goes through a sequence of requirements. As required by CMS, each event is first checked against the list of certified luminosity sections, also provided with the datasets, and discarded if its run or luminosity sections are not certified for physics. Then, the event is required to have triggered at least one of the `HLT_AK8PFJet` trigger paths. After passing these two conditions, the jets of the `slimmedJetsAK8` collection are read off and those satisfying the transverse-momentum and pseudorapidity requirements of Table 3.4 are extracted. For each extracted jet, its kinematics are stored together with the list of its constituents, with $p_{\mathrm{T}} > 1\mathrm{GeV}$, accompanied by their kinematics and electric charge. Finally, the event is kept only if it contains at least two extracted jets separated in azimuth by more than 2 radians, in which case it is written to the output tree.

The simulated sample is processed in a similar way with two differences. The run and luminosity certification and trigger requirements are not applied, since they concern real conditions of data taking and have no counterpart in simulation. Because the selected jets lie in the region where the trigger is almost fully efficient, omitting it in simulation does not introduce a difference between the two samples. In addition, a reconstructed jet and its constituents are only extracted if it has an associated generator-level jet, so that every simulated jet entering the analysis has a counterpart at generator level. Furthermore, for each event, the final state partons are also stored. These partons are quarks and gluons of the generator level record that do not split into other partons. In other words, they are the last partons of the shower before hadronisation. These final state partons are stored for the event as a whole, with their kinematics and identity, and are not associated to any reconstructed jet. This step is only done at a later stage described in Section 3.4.3.

The output of this stage is then a ROOT tree per sample, in which each entry corresponds to an event and holds the selected jets, their constituents and, in simulation, the generated partons, as listed in Table 3.3. After acquiring this tree, the CMS software is no longer needed for the remaining stages of this work's pipeline, since it is not in a MiniAOD data format and instead a simple ROOT tree.

3.4.2 Declustering and Lund observables

The second stage of this pipeline picks up the output ROOT tree, from the previous stage, and computes, for each jet present, the Lund declustering and the observables built from it. This script is written in C++ and uses ROOT and FastJet [11], running outside the CMS container. Each ntuple file can be processed

independently, so the stage is parallelised across files in the same way as the extraction, although it can also be run on a single merged output file. The variables computed by this script are stored in the same tree per jet as new branches, so that a single file holds, for every jet, both the extracted information and the observables derived from it.

For each event inside the ROOT tree, the script processes the jets in decreasing order of their transverse momentum, and each jet contributes to the analysis. The stored constituents of the jet are first converted into four-momenta from their transverse momentum, pseudorapidity, azimuth and mass, following equations 3.1-3.2, and each is tagged with its electric charge so that it can be traced back after the clustering process. The constituents are then reclustered with the C/A algorithm of Section 2.2.3, using $R = 2$. This value is far larger than the angular size of an AK8 jet (0.8), since it guarantees that all the constituents of a single anti-k_T jet are gathered into a single C/A object and that no part of the jet is lost during the reclustering. Jets with fewer than two constituents cannot be declustered and are flagged as failures.

$$p_{x,i} = p_{\mathsf{T},i} \cos\phi_i, \qquad p_{y,i} = p_{\mathsf{T},i} \sin\phi_i, \qquad p_{z,i} = p_{\mathsf{T},i} \sinh\eta_i, \tag{3.1}$$

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_i^2}. \tag{3.2}$$

After the reclustering, the full history is then declustered with the LundGenerator of the FastJet contrib [1], which follows the harder branch at each step and returns the ordered list of primary declusterings. For each declustering it provides the angular separation ΔR, the relative transverse momentum k_T, the momentum fraction z and the Lund coordinates of equation (2.12). Each primary declusterings, in order to be recorded, has to pass the SoftDrop condition of equation 2.14 with $z_\mathsf{cut} = 0.1$ and $\beta = 0$, reducing it to $z > 0.1$. With this list of primary declusterings, the primary splitting is chosen by selecting the declustering with largest k_T, with its harder and softer branches denoted as p_1 and p_2. The softer branch p_2 is then declustered in the same way, producing a second list of secondary declusterings. These secondary declusterings that satisfy, again, the SoftDrop condition, form the secondary Lund plane, and the one with largest k_T is selected as the secondary splitting, with harder and softer branches denoted as p_3 and p_4. If among the list of primary or secondary declusterings none satisfies the SoftDrop condition, the jet is flagged as a failure and no $\Delta\psi_{12}$ is computed.

Following the selection of the primary and secondary splittings, the azimuthal angle between the two splitting planes is computed from the three-momenta of the four branches, following the convention of the CMS measurement [2]. The normals of the two planes are

$$\hat{n}_1 = \frac{\vec{p}_1 \times \vec{p}_2}{|\vec{p}_1 \times \vec{p}_2|}, \qquad \hat{n}_2 = \frac{\vec{p}_3 \times \vec{p}_4}{|\vec{p}_3 \times \vec{p}_4|}, \tag{3.3}$$

and the angle between them is

$$\Delta\psi_{12} = \arccos(s\,\hat{n}_1 \cdot \hat{n}_2)\,, \qquad s = \mathrm{sign}[(\hat{n}_1 \times \hat{n}_2) \cdot \vec{p}_\mathsf{h}]\,, \tag{3.4}$$

where $\vec{p}_\mathsf{h}$ is the more energetic of $\vec{p}_1$ and $\vec{p}_2$. The sign s orients the two normals consistently, so that $\Delta\psi_{12}$ is defined in the range $[0, \pi]$. Since the two branches of a declustering are always separated in angle, both planes are well defined. Nevertheless, as a safeguard, the script returns no value in the case of exactly collinear branches, for which the cross products of equation (3.3) vanish.

To conclude this process, a set of shape variables is computed for each of the four branches p_1 to p_4 from the constituents it contains. These are the number of constituents n_{all}, the number of charged constituents n_{ch}, the transverse-momentum-weighted width σ and the transverse-momentum-weighted mean constituent momentum $\langle p_{\text{T}} \rangle$, defined as

$$\sigma = \frac{\sum_i p_{\text{T},i}\, \Delta R_i}{\sum_i p_{\text{T},i}}, \qquad \langle p_{\text{T}} \rangle = \frac{\sum_i p_{\text{T},i}^2}{\sum_i p_{\text{T},i}}, \tag{3.5}$$

where the sums run over the constituents of the branch and ΔR_i is the angular distance of constituent i from the branch axis. These variables play no role in the computation of $\Delta\psi_{12}$. They characterise the internal structure of each branch, which differs between branches initiated by quarks and by gluons, and they serve as the inputs to the flavour selection of Section 3.5. In collision data no truth information is available, so the channel of the second splitting is left undetermined at this stage. The same computation, together with the determination of the splitting channel, is carried out in simulation data and it is described on the following section.

3.4.3 Generator-level channel labelling

In simulated data, the reconstructed jets are processed exactly as described in Section 3.4.2, so that the Lund observables obtained are identical to the ones obtained in collision data. However, the channel label of the primary and secondary splittings is also assigned. This label is built from the final-state partons stored during extraction and its construction is made in three steps. Flavoured jets are first built from the partons, these generator-level jets are then matched to the reconstructed jets, and finally the splittings of the two are matched and labelled.

Firstly, the final-state partons of each event are converted into four-momenta in the same way as the reconstructed constituents, through equation (3.1) and (3.2), but, in addition, each parton is also assigned a flavour from their identity. A quark carries one unit of its flavour, an antiquark minus one unit, and a gluon is flavourless. The partons are then clustered using the anti-k_{T} algorithm with $R = 0.8$, mimicking the same definition as the reconstructed jets. Afterwards, it is necessary to match these new generated jets with their respective reconstructed jet. This step is performed in their C/A reclustering forms. On the reconstructed side, this is the object obtained in Section 3.4.2 by reclustering the constituents of the jet with the Cambridge/Aachen algorithm and $R = 2$. On the generator side, the partons of each anti-k_{T} jet containing at least two partons are reclustered in the same way, with the Cambridge/Aachen algorithm and $R = 2$, combined with the IFN procedure of Section 2.2.4 [15] with its parameter α set to 2. Each reconstructed jet, taken in order of decreasing transverse momentum, is matched to the closest generator-level object in the (η, ϕ) plane among those not already matched, provided that

$$\Delta R\big(j_{\text{C/A}}^{\text{reco}},\, j_{\text{C/A}}^{\text{gen}}\big) < 0.4, \tag{3.6}$$

a threshold equal to half the radius of the anti-k_{T} jets. This matching process is therefore one to one. Since each C/A object contains all the constituents of the anti-k_{T} jet from which it was built from, its axis should coincide with the axis of the original jet, making it equivalent to matching the anti-k_{T} jets themselves. The generator-level object selected in this way provides a Cambridge/Aachen history in which every branch carries an infrared- and collinear-safe flavour. A reconstructed jet with no generator-level counterpart satisfying equation (3.6) receives no truth label and is not used in the training of the flavour selection.

With the jet matching complete, the generator-level history is then declustered via the same process as the reconstructed one. The harder branch is followed at each step, each splitting is tested against the condition $z > 0.1$, and among the splittings that satisfy it the one with the largest k_{T} is taken as the primary generator-level splitting. Its softer branch is declustered in turn to obtain the secondary generator-level splitting. For each of these two splittings, the flavours of their branches are read from the IFN history and the channel of their splitting is labelled according to Table 3.5.

Table 3.5 – Assignment of the splitting channel from the IFN flavours of the two branches of a generator-level splitting.

Flavours of the two branches	Channel
both flavourless	$g \to gg$
one quark or antiquark and one flavourless	$q \to qg$
a quark and an antiquark of the same flavour	$g \to q\bar{q}$
any other combination	rest

Note that during this work, the channel of a splitting is denoted as the one inferred from the IFN flavours of its own branches and, although it is an infrared and collinear safe proxy, it may not coincide with the real truth channel of the splitting. The flavour of a branch reflects the whole of the radiation it contains, and the matching to the reconstructed jet and the hadronisation introduce further differences. Therefore, the modulation obtained for a given label in simulation does not reproduce the theoretical value of [18] for the corresponding channel.

In order to then label the primary and secondary reconstructed splittings, the primary and secondary generator-level splittings have to correspond geometrically to them. For a reconstructed splitting with branches p_a and p_b and opening angle Δ_{ab}, and the generator-level splitting with branches g_a and g_b, the requirement is

$$\Delta R(p_a, g_a) < \tfrac{1}{2}\,\Delta_{ab} \quad \text{and} \quad \Delta R(p_b, g_b) < \tfrac{1}{2}\,\Delta_{ab}, \tag{3.7}$$

where the same condition is also accepted with g_a and g_b interchanged. This is applied separately to the primary splitting, with branches p_1 and p_2, and to the secondary splitting with branches p_3 and p_4. When this condition is satisfied, the reconstructed splitting gets labelled with the channel of Table 3.5. When the condition fails to give any channel, the splitting is labelled as unmatched. An unmatched splitting is one for which the reconstructed declustering does not reproduce the partonic branching, typically because hadronisation or detector effects have altered the clustering sequence. This type of splitting carries no reliable flavour information, so it acts as a background which must be rejected.

Subsequently, each simulated jet carries a label for its first and second splittings, taking one of the values unmatched, $g \to q\bar{q}$, $q \to qg$, $g \to gg$ or rest. The label of the second splitting is the truth information on which the flavour selection of Section 3.5 is trained. It is used only as the target of that selection, while its inputs are the reconstructed quantities of Section 3.4.2, which are available in data as well.

3.4.4 The energy correlator

The three-point energy correlator of Section 2.4.4 is computed separately from the declustering, using directly the jet constituents stored after the extraction. This process does not involve any declustering or grooming, so that it probes the spin correlation through a different route from the Lund observables. It is evaluated both for all jets and for the jets selected by the flavour selection of Section 3.5. The definition

of the three-point energy correlator follows that of Ref. [18], based on the original proposal of Ref. [19], with the particle energies replaced by transverse momenta and the opening angles between particles replaced by their distances in the (η, ϕ) plane, as appropriate for a hadron collider.

For each jet, its constituents with non-zero transverse momentum are converted into three-momenta through equation (3.1), and jets with fewer than three such constituents are skipped. The angular distance between every pair of constituents is computed as

$$\Delta R_{ij} = \sqrt{(\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2}, \tag{3.8}$$

with the azimuthal difference computed in the range $(-\pi, \pi]$. Again, this distance is invariant under boosts along the beam axis, making it the natural measure of angular separation at a hadron collider, and it is the same measure used to define the jets themselves. An opening angle in three dimensions would instead shrink with the pseudorapidity of the jet, so that a fixed window in that angle would select different regions of the jet for central and forward jets.

This computation then runs over every unordered triplet of constituents. The three distances of a triplet are ordered as $\Delta R_S \leq \Delta R_M \leq \Delta R_L$, and the triplet is retained only if

$$\sqrt{0.1} < \Delta R_L < 1 \qquad \text{and} \qquad 0.01 < \Delta R_S < 0.1. \tag{3.9}$$

These are the values used for the opening angles in Ref. [18], but for the (η, ϕ) plane. These two definitions coincide at small angles for central jets. The window selects configurations with a strong hierarchy of angles. Two of the constituents lie close together and resolve a narrow splitting, while the third lies at a much wider angle. This configuration then approximates two successive splittings with strongly ordered opening angles, which is the regime where the correlator is sensitive to the spin of the intermediate gluon.

For any retained triplet, let a and b be the two constituents separated by the smallest distance ΔR_S and c the third. The azimuthal angle ψ is the angle between the plane containing $\vec{p}_a$ and $\vec{p}_b$ and the plane containing their sum and $\vec{p}_c$. It is computed from the three-momenta of equation (3.1), through the normals

$$\hat{n}_1 = \frac{\vec{p}_a \times \vec{p}_b}{|\vec{p}_a \times \vec{p}_b|}, \qquad \hat{n}_2 = \frac{(\vec{p}_a + \vec{p}_b) \times \vec{p}_c}{|(\vec{p}_a + \vec{p}_b) \times \vec{p}_c|}, \tag{3.10}$$

as the signed angle $\psi \in (-\pi, \pi]$ defined by

$$\cos\psi = \hat{n}_1 \cdot \hat{n}_2, \qquad \sin\psi = \hat{u} \cdot (\hat{n}_1 \times \hat{n}_2), \qquad \hat{u} = \frac{\vec{p}_a + \vec{p}_b}{|\vec{p}_a + \vec{p}_b|}. \tag{3.11}$$

Since both normals are perpendicular to $\hat{u}$, these two relations fix ψ uniquely, its sign being given by the sense of the rotation from $\hat{n}_1$ to $\hat{n}_2$ about $\hat{u}$. Exchanging a and b will reverse $\hat{n}_1$ and shift ψ by π, leaving $\cos 2\psi$ unchanged. The two constituents are taken in the order in which they are stored. Each triplet enters the distribution of ψ with the weight

$$w_{abc} = \frac{p_{\mathrm{T},a}\, p_{\mathrm{T},b}\, p_{\mathrm{T},c}}{p_{\mathrm{T,jet}}^3}, \tag{3.12}$$

in accordance with equation (2.22). Since the sum runs over unordered triplets, it will differ from a sum over all orderings of i, j and k only by a constant factor. This will affect the normalisation of the distribution, but not its shape and therefore it will not affect the factor a_2/a_0. The weighted distribution is accumulated in a histogram of 100 bins spanning $(-\pi, \pi]$. In the simulated datasets, the distribution is also accumulated separately for jets in each channel of the second splitting, as labelled in Section 3.4.3, which provides a

channel-by-channel reference for the modulation. Since the triplets of a jet share its constituents, they are not independent, and the statistical uncertainties are computed treating the jets as the independent units.

The main difficulty of this stage is the computational cost, since each jet may have many constituents and the number of triplets grows as the cube of their number. To make the computation manageable, the pairwise distances are computed once per jet, the window of equation (3.9) is applied before any other quantity is evaluated, so that most triplets are rejected at negligible cost, and the loop over triplets is compiled to machine code with the Numba just-in-time compiler [33]. The correlator of each jet is also computed only once and added to every category of jets to which the jet belongs, and the events are processed in parallel in independent blocks.

3.5 Flavour selection

The core measurement of this work requires isolating the secondary splitting channel $g \to q\bar{q}$. As discussed in Section 2.4.2, the modulation of this channel carries opposite sign compared to the $g \to gg$, so that an inclusive measurement dilutes the effect, while that of the other channel $q \to qg$ vanishes. However, the channel of a splitting is only defined in simulation, through generator-level labelling of Section 3.4.3, while in collision data it remains undefined.

The approach adopted here in this section is to learn, in simulation, the relation between this true channel and quantities that are measured in the same way in data. For that, a classifier was trained on the reconstructed features of each jet, with the truth label as its target, and is then applied unchanged to the collision data. An important feature is that its output is not a definite label but a score, whose value gives how likely a certain splitting is to belong to the $g \to q\bar{q}$ channel. With that score, one can require it to exceed a threshold and obtain an enriched sample of that channel. The $\Delta\psi_{12}$ observable is then measured as a function of this threshold, with the fraction of $g \to q\bar{q}$ splittings in each selected sample, its purity, determined from simulation.

This section is organised as follows. Section 3.5.1 describes the overall strategy and the classes into which the splittings are divided. Section 3.5.2 presents the input features of the classifier, and Section 3.5.3 the architecture of the networks and the procedure used to train them. Section 3.5.4 describes the alternative approaches that were considered, and Section 3.5.5 explains how the trained networks are applied to the data and how the purity of the selected samples is computed in simulation.

3.5.1 Strategy and classes

The generator level-labelling of section 3.4.3 divides the simulated jets into five classes according to their secondary splitting channel. A splitting can either be unmatched, if the reconstructed declustering does not reproduce the partonic branching, or be assigned with the channels $g \to q\bar{q}$, $q \to qg$, $g \to gg$, or with a residual class, denoted as rest. The composition of the simulated sample in these classes is given in Table 3.6. As one can notice, the unmatched label represents the majority of the dataset, about three quarters of it, but it carries no reliable flavour information, while, among the matched splittings, the $g \to q\bar{q}$ channel is a minority, outnumbered by both $q \to qg$ and $g \to gg$.

The selection procedure is therefore divided into two stages, each addressing a part of the dataset. The first stage tries to separate the matched splittings, the channels $g \to q\bar{q}$, $q \to qg$ and $g \to gg$, from the unmatched ones. Its output is a matched score, which is the classifier's confidence level that it is a

Table 3.6 – Composition of the labelled simulated sample by class of the second splitting, for jets with a valid $\Delta\psi_{12}$.

Class	Jets	Fraction (%)
unmatched	22 689 428	75.95
$g \to q\bar{q}$	896 936	3
$q \to qg$	1 736 862	5.81
$g \to gg$	4 298 461	14.39
rest	253 299	0.85
total	29 874 986	100.0

matched labelled channel. Only the jets with matched score exceeding 0.8 are passed to the second stage. The second stage is only trained on matched splittings and separates the $g \to q\bar{q}$ channel from the $q \to qg$ and $g \to gg$ channels taken together. This classifier's output is the $g \to q\bar{q}$ score on which the threshold of the measurement is placed. As suggested in [2], the residual category is excluded from the training of both stages, since it represents less than one percent of the dataset and represents other flavour configurations that do not correspond to any channel.

Dividing the problem in this way makes the two classifiers concentrate on a single binary question. The first stage removes the unmatched background, in which the flavour label is not useful. The second stage then has to solve only the separation that matters for the measurement, without spending capacity on the distinction between $q \to qg$ and $g \to gg$, which plays no role in the selection. The strong imbalance between the classes in both stages is compensated during training, as described in Section 3.5.3.

3.5.2 Input features

The inputs of the classifiers are built from the primary and secondary splittings of each jet and from the four branches computed in section 3.4.2. They fall into two groups, the shapes of the branches of the second splitting and the kinematics of the two selected splittings, and are listed in Table 3.7.

Each of these variables exploits the different radiation patterns of quarks and gluons. A gluon carries the larger colour factor, $C_A = 3$ against $C_F = 4/3$ for a quark, as introduced in Section 2.1, and therefore radiates more. Because of that, a branch initiated by a gluon tends to have more constituents, to be wider, and to share its momentum among softer constituents than a branch initiated by a quark. For each of the two branches p_3 and p_4 of the secondary splitting, this is captured by the number of constituents n_{all}, the number of charged constituents n_{ch}, the width σ and the weighted mean constituent momentum $\langle p_{\text{T}} \rangle$ of equation (3.5). This is supposed to capture the nature of these two branches. A $g \to q\bar{q}$ splitting produces two quark branches, a $g \to gg$ splitting two gluon branches, and a $q \to qg$ splitting one of each.

The kinematic variables exploit the different momentum sharing of the channels. From the splitting functions of equations (2.4) and (2.3), it can be noted that the channels $g \to gg$ and $q \to qg$ have increased probability when one of the two products is soft, while the $g \to q\bar{q}$ channel of equation (2.5), has increased probability with a more even sharing of the momentum. The momentum fraction z_2 of the secondary splitting is, therefore, sensitive to its channel as is shown in section 4.4.1 precisely that. It is also complemented by the angular separation ΔR_{34} and the relative transverse momentum $k_{\text{T},2}$ of the second splitting, and by the momentum fraction z_1 of the first.

The first stage uses these twelve variables directly while the second stage is also supplemented by combinations that do not depend on the ordering of the two branches. For each pair of variables and for the pair z_1 and z_2, the sum the absolute difference, the minimum, the maximum and the ratio of the

Table 3.7 – Input features of the two stages of the flavour selection. The superscripts (3) and (4) refer to the harder and softer branches of the second splitting.

Feature	Description	Stage 1	Stage 2
z_1, z_2	momentum fractions of the two splittings	yes	yes
ΔR_{34}	angular separation of the second splitting	yes	yes
$k_{\mathrm{T},2}$	relative transverse momentum of the second splitting	yes	yes
$n_{\mathrm{all}}^{(3)}$, $n_{\mathrm{all}}^{(4)}$	number of constituents	yes	yes
$n_{\mathrm{ch}}^{(3)}$, $n_{\mathrm{ch}}^{(4)}$	number of charged constituents	yes	yes
$\sigma^{(3)}$, $\sigma^{(4)}$	width	yes	yes
$\langle p_{\mathrm{T}} \rangle^{(3)}$, $\langle p_{\mathrm{T}} \rangle^{(4)}$	weighted mean constituent momentum	yes	yes
symmetric combinations	sum, difference, minimum, maximum and ratio	no	yes
total		12	37

minimum to the maximum are computed and added to the second classifier. This is motivated by the symmetry of the problem. Since in a $g \to q\bar{q}$ splitting both branches are quarks and in a $q \to qg$ splitting the quark is either the harder or the softer branch, this should help the classifier see the flavour information as a whole and not by its ordering. In the second stage the variables $\langle p_{\mathrm{T}} \rangle$ and $k_{\mathrm{T},2}$, whose distributions have long tails, are also replaced by their logarithms, $\ln(1 + x)$. In both stages every feature is finally standardised to zero mean and unit variance, using the statistics of the training sample.

Moreover, to avoid biasing the measurement, two quantities available for each jet are excluded from the inputs. The first is the observable itself, $\Delta\psi_{12}$. A classifier with access to it could select jets according to their azimuthal angle, and would thereby shape the very distribution that this work measures. The second is the transverse momentum of the jet. The flavour composition of the sample changes with the jet transverse momentum, and the selection is trained and calibrated in different intervals of $\hat{p}_{\mathrm{T}}$. Excluding the jet transverse momentum prevents the classifier from learning this composition, and keeps its output tied to the internal structure of the splitting rather than to the kinematics of the jet. It also allows the same classifier to be applied, in future work, to jets in other ranges of transverse momentum.

3.5.3 Network architecture and training

The two stages of the selection are feedforward neural networks, implemented with Keras [34] on top of TensorFlow [35], with the preprocessing of the inputs performed with scikit-learn [36]. The networks share the same structure with three hidden layers of 256, 128 and 64 neurons with rectified linear activations, and are regularised through a penalty on the squared magnitude of its weights, batch normalisation and dropout. They differ in the strength of the regularisation, in their output layer and in the loss function minimised during training. The complete configuration of the two networks is given in Table 3.8.

The first stage neural network has two output neurons with a softmax activation, thus returning the probabilities that the secondary splitting is matched or unmatched, and the first one is the matched score used in this selection. This network was trained by minimising the categorical cross-entropy and, since the unmatched splittings make about three quarters of the dataset, each class is weighted in the loss by a factor inversely proportional to its frequency. These weights are softened by raising them to the power 0.75, which prevents the network from being dominated by the majority class without overcompensating for it.

The second stage has a single output neuron with a sigmoid activation, whose value is the $g \to q\bar{q}$ score. It is trained with the binary focal loss [37],

Table 3.8 – Architecture and training configuration of the two stages of the flavour selection.

	Stage 1	Stage 2
Hidden layers	256, 128, 64	256, 128, 64
Activation	ReLU	ReLU
L2 penalty	10^{-4}	10^{-5}
Batch normalisation	after every layer	after every layer
Dropout	0.25, 0.20, 0.10	0.20, 0.20, 0.10
Output	two neurons, softmax	one neuron, sigmoid
Loss	categorical cross-entropy	binary focal loss
Class weights	balanced, power 0.75	balanced
Optimiser	Adam, rate 10^{-3}	Adam, rate 10^{-3}
Batch size	1024	1024
Maximum epochs	100	300
Monitored metric	macro-averaged F1 score	area under the PR curve

$$\mathcal{L}_{\text{focal}} = -\,\alpha_t\,(1-p_t)^{\gamma}\ln p_t, \tag{3.13}$$

where p_t is the probability that the network assigns to the correct class, and α_t equals α for splittings of the $g \to q\bar{q}$ channel and $1-\alpha$ for the others. The values $\gamma = 2$ and $\alpha = 0.5$ are used. The factor $(1-p_t)^{\gamma}$ reduces the contribution of splittings that are already classified with confidence, and concentrates the training on the ambiguous ones. This is very fitting to the problem since the channels overlap each other in the space of the input features and most splittings are hard to classify. Balanced class weights are applied furthermore, to compensate for the minority of the $g \to q\bar{q}$ channel among the other splittings.

The difference between the two networks is due to the different nature of their task. The first stage network separates classes which differ more clearly in the kinematics of the splittings, since unmatched splittings are those in which the reconstructed branching departs from the partonic one, and the separation is learned with relative ease. Therefore, this model has a stronger regularisation, with larger weight penalty and more dropout in its first layer, preventing it from learning accidental patterns or noise that exists only in the training dataset and encouraging it to more general patterns that are more likely to work on general unseen data. Meanwhile, the second stage must instead extract the smaller differences between the shapes of quark and gluon branches. It uses lighter regularisation and a longer maximum training time so it can learn these weaker features, while early stopping prevents overtraining in both cases. The class imbalance treatment also differs. In the first stage, neither class is of particular interest, and the softened weights keep the matched score closer to a probability, so that the fixed requirement of 0.8 retains its meaning. In the second stage the underrepresented group is the one of interest, and fully balanced weights ensure that it is not neglected. Finally, the metric used to select the best epoch matches how each output is used. The macro-averaged F1 score weights the two classes of the first stage equally, while the precision of the second stage is precisely the purity of the selected sample, so that the area under the precision–recall curve measures directly the purity achievable at each efficiency.

Both networks are trained on the simulated dataset with $800 < \hat{p}_{\text{T}} < 1000$ GeV, using the jets whose secondary splitting has been labelled, but excluding the residual class. The first stage network was trained on all such jets while the second stage network only on the matched ones. The sample is divided into training, validation and test subsets containing about two thirds, one sixth and one sixth of the events. The division was made by event, so that the jets of a given event never appear in different subsets. This step was also tested with a division made by jet and the performance was essentially the same, which was expected since the jets supposedly originate from independent showers. The preprocessing

of Section 3.5.2, is determined on the training subset alone and then applied unchanged to the other subsets and to the data.

The networks are optimised with the Adam algorithm [38] in batches of 1024 jets. After each epoch, the performance is evaluated on the validation subset, using the macro-averaged F1 score for the first stage and the area under the precision-recall curve for the second. The latter is better suited than the area under the Receiver Operating Characteristic (ROC) curve to a problem in which the class of interest is a minority. The learning rate is halved whenever this metric fails to improve for five consecutive epochs, down to a minimum of 10^{-6}, and training stops after fifteen epochs without improvement, at which point the weights of the best epoch are restored. The test subset is used only for the final evaluation of the networks.

3.5.4 Alternative approaches

The double stage selection described in the previous sections was used after comparison with three alternative strategies. All these different approaches were built on the same input features and the same preprocessing, and, again, they all exclude the $\Delta\psi_{12}$ variable from their inputs.

The first alternative is a single neural network that classifies each jet into the four categories unmatched, $g \to q\bar{q}$, $q \to qg$ and $g \to gg$. Its structure is the same as that of the double stage selection with the same hidden layers described in Table 3.8, a softmax output with four neurons, and is trained with the categorical cross-entropy and class weights softened with the power 0.75. This strategy attempts to solve the two problems that the two stage neural networks approach individually. A network of this kind is used by CMS in [2], where it was trained on a different generator and it tries to reject the unmatched splittings and identify the $g \to q\bar{q}$ channel.

The second approach replaces the binary second stage by a network with three outputs, trained on the matched splittings to separate $g \to q\bar{q}$, $q \to qg$ and $g \to gg$, with the same loss and class weights as the first alternative. It is applied after the same first stage, and its $g \to q\bar{q}$ output is used for the selection.

The third alternative is an unsupervised neural network. The matched splittings are described by the same features, reduced with a principal component analysis, and a Gaussian mixture model with full covariance matrices is fitted to them for several numbers of components, without any use of the flavour labels. The resulting clusters are then compared with the channel composition, in order to test whether the structure of the feature space separates the channels by itself.

Although these approaches seem fair, none of them could separate the $g \to q\bar{q}$ channel as effectively as the two-stage selection, in an high purity region, as shown in Chapter 4. The reason behind this may be because the $g \to q\bar{q}$ channel is a minority of the sample, and a network that must also separate the channels $q \to qg$ and $g \to gg$, or reject the unmatched splittings at the same time, spends part of its power on distinctions that this measurement does not require. In the unsupervised case, the channels overlap strongly in the space of the input features, so that there is no reason for the clusters of a Gaussian mixture to align with them, and the clusters found indeed mix the channels. The two-stage selection is therefore the one used in the remainder of this work.

3.5.5 Application and purity calibration

After training, the two stage networks are applied unchanged to every jet with a valid $\Delta\psi_{12}$, both in the collision data and in the simulated calibration sample. The inputs of each jet are built and preprocessed exactly as in training, using the same engineered features and the same imputation and standardisation, determined on the training subset. Thus, each jet receives two scores, the matched score of the first stage and the $g \to q\bar{q}$ score of the second, which are then stored alongside the other jet quantities.

Afterwards, the selection is made by picking jets with a matched score above 0.8, and a $g \to q\bar{q}$ score above a threshold t. The first requirement is fixed and removes most of the unmatched splittings. The threshold t varies, and each of its values defines a selected sample. As t increases, the samples become smaller and progressively enriched in the $g \to q\bar{q}$ channel.

For each threshold t, the composition of the selected sample is computed on the simulated sample with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV, which is independent of the one used in training. The purity and the efficiency of the selection are defined as

$$P(t) = \frac{N_{q\bar{q}}(t)}{N(t)}, \qquad \varepsilon(t) = \frac{N_{q\bar{q}}(t)}{N_{q\bar{q}}}, \tag{3.14}$$

where $N(t)$ is the number of selected jets, $N_{q\bar{q}}(t)$ the number of them whose secondary splitting is labelled $g \to q\bar{q}$, and $N_{q\bar{q}}$ the total number of jets carrying that label before the selection. When increasing the threshold, the purity grows while the efficiency declines, and the comparison between the two is shown in Chapter 4 summarising the performance of the selection.

At each threshold, the mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of equation (3.15) is computed on the selected jets, separately in data and simulation, with the statistical uncertainty of equation (3.16). The measurement made is presented as a function of the threshold, with the purity of equation (3.14) given for each value. The two references, computed from simulation and data, are the mean over all jets labelled $g \to q\bar{q}$ and the mean over those labelled $q \to qg$ or $g \to gg$, which indicates the value expected for the labelled channel and for the background before any selection. The evaluated threshold range is from 0.3 up to 0.6 and beyond this value the number of jets falls below a few hundred and the relative statistical uncertainty exceeds 10%, making these points carry little information.

The main assumption of this procedure is that the purity computed in simulation samples also describes the samples selected from data at the same threshold. The networks are identical in the two cases, but the composition of the selected samples depends on how well the simulation reproduces the data, as discussed in Section 3.6.2.

3.6 Systematic uncertainties

This work's quantity is the mean value $\langle \cos 2\Delta\psi_{12} \rangle$, given per threshold of the $g \to q\bar{q}$ score organized in samples, accompanied by the purity assigned to each sample, according to simulation. The $\langle \cos 2\Delta\psi_{12} \rangle$ and purity quantities are both subject to uncertainties. The statistical uncertainty comes from the finite number of jets in each sample and it is evaluated at every point of this measurement. The systematic uncertainties arise from the modelling of the simulation on which the selection is trained and calibrated, from the response of the detector, and from some choices made in this analysis. However, unlike the

statistical uncertainties, they do not decrease when extracting more jets and they may shift the result in a definite direction.

A complete quantitative evaluation of every systematic uncertainty was not done in this work. This section just identifies the relevant sources to the measurement, describes how each of them is managed by this analysis, and states which ones are not evaluated. Section 3.6.1 describes the statistical uncertainty. Section 3.6.2 discusses the modelling of the purity, which is the dominant systematic source. Section 3.6.3 addresses the effects of the detector, and Section 3.6.4 the trigger and the choices of the analysis. An overview of the sources considered and of the treatment of each is given in Table 3.9, and each source is discussed in turn in the subsections that follow.

Table 3.9 – Overview of the sources of uncertainty and of their treatment in this work. Each source is discussed in the section indicated.

Source	Treatment	Section
Statistical	Evaluated for every selected sample	3.6.1
Modelling of the purity	Not evaluated, single generator	3.6.2
Kinematic dependence of purity	Computed on an independent sample	3.6.2
Detector resolution	Present in data and simulation, not unfolded	3.6.3
Jet energy scale	Expected small, not evaluated	3.6.3
Pile-up	Suppressed by $z > 0.1$, overlaid in simulation	3.6.3
Trigger	Negligible above 700 GeV	3.6.4
Analysis choices	Fixed, not varied	3.6.4

3.6.1 Statistical uncertainty

For a selected sample of N jets, the measured quantity is the mean computed as

$$\langle \cos 2\Delta\psi_{12} \rangle = \frac{1}{N} \sum_{j=1}^{N} \cos 2\Delta\psi_{12}^{(j)}, \tag{3.15}$$

and its statistical uncertainty is the standard error of the mean,

$$\delta_{\text{stat}} = \frac{s}{\sqrt{N}}, \qquad s^2 = \frac{1}{N-1} \sum_{j=1}^{N} \left(\cos 2\Delta\psi_{12}^{(j)} - \langle \cos 2\Delta\psi_{12} \rangle \right)^2 . \tag{3.16}$$

For a distribution of the form of equation (2.21), the mean value is $\langle \cos 2\Delta\psi_{12} \rangle = \frac{1}{2}\, a_2/a_0$, so that the uncertainty on a_2/a_0 is twice that of equation (3.16).

This expression assumes that each jet entering the sample is independent. In each event after selection, there are at least two jets and all of them enter the analysis, although, events with more than two jets are rare. In that case, the jets of an event originate from separate showers, and their second splittings are essentially uncorrelated. This statement is supported by the behaviour of the flavour selection, because its performance did not change, when training and test samples were separated by event rather than by jet.

The purity given for each sample is the fraction of selected simulated jets whose secondary splitting is labelled $g \to q\bar{q}$ by the IFN procedure. Its statistical uncertainty follows from the binomial distribution,

$$\delta P = \sqrt{\frac{P\,(1-P)}{N_{\mathrm{MC}}}}, \tag{3.17}$$

where N_{MC} is the number of selected simulated jets. The same expression of equation (3.16) gives the statistical uncertainty of the mean values obtained in simulation. In this work, when showing the results as a function of the threshold, the threshold itself is an exact choice, given by the classifiers, and carries no uncertainty, so that only the vertical uncertainties of equation (3.16) are presented.

There are two features of the statistical uncertainties that should be kept in mind while reading the results. The first is that they grow with the threshold, since raising it catches fewer jets, making the threshold's range limited by the number of jets that survive. The second is that each sample is a subset of the samples selected at lower thresholds, so that consecutive points share most of their jets and their fluctuations are correlated. A trend across thresholds should therefore be judged against the uncertainty of the individual points rather than against the smoothness of the curve.

3.6.2 Modelling of the purity

As said on the previous subsection, the purity is the fraction of the selected simulated jets whose secondary splitting is labelled $g \to q\bar{q}$ by the IFN procedure. It is taken from the simulation and transferred to the data, and it is therefore correct for the data only if the simulation reproduces two of their features. The first is the flavour composition of the secondary splitting, in particular the rate of $g \to q\bar{q}$ splittings relative to $q \to qg$ and $g \to gg$. The second is the distribution of the input features of Table 3.7 within each channel. If one of them is mismodelled, the same threshold selects a sample of different purity in data than in simulation.

There is direct evidence that this effect is not negligible. In the CMS measurement on which this analysis is based [2], neither Pythia 8 nor Herwig 7 describes the distribution of the classifier output in data, owing to the mismodelling of several input variables, and the simulation is reweighted to reproduce the data before the purity is determined, with an associated uncertainty derived from the difference in flavour composition between the two generators. No such reweighting is applied in this work, and the purities quoted are those of Pythia 8 with the CP5 tune. A repetition of the calibration with a generator based on a different parton shower and hadronisation model was not performed, and the modelling of the purity is therefore the largest source of systematic uncertainty that is not quantified here.

The labelling done in Section 3.5.1, provides a partial consistency check. The composition shown in Table 3.6 agrees closely with the one obtained in [2]. This confirms that the labels are constructed correctly, but it does not prove that the simulation describes the data.

Another aspect concerning the purity is its dependence on the kinematics of the jets. The selection is trained on simulation with $800 < \hat{p}_{\mathrm{T}} < 1000$ GeV and computed on the independent interval $600 < \hat{p}_{\mathrm{T}} < 800$ GeV, which secures that the purity is not biased by the training. Since $\hat{p}_{\mathrm{T}}$ is the transverse momentum of the partonic hard process, each sample contains jets over a broader range of transverse momenta than its nominal interval, and both contain jets above the selection threshold of 700 GeV. The distribution of jet transverse momenta in a single interval of $\hat{p}_{\mathrm{T}}$ still differs from the data, which receives contributions from all intervals, weighted by their cross sections. Although the jet transverse momentum is not a variable which the classifier uses, the relation between threshold and purity may still vary with it, so that a calibration performed on a single interval may not describe the full data sample exactly. Combining several intervals,

each weighted by its cross section, would reproduce the spectrum of the data and remove this limitation.

In addition, the purity depends on the definition of the flavour label. In this work, it is defined as the matching of the branches of the second splitting, p_3 and p_4, to their generator-level counterparts. However, other choices are possible and, for example, the purities quoted in [2] are obtained by matching the branches p_2 and p_3 instead, yielding far fewer unmatched splittings, thus a different result. Purities obtained with different definitions are, of course, not directly comparable.

3.6.3 Detector effects

This measurement is performed on reconstructed particles which are not corrected to the level of stable particles, making every quantity affected by the finite resolution of the detector. The most relevant effect for an angular observable is the resolution on the directions of the constituents. For charged particles, the direction is measured by the inner tracker with high precision. For neutral particles, it is determined only by the calorimeters, whose granularity in the barrel corresponds to cells of about 0.0174×0.0174 in (η, ϕ) for the electromagnetic calorimeter and 0.087×0.087 for the hadron calorimeter [23]. Splittings resolved at angles comparable to these sizes, are thus measured less precisely which is particularly relevant for the lower edge $\Delta R_S > 0.01$ of the window of the energy correlator. The smearing of the directions blurs the orientation of the reconstructed planes and dilutes the modulation, so that the measured values of $\langle \cos 2\Delta\psi_{12} \rangle$ are expected to be somewhat smaller in magnitude than at particle level.

The simulation dataset is done with a full simulation of the CMS detector, so that data and simulation are compared at the same level and this dilution is present in both. Therefore, comparing both the datasets is consistent. However the values obtained are specific to the CMS detector and cannot be compared with calculations at particle level. Such a comparison would require an unfolding of the detector response, as performed in [2], which is not done in this work.

The jets come with the standard energy corrections done by CMS, whose residual uncertainty moves jets across the selection threshold of 700 GeV. Since the measured observable is an angle averaged by many jets, and the jets near the threshold form a small part of the sample, this effect is expected to be small, and it is not evaluated.

Particles from additional proton–proton interactions in the same bunch crossing, known as pile-up, enter the jets as soft constituents spread over wide angles. Part of this contribution is already removed in the reconstruction of the jets, since charged particles originating from pile-up vertices are discarded before the clustering, a procedure known as charged-hadron subtraction. Neutral pile-up particles may nevertheless remain among the constituents, with their full momenta. The CMS measurement goes further and also rescales the momenta of neutral particles by their probability of originating from the primary interaction [2], which is not done in this work. The effect of the remaining pile-up on the observable is limited by the condition $z > 0.1$ imposed on the selected splittings, which rejects the soft branches in which pile-up particles are mostly found. The simulated samples include an overlay of pile-up interactions, so that its effect is present in data and simulation alike.

3.6.4 Trigger and analysis choices

The trigger is not expected to introduce a significant uncertainty, since, in CMS, a trigger requiring a jet with a transverse momentum above 550 GeV was found to be more than 99.5% efficient for jets above

700 GeV [2]. Thus, the trigger paths used here in this analysis, are expected to have almost full efficiency, but this efficiency is not measured in this work.

Regarding some analysis choices done along the development of this work, several parameters were fixed. These are the grooming condition $z > 0.1$ applied to the selected splittings, the requirement of a matched score above 0.8 in the first stage of the selection, the angular window of the energy correlator, and the criteria used to match the generator-level jets and splittings to the reconstructed ones. Playing with these conditions would probe the stability of this work's result, however, this was not done here.

Of the sources discussed in this section, the two related to the purity are the ones that most limit the interpretation of the result. They are the reason for presenting the measurement as a function of the threshold, with the purity from simulation given as additional information rather than as a calibrated axis. A full evaluation of these sources, based on an alternative generator and on the variations of the analysis choices listed above, is the most important extension of the present analysis and is discussed in Chapter 5.

4 RESULTS AND DISCUSSION

This chapter presents the results obtained from the analysis described in Chapter 3. They are revealed in the order in which the analysis is built, starting by showing that the selected jets and their substructure are described consistently in data and simulation, followed by the results of the flavour labelling and selection, and finally by the measurement of the spin-sensitive observables.

Section 4.1 presents the selected jets and their kinematic distributions. Section 4.2 shows the primary and secondary Lund jet planes, and Section 4.3 the distribution of $\Delta\psi_{12}$ before any flavour selection. Section 4.4 describes the composition of the simulated sample in the channels defined by the IFN labelling and the modulation associated with each of them. Section 4.5 evaluates the performance of the flavour selection and compares it with the alternative approaches of Section 3.5.4. Section 4.6 presents the central result of this work, the mean value $\langle \cos 2\Delta\psi_{12} \rangle$ as a function of the threshold of the flavour selection, in data and in simulation. Section 4.7 presents the three-point energy correlator, and Section 4.8 discusses the results in the light of the theoretical expectations and of the CMS measurement.

The achieved results were obtained at the level of reconstructed particles, and data and simulation are compared at this same level. The collision data are those of the JetHT dataset of Table 3.2, and the two simulation datasets are the sample with $800 < \hat{p}_\mathrm{T} < 1000$ GeV, on which the flavour selection is trained, and the sample with $600 < \hat{p}_\mathrm{T} < 800$ GeV which is used for the calibration of the purity and for every comparison with data. The sample used in each figure is indicated in its caption and, unless stated otherwise, the distributions shown are normalised to unit area, so that their shapes are compared, and the uncertainties are statistical only. The systematic uncertainties discussed in Section 3.6 are not represented in the figures.

4.1 Event selection and jet kinematics

The number of events and jets, of collision and simulation data, remaining after the selection of Table 3.4 is shown in Table 4.1. Of all the selected jets, only those whose primary and the secondary declusterings both contain a splitting satisfying $z > 0.1$ yield a valid value of $\Delta\psi_{12}$. These jets define the sample on which all subsequent results are based.

Table 4.1 – Number of events, selected jets and jets with a valid $\Delta\psi_{12}$, in the collision data and in the simulated calibration sample.

	Data	Simulation $600 < \hat{p}_\mathrm{T} < 800$	Simulation $800 < \hat{p}_\mathrm{T} < 1000$
Events	30 241	5 737 037	29 652 066
Selected jets	60 566	11 476 817	59 389 760
Jets with a valid $\Delta\psi_{12}$	45 755	8 495 207	43 071 113

The kinematic distributions of the selected jets are shown in Figure 4.1, for collision and simulation data. The three distributions start at 700 GeV, because of the imposed selection threshold. It is possible to see that in data, the distribution falls smoothly and it extends to higher values, since it receives contributions from all intervals of $\hat{p}_\mathrm{T}$. However, in simulation, the distributions fall much more quickly and they struggle to reach the same higher values as in collision data, since their jets originate from an interval of $600 < \hat{p}_\mathrm{T} < 800$ GeV and $800 < \hat{p}_\mathrm{T} < 1000$ GeV. This difference is the one discussed in Section

3.6.2, and it should be kept in mind when comparing the other distributions. Now, comparing the two simulated distributions it is worth noting that the $800 < \hat{p}_{\mathrm{T}} < 1000$ GeV distribution extends further above the selection threshold. Although the flavour selection is trained, calibrated and applied on jets with partly different distributions, the measurement is still possible since the jet transverse momentum is excluded from its inputs, as discussed in Section 3.5.2, while the residual effect of this difference on the purity is discussed in Section 3.6.2.

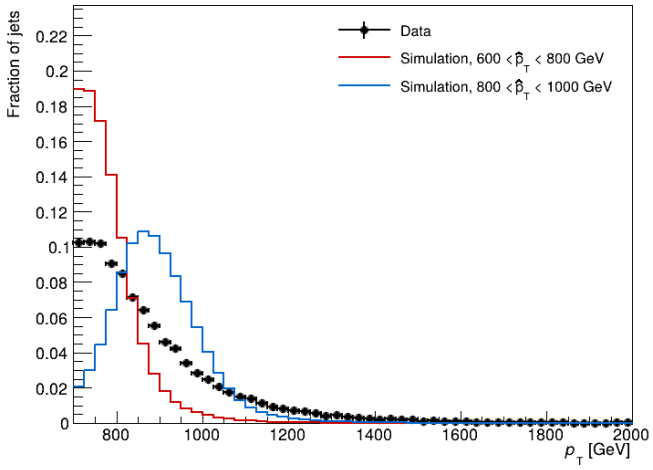


Figure 4.1 – Transverse-momentum distribution of the selected jets in the collision data and in the simulated samples with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV and $800 < \hat{p}_{\mathrm{T}} < 1000$ GeV, normalised to unit area.

The other distribution which is relevant for the comparison of the different datasets is the number of constituents of the selected jets. This is the extracted quantity most related to the substructure observables, since it shows the amount of radiation inside a jet, and it contributes to the flavour selection procedure through the multiplicities of the branches of the second splitting. Its distribution is shown in Figure 4.2 for the collision data and for the two simulated samples, normalised to unit area. The three distributions are very similar, peaking at about 35 constituents and extending up to about 130. The simulated samples reproduce the shape of the data very closely.

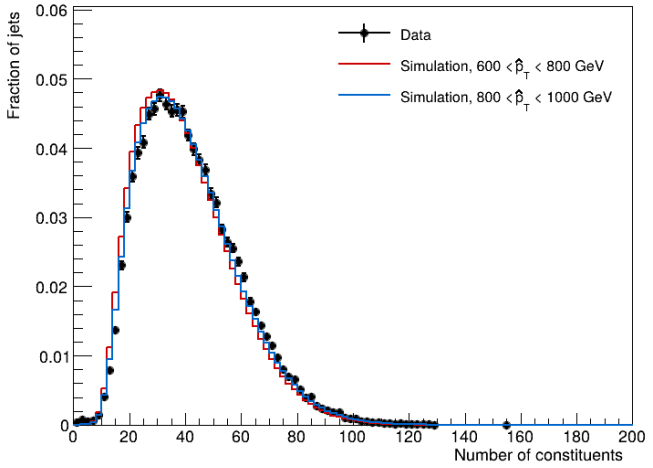


Figure 4.2 – Number of constituents of the selected jets in the collision data and in the simulated samples with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV and $800 < \hat{p}_{\mathrm{T}} < 1000$ GeV, normalised to unit area.

The very small difference between the samples represents well the pattern from their transverse momenta.

The sample with $600 < \hat{p}_\mathrm{T} < 800$ GeV, whose jets are the softest, is slightly shifted towards lower number of constituents while the sample with $800 < \hat{p}_\mathrm{T} < 1000$ GeV lies closest to the data. This difference is very small despite the very different spectra of Figure 4.1, reflecting a slow growth of the number of constituents with the jet's transverse momentum. The data lies slightly above both simulated samples between 50 and 80 constituents, which may reflect the modelling of the parton shower and of hadronisation, as well as differences in the pile-up conditions between data and simulation, since neutral pile-up particles remain among the constituents. Nevertheless, the number of constituents of the jets is well described by the simulation. Since the multiplicities of the branches are among the inputs of the flavour selection, this agreement supports, although it does not by itself establish, the transfer of the selection from simulation to data discussed in Section 3.6.2.

4.2 Lund jet planes

The average primary and secondary Lund planes obtained from the collision and simulated calibration samples respectively, are shown in Figure 4.3. Each plane shows the density ρ of equation (2.13), which is the number of emissions per jet and per unit area in the coordinates $\ln(1/\Delta R)$ and $\ln k_\mathrm{T}$. Since the observable $\Delta\psi_{12}$ is the final measurement of this work, only the declusterings that satisfy the condition $z > 0.1$ are included.

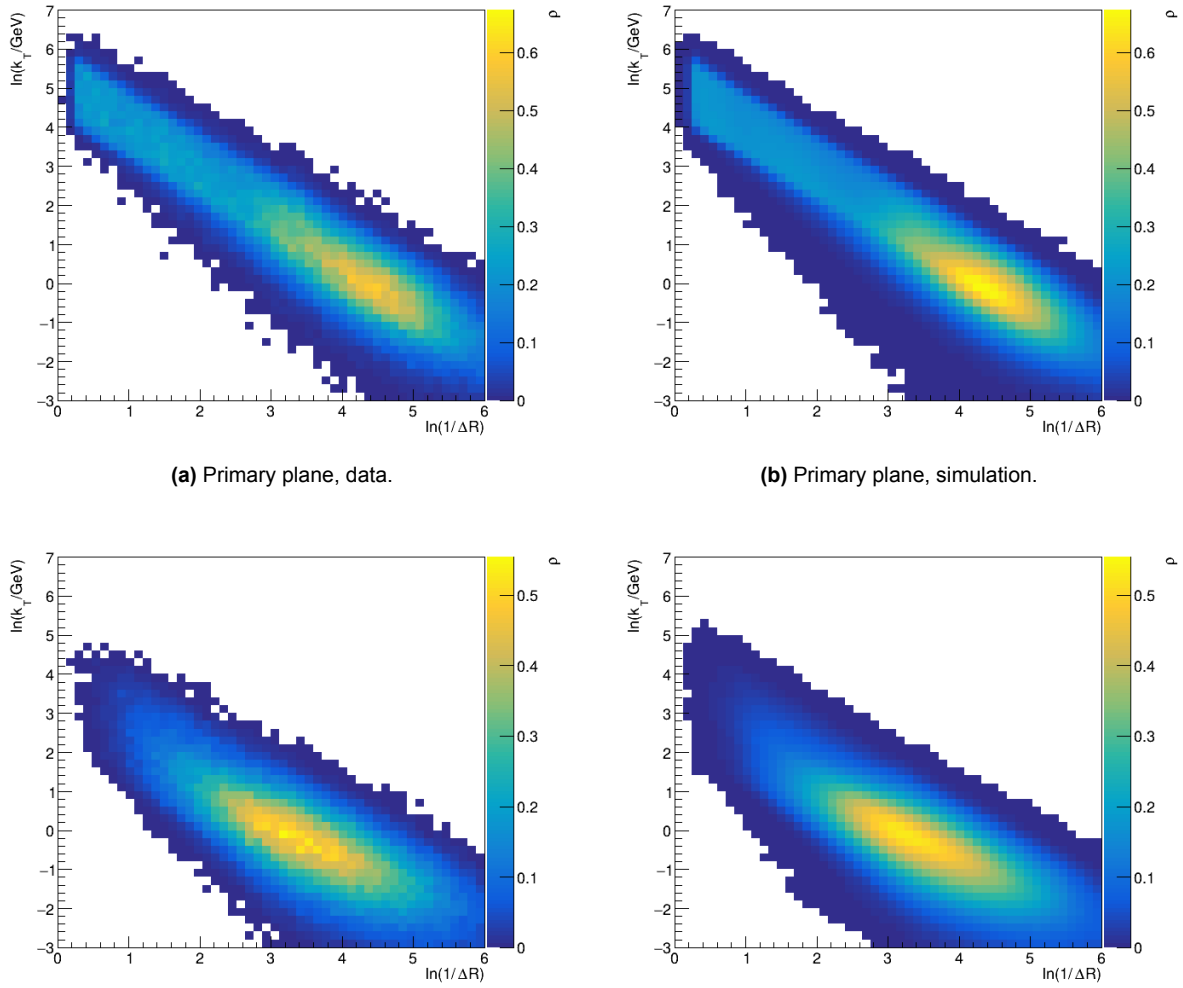
The shape of the populated region follows from the definition of section 2.2.3. Since $k_\mathrm{T} = z\,(p_{\mathrm{T},a}+p_{\mathrm{T},b})\,\Delta R$, the emissions of a given branch with fixed momentum fraction lie on straight lines of slope -1 in the plane,

$$\ln k_\mathrm{T} = \ln z + \ln(p_{\mathrm{T},a} + p_{\mathrm{T},b}) - \ln \frac{1}{\Delta R}. \tag{4.1}$$

The upper edge of the plane corresponds to the limit of $z = 1/2$, because it is the largest momentum fraction allowed when the softer branch is taken as the emission, and the condition $z > 0.1$ imposes a parallel lower edge. Therefore, for a parent with fixed momentum, the emissions should occupy a diagonal band of width $\ln 5$ in $\ln k_\mathrm{T}$ at fixed angle. However, in practice, the momentum of a parent varies along the declustering sequence and from jet to jet, which broadens the band and smooths its edges. The left side of the plane is bounded by the radius of the jets, because the jet's widest splittings separate two subjets whose axes lie within the jet, so that their separation rarely exceeds $R = 0.8$, corresponding to $\ln(1/\Delta R) \approx 0.2$.

In the primary Lund plane, the density along the band is approximately uniform at large angles and large transverse momenta, at about 0.2 emissions per jet and per unit area which is the expected behaviour in the perturbative region from Section 2.3.1. In addition, towards smaller angles the density rises, reaching its maximum around $\ln(1/\Delta R) \approx 4.5$ and $\ln k_\mathrm{T} \approx 0$, that is for k_T of order 1 GeV, and then falling at lower transverse momenta. This maximum lies where k_T approaches the scale of hadronisation, and marks the transition to the non-perturbative region of the plane.

The secondary Lund plane is populated by emissions of the softer branch of the primary splitting, but it shows the same properties. However, since this branch carries only a fraction of momentum of the jet, the band is shifted towards lower transverse momenta, by about one unit in $\ln k_\mathrm{T}$, and it is broader, as this fraction varies from jet to jet and its maximum lies at larger angles than in the primary plane, at around $\ln(1/\Delta R) \approx 3.3$. This comes from equation (4.1), where a branch with lower momentum has the region of $k_\mathrm{T} \approx 1$ GeV, where the density peaks, at a larger angular separation.



(a) Primary plane, data.

(b) Primary plane, simulation.

(c) Secondary plane, data.

(d) Secondary plane, simulation.

Figure 4.3 – Average primary (top) and secondary (bottom) Lund jet planes of the jets with a valid $\Delta\psi_{12}$, in the collision data (left) and in the simulated sample with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV (right). Only declusterings satisfying $z > 0.1$ are included. The colour scale shows the density of emissions per jet and per unit area, with the same scale for data and simulation in each row.

Now, comparing the collision data to the simulation, it is notable that they have a very close structure. The position and extent of the bands, and the location of their maxima, agree between data and simulation within the precision of the maps, which for the data show larger fluctuations owing to their smaller number of jets. This similarity shows that the reconstruction and declustering of the jets behave consistently in data and simulation. This also involves the quantities z, k_T and ΔR of the selected splittings, which enter the flavour selection as inputs, and thus supports the use of the selection trained in simulation on the collision data.

4.3 Inclusive $\Delta\psi_{12}$ distribution

The obtained distribution of $\Delta\psi_{12}$ for all jets with a valid value, before any flavour selection, is shown in Figure 4.4 for the collision data and for the simulated calibration sample. With the convention of equation (3.4), the angle is defined in the range of $[0, \pi]$, over which $\cos 2\Delta\psi_{12}$ completes one full period. In this figure it is notable that the form of equation (2.21) has a negative coefficient a_2/a_0 with an excess of jets around $\Delta\psi_{12} = \pi/2$, where the two splitting planes are perpendicular, and a depletion near $\Delta\psi_{12} = 0$ and π, where they are aligned.

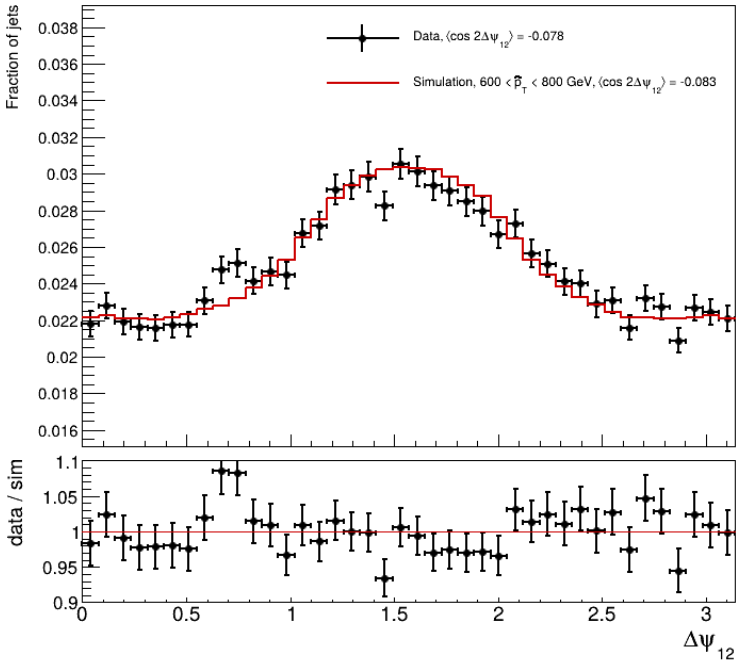


Figure 4.4 – Distribution of $\Delta\psi_{12}$ for all jets with a valid value, before any flavour selection, in the collision data and in the simulated sample with $600 < \hat{p}_\mathrm{T} < 800$ GeV, normalised to unit area.

The distribution shows a clear modulation of this kind. The fraction of jets rises from about 0.022 per bin near $\Delta\psi_{12} = 0$ and π to about 0.030 around $\pi/2$. The mean values are

$$\langle\cos 2\Delta\psi_{12}\rangle_\mathrm{data} = -0.0780 \pm 0.0033, \qquad \langle\cos 2\Delta\psi_{12}\rangle_\mathrm{sim} = -0.0831 \pm 0.0002, \tag{4.2}$$

where the uncertainties are statistical. Through the relation $a_2/a_0 = 2\langle\cos 2\Delta\psi_{12}\rangle$, they correspond to coefficients of -0.156 ± 0.007 and -0.166 ± 0.001 respectively. The two values differ by about 1.5 standard deviations, and the simulation describes the shape of the distribution in data within about 5% in almost

every bin, the few larger deviations being compatible with statistical fluctuations. However, the shape is not of a pure $\cos 2\Delta\psi_{12}$ modulation since the distribution is nearly flat close to $\Delta\psi_{12} = 0$ and π, and its central maximum is narrower than that of a single harmonic modulation, which may indicate the presence of higher harmonics in addition to the $\cos 2\Delta\psi_{12}$ term.

This inclusive result is dominated by splittings that are unmatched or belong to channels other than $g \to q\bar{q}$, as shown in Table 3.6. Therefore, this modulation is a superposition of all contributions of different channels and it is diluted accordingly. Thus, this distribution serves as baseline for evaluating the impact of the flavour selection in Section 4.5, while the significance of its magnitude is examined in Section 4.6.

4.4 Channel labelling

The composition of the simulated training sample in the channels defined by the IFN labelling of Section 3.4.3 is given in Table 3.6 while the of the simulated calibration sample is given by the following Table 4.2. The composition obtained also agrees closely with the one obtained in the CMS measurement with Pythia 8, the same flavour definition and the same matching of the branches p_3 and p_4, namely 72%, 16%, 8%, 3% and 1% [2]. The agreement confirms that the labelling was constructed correctly, although it does not prove whether the simulation describes the data.

The mean value $\langle \cos 2\Delta\psi_{12} \rangle$ obtained from the simulated calibration sample is also given in Table 4.2, together with the value predicted at second order in the collinear limit for the corresponding channel [18]. This prediction was computed on electron-positron collisions without hadronisation or detector effects, and, in this work, it serves only as a reference point for the sign and the order of magnitude of each contribution.

Table 4.2 – Mean value $\langle \cos 2\Delta\psi_{12} \rangle$ for the jets of each channel of the IFN labelling, in the simulated calibration sample, with statistical uncertainties. The last column gives the prediction of Ref. [18] for the corresponding channel, obtained for electron-positron collisions with $z_1, z_2 > 0.1$.

Class	Jets	Fraction (%)	$\langle \cos 2\Delta\psi_{12} \rangle$	Prediction
$g \to q\bar{q}$	179 810	3.1%	-0.167 ± 0.002	-0.31
$g \to gg$	857 140	14.7%	-0.077 ± 0.001	$+0.025$
$q \to qg$	358 498	6.1%	-0.091 ± 0.001	0
unmatched	4 397 983	75.2%	-0.0585 ± 0.0003	—
rest	53 911	0.9%	-0.161 ± 0.003	—
all labelled jets	5 847 342	100%	-0.0674 ± 0.0003	—

The $g \to q\bar{q}$ channel shows a modulation with $\langle \cos 2\Delta\psi_{12} \rangle = -0.167 \pm 0.002$, while the $q \to qg$ and $g \to gg$ channels, which form the background of the flavour selection, give together -0.081 ± 0.001. Thus, the modulation of the $g \to q\bar{q}$ channel is about twice as strong as that of the background, and it is this difference that the flavour selection exploits. In Figure 4.5, it is possible to see the difference between the $g \to q\bar{q}$ channel and the whole sample. It is important to note that the sign of the $g \to q\bar{q}$ class agrees with the prediction, while its magnitude is smaller which is expected for a flavour based label that does not coincide splitting by splitting with the channel of the parton shower and for an observable built from reconstructed particles. More importantly, the $q \to qg$ and $g \to gg$ classes also show a negative sign in contrast with the vanishing and slightly positive values predicted for the corresponding channels. A negative contribution is thus present in every class, including those in which no spin correlation is expected and its source is discussed in Section 4.8.

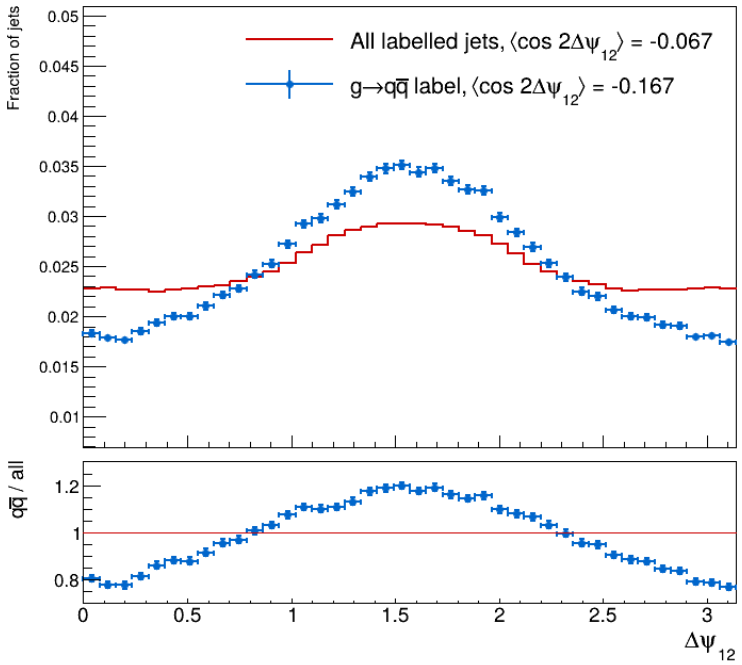


Figure 4.5 – Distribution of $\Delta\psi_{12}$ for all labelled jets and for the jets whose second splitting is labelled $g \to q\bar{q}$, in the simulated sample with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV, normalised to unit area. The lower panel shows the ratio of the $g \to q\bar{q}$ distribution to that of all labelled jets.

4.4.1 Momentum fractions of the two splittings

The momentum fractions z_1 and z_2 of the primary and secondary selected splittings are two of the features used by the flavour selection of Section 3.5. Each is the fraction carried by the softer branch and lies between the SoftDrop grooming condition of $z = 0.1$ and $z = 1/2$. Their complete distribution is shown in Figure 4.6 for all jets with a valid $\Delta\psi_{12}$, in data and in simulation, and for the simulated jets in each channel. The ratio of each distribution to that of all simulated jets is shown in Figure 4.7, where the differences between the channels can be distinguished more easily.

The distribution of all jets is concentrated at small z_1 and z_2, so that asymmetric splittings with soft branch dominate the sample. The simulation describes the collision data well with mean values of $\langle z_1 \rangle = 0.270$ and $\langle z_2 \rangle = 0.283$ in data, against 0.268 and 0.283 in simulation, and the ratio of the two distributions shows no systematic structure, only fluctuations compatible with the statistical precision of the data.

The channels populate the plane differently. The unmatched jets, which make up about half of the sample, are concentrated in the corner of small z_1 and z_2, with $\langle z_1 \rangle = 0.264$ and $\langle z_2 \rangle = 0.275$. Soft branches are the most affected by hadronisation and by the detector, so their splittings are the least often matched at generator level. The labelled channels instead populate the interior of the plane. The jets labelled $g \to q\bar{q}$ have the largest $\langle z_2 \rangle$, 0.315. Relative to all jets they are enhanced by 50% or more for $0.3 \lesssim z_2 \lesssim 0.45$ and intermediate z_1, and suppressed by more than a third for $z_2 < 0.15$. This reflects the splitting function of Eq. (2.5), which has no enhancement for soft splittings, in contrast with those of $g \to gg$ and $q \to qg$. The jets labelled $g \to gg$ show a similar but weaker pattern, with $\langle z_2 \rangle = 0.301$. The jets labelled $q \to qg$ differ mainly in z_1, with the largest mean value, $\langle z_1 \rangle = 0.294$, and an enhancement at $z_1 \gtrsim 0.3$. For the second splitting to be $q \to qg$, the softer branch of the first splitting must be a quark, and a quark is produced without the soft enhancement associated with an emitted gluon. Therefore, the first splitting of these jets is more symmetric.

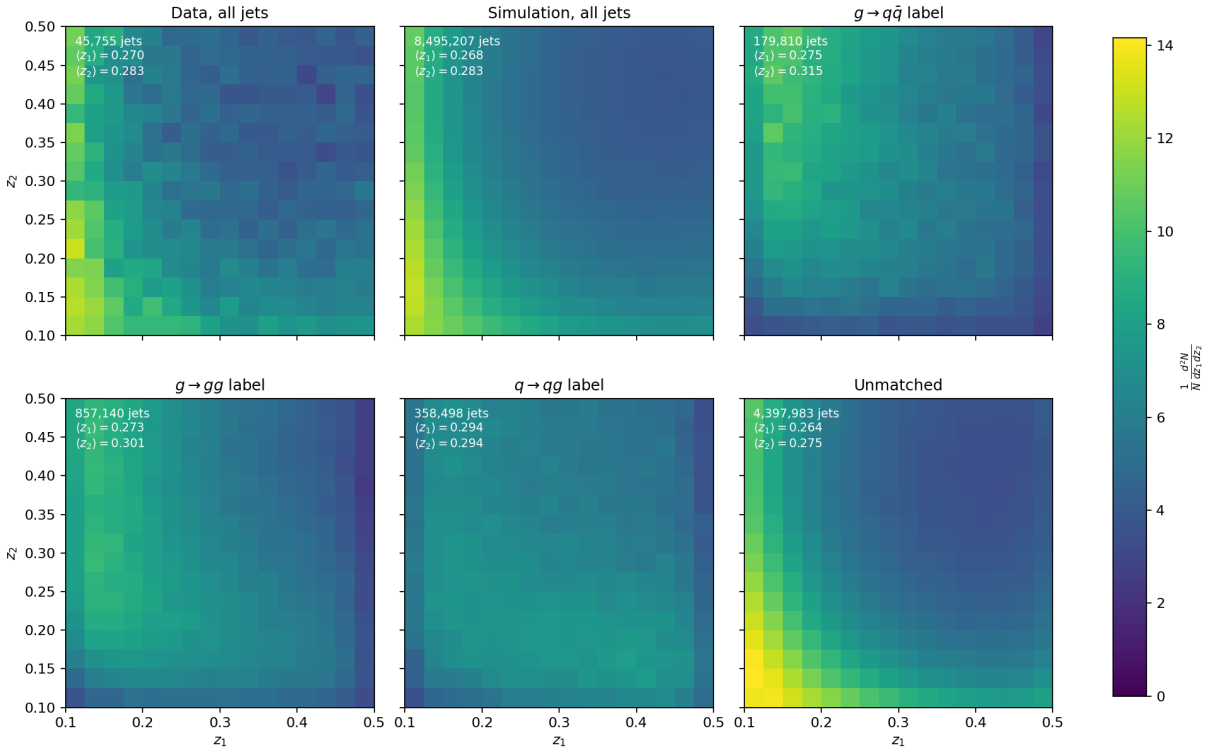


Figure 4.6 – Distribution of the momentum fractions z_1 and z_2 of the first and second selected splittings, normalised to unit area, for all jets with a valid $\Delta\psi_{12}$ in the collision data and in the simulated calibration sample, and for the simulated jets labelled $g \to q\bar{q}$, $g \to gg$ and $q \to qg$ and those unmatched. Each panel gives the number of jets and the mean values of z_1 and z_2.

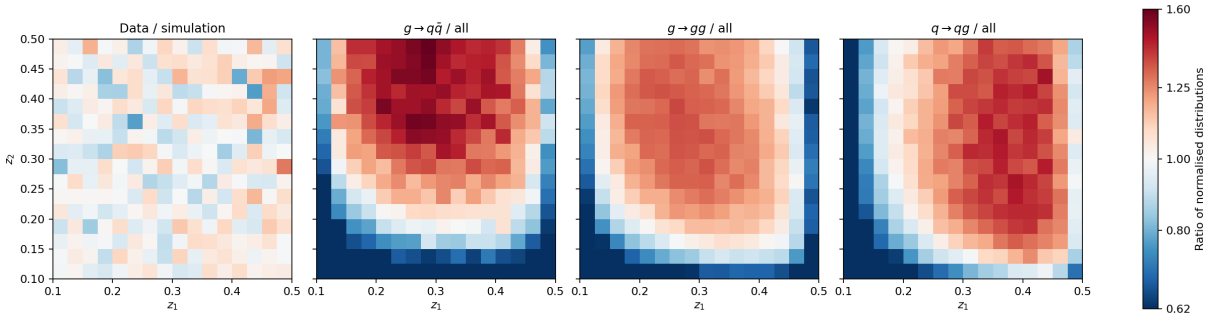


Figure 4.7 – Ratio of the normalised distributions of Figure 4.6 to that of all simulated jets, for the collision data and for the simulated jets labelled $g \to q\bar{q}$, $g \to gg$ and $q \to qg$. The colour scale is logarithmic and symmetric around unity, and ratios beyond its range are shown at its limits.

As one can notice, all channels have a depletion near the edge of $z_1 \to 1/2$. When the primary splitting is nearly symmetric, its softer branch is decided by small differences in the transverse momentum of the splitting's daughters. So, the generator-level and the reconstructed declusterings can then follow different branches to the secondary splitting, making the algorithm label the secondary splitting as unmatched.

These differences are the information in z_1 and z_2 that the flavour selection can exploit. The $g \to q\bar{q}$ channel is distinguished mainly by its more symmetric second splitting, and the effect of the selection on the distribution of z_2 is examined in Section 4.6.2.

4.5 Flavour selection performance

In order to measure the spin-sensitive observable using the flavour selection, discussed in Section 3.5, it must first be shown that it can identify the $g \to q\bar{q}$ channel effectively. This section evaluates its performance in simulation, where the IFN label of every splitting is known, and compares it with the alternative approaches of Section 3.5.4.

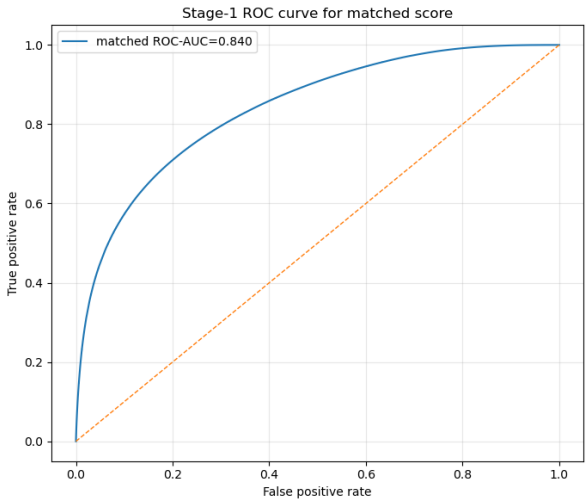
As stated before, two simulated samples were used, for different purposes. The separation achieved by each network is assessed on the test subset of the training sample, with $800 < \hat{p}_\mathrm{T} < 1000$ GeV, which was not used during the training. It is characterised by the distributions of the scores for the different channels and by the area under the ROC curve, which measures the ability of the network to rank a splitting of the signal channels above one of the background. For the second stage, in which the $g \to q\bar{q}$ channel is a minority, the area under the precision-recall curve is also given, since it reflects more directly the purity that can be achieved. The performance of the complete selection, expressed through the purity and the efficiency of equation (3.14), is determined instead on the independent calibration sample, with $600 < \hat{p}_\mathrm{T} < 800$ GeV. This is the sample on which the purity of the measurement is calibrated, and it allows all methods to be compared without any bias from their training.

Section 4.5.1 presents the performance of the two stages of the selection and the purity reached as a function of the threshold. Section 4.5.2 compares the two-stage selection with the alternative approaches.

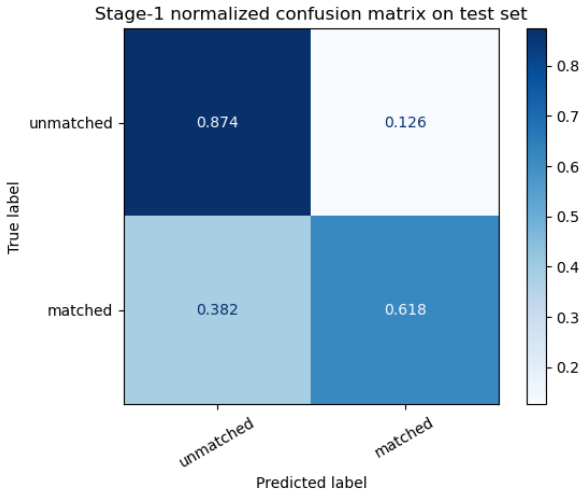
4.5.1 Two-stage selection

The first stage network separates matched from unmatched splittings with an area under the ROC curve of 0.840, as shown in Figure 4.8a. This network performance is sufficient in order to remove the majority of unmatched secondary splittings before the flavour of the second splitting is examined. In Figure 4.8b, the confusion matrix of the first stage classifier, obtained from the test subset, is shown. Since unmatched splittings form the bulk of the data sample, a classifier trained on this imbalance could assign almost all splittings to this class. The matrix presented reveals that this is not the case. The results revealed that of the unmatched splittings, 87.4% are correctly identified and 12.6% are classified as matched, while 61.8% of the matched splittings are correctly identified and 38.2% are classified as unmatched. Therefore, both classes were recognised as intended by the class weights during training. However, since the unmatched splittings are about three times more numerous, the 12.6% of them classified as matched still make up about 40% of all the splittings assigned to the matched class. This is why the matched score is set to be above 0.8, rather than the assignment to the most probable class on which the matrix is based.

The performance of the second stage is evaluated on the population on which it was trained, namely splittings labelled as $g \to q\bar{q}$, $q \to qg$ or $g \to gg$. The distribution of its score for the $g \to q\bar{q}$ class and



(a) ROC curve.



(b) Confusion matrix.

Figure 4.8 – Performance of the first stage on the test subset of the simulated sample with $800 < \hat{p}_{\mathrm{T}} < 1000$ GeV. (a) ROC curve of the matched score. (b) Confusion matrix, in which each row is normalised to the number of splittings of the corresponding true class and each splitting is assigned to the class of highest probability.

for the other two classes together is shown in Figure 4.9. Both distributions peak at about 0.3, and the background is concentrated in the range between 0.2 and 0.4. The main and most important difference between these two distributions is that the $g \to q\bar{q}$ class has a long tail extending to scores of about 0.65, in which it dominates. These scores are, however, confined between 0.15 and 0.7, which reflects the strong overlap of the channels in the space of the input features and the use of the focal loss, which does not push the outputs towards the extreme values.

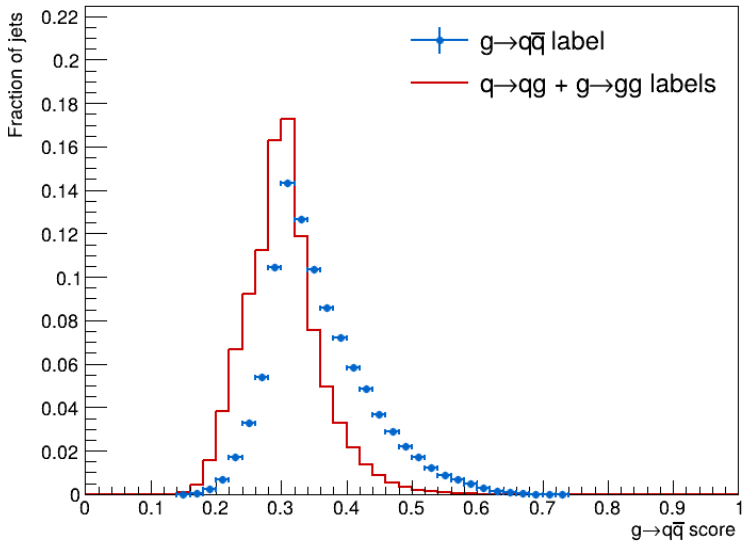


Figure 4.9 – Distribution of the $g \to q\bar{q}$ score of the second stage for the $g \to q\bar{q}$ class and for the $q \to qg$ and $g \to gg$ classes together, on the simulated calibration sample with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV, normalised to unit area.

The ROC and precision-recall curves are shown in Figure 4.10. The area under the ROC curve is 0.714, which is worse compared to the first stage classifier, and the area under the precision-recall curve is 0.314, compared with 0.129 for a classifier without discriminating power, which corresponds to the fraction of the $g \to q\bar{q}$ class among the matched splittings. The same areas were computed on the simulated calibration sample and the values of 0.715 and 0.314 were obtained for the ROC and precision-recall

curves respectively. These values show that this network generalises to jets outside its training sample and the precision-recall curve also shows that a high purity is reached only at a small efficiency, the purity exceeding 0.6 only for efficiencies below about 10%.

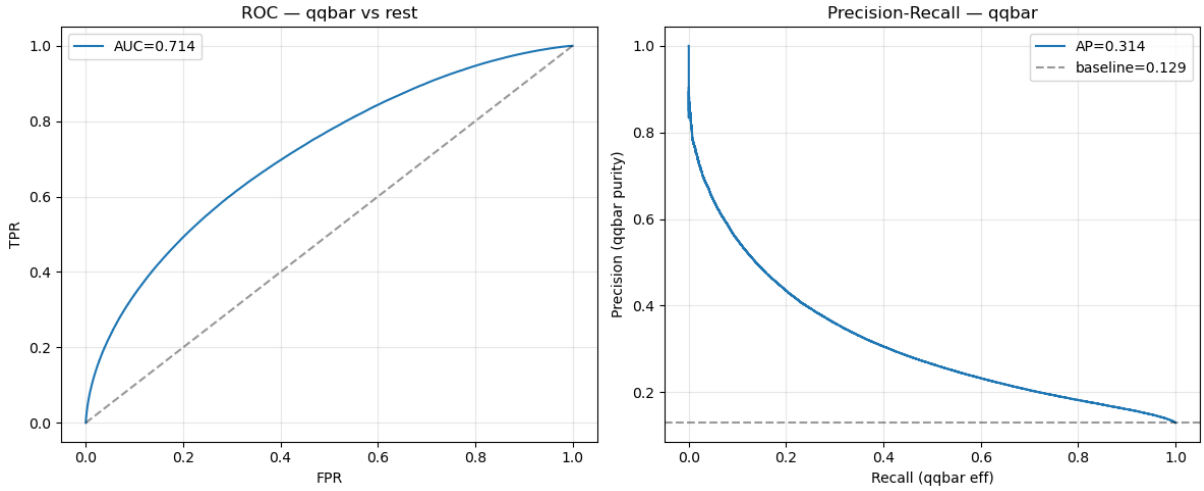


Figure 4.10 – ROC curve (left) and precision-recall curve (right) of the $g \to q\bar{q}$ score of the second stage. The dashed line in the right panel shows the precision of a classifier without discriminating power.

The purity and the efficiency of the second stage on the test sample are given in Table 4.3 and shown in Figure 4.11 as a function of the threshold. As the threshold increases from 0.30 to 0.60, the purity rises steadily from 0.18 to 0.80, while the efficiency falls from 81% to below 1%. Above 0.60 the purity continues to rise, but the number of selected jets falls to a few hundred and then even less, meaning that these thresholds carry little statistical weight.

Table 4.3 – Number of selected jets, purity and efficiency of the second stage as a function of the threshold on the $g \to q\bar{q}$ score, for the splittings labelled $g \to q\bar{q}$, $q \to qg$ or $g \to gg$ in the simulated calibration sample with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV. The uncertainties on the purity are statistical, and the efficiency is relative to all splittings labelled $g \to q\bar{q}$.

Threshold	Selected jets	Purity	Efficiency (%)
0.30	$1\,319\,798$	0.1807 ± 0.0003	80.6
0.35	$534\,802$	0.270 ± 0.001	48.7
0.40	$211\,459$	0.379 ± 0.001	27.1
0.45	$81\,911$	0.500 ± 0.002	13.8
0.50	$30\,446$	0.616 ± 0.003	6.3
0.55	$10\,111$	0.716 ± 0.005	2.5
0.60	$2\,455$	0.802 ± 0.008	0.67
0.65	319	0.865 ± 0.019	0.09
0.70	16	0.94 ± 0.06	0.01

4.5.2 Comparison with alternative approaches

The three alternative approaches of Section 3.5.4 are compared with the previous two-stage selection on the simulated calibration sample. A first comparison was made regarding the performance of the networks, evaluated on the splittings labelled as $g \to q\bar{q}$, $q \to qg$ or $g \to gg$. On this population, the areas under the ROC and precision–recall curves are 0.715 and 0.314 for the second stage of the two-stage selection, 0.710 and 0.309 for the three-class network, and 0.693 and 0.300 for the four-class network. Thus, the three different networks reach a very similar performance, meaning that the selection of the

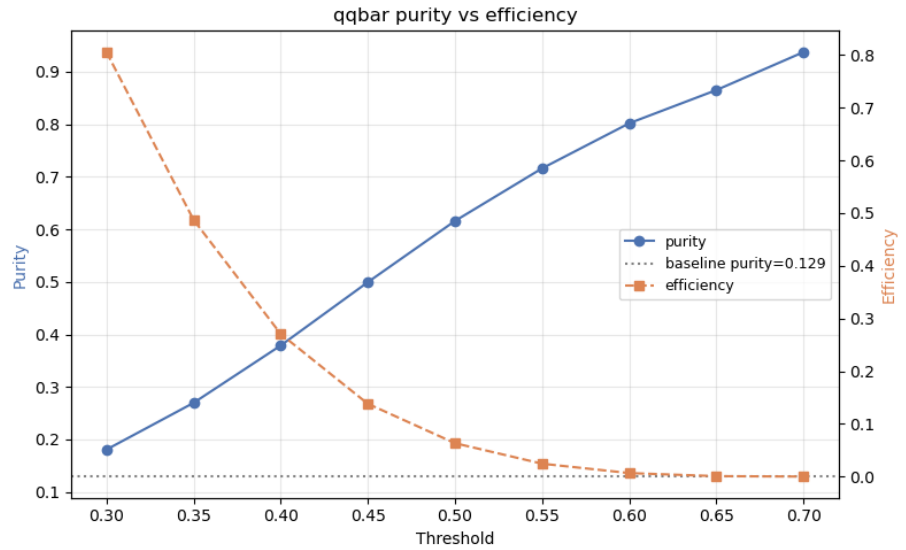


Figure 4.11 – Purity (left axis) and efficiency (right axis) of the second stage as a function of the threshold on the $g \to q\bar{q}$ score, for the splittings labelled $g \to q\bar{q}$, $q \to qg$ or $g \to gg$ in the simulated calibration sample. The dotted line shows the fraction of $g \to q\bar{q}$ splittings among these, before any selection.

$g \to q\bar{q}$ channel is limited by the input features rather than the architecture of the network. The Gaussian mixture model provides no useful separation, the purities of its clusters lying between about 0.10 and 0.16, close to the fraction of $g \to q\bar{q}$ splittings among the matched ones.

However, this first comparison totally ignores the unmatched splittings, which the four-class network rejects at the same time as it identifies the $g \to q\bar{q}$ channel. The more relevant comparison for the measurement of this work is that of the complete selections, applied to all labelled splittings. The two-stage selection requires a matched score above 0.8 and a threshold on the second-stage score. The second selection replaces the second stage by the three-class network after the same requirement on the matched score. The third applies a threshold on the output of the four-class network alone. Thus the purity of each selection, including the unmatched splittings among the selected jets, is shown in Figure 4.12 as a function of its efficiency, and its value at different efficiencies is given in Table 4.4.

Table 4.4 – Purity of the complete selections at fixed efficiencies for the $g \to q\bar{q}$ channel, on all labelled splittings of the simulated calibration sample.

Efficiency (%)	Two-stage	Stage 1 + three-class	Four-class
0.5	0.622	0.595	0.606
1	0.608	0.579	0.587
2	0.556	0.547	0.553
5	0.474	0.475	0.471
10	0.383	0.384	0.386
20	0.263	0.262	0.279
30	0.168	0.168	0.211

The three selections behave differently in different regions. At efficiencies below about 2%, where the purity is highest, the two-stage selection reaches the largest purity, by about 0.02 to 0.03 at an efficiency of 1%, a difference of two to three times the statistical uncertainty of the purity. Between 5% and 10% the three selections are equivalent. At larger efficiencies the four-class network performs better, and it alone can exceed the efficiency of 35.7% to which the requirement on the matched score limits the other two. This is the advantage of treating the rejection of the unmatched splittings and the identification of the $g \to q\bar{q}$ channel in a single step.

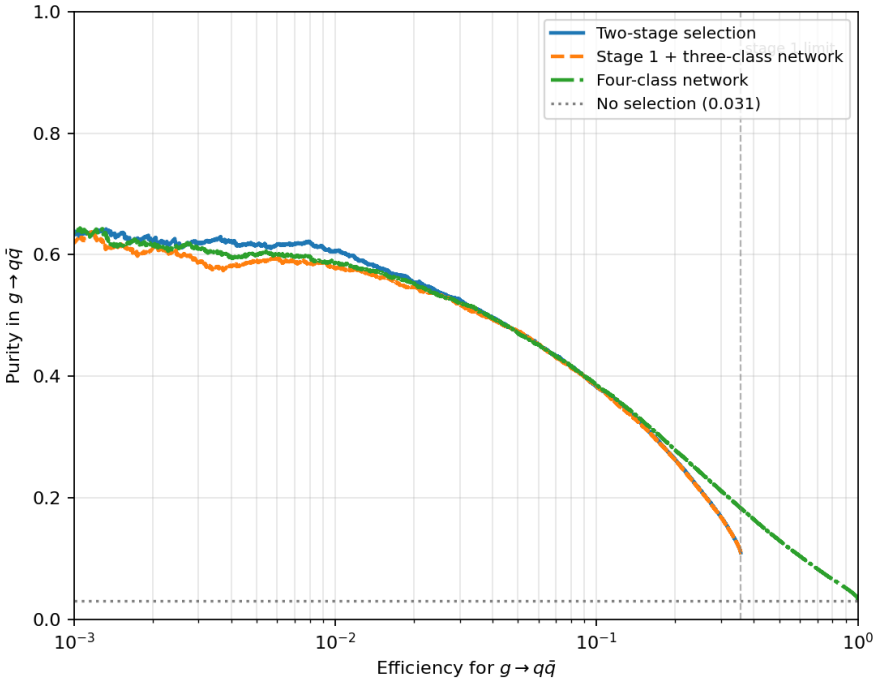


Figure 4.12 – Purity as a function of efficiency for the $g \to q\bar{q}$ channel, for the complete two-stage selection, for the first stage followed by the three-class network, and for the four-class network, applied to all labelled splittings of the simulated calibration sample. The vertical line marks the largest efficiency allowed by the requirement on the matched score, and the dotted line the purity before any selection.

However, for this measurement, the region of interest is the one with the highest purity possible while keeping enough statistics. The thresholds used in Section 4.6 correspond to efficiencies between about 28% and 0.5%, and the most enriched samples lie below 5%, where the two-stage selection performs best. The two-stage selection is therefore retained for the measurement.

4.6 Spin modulation as a function of the flavour selection

Finally, this section presents the main result of this work, the mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of the jets selected by the two-stage selection, as a function of the threshold on the $g \to q\bar{q}$ score. Again, every jet is required to have a matched score greater than 0.8 and the threshold on the second score goes from 0.30 to 0.60 in steps of 0.05. Because of statistics, the function does not continue past 0.60, since the number of selected jets becomes too small for the mean value to carry useful information, as discussed in Section 3.5.5. Each threshold defines a sample increasingly enriched in the $g \to q\bar{q}$ channel, and the evolution of the modulation along this sequence of samples is the quantity of interest.

The purity computed for each threshold is that of a complete selection, determined in the simulation calibration sample. Its value depends on the treatment of the jets that carry no channel label, for which no generator-level counterpart was found or the generator-level declustering failed. These jets make up about 31% of the simulated jets with a valid $\Delta\psi_{12}$, and they are selected in data in the same way as the others. Counting them as background gives a lower bound on the purity, since some of them originate from genuine $g \to q\bar{q}$ splittings. Excluding them gives the purity under the assumption that their composition is the same as that of the labelled jets. Both values are quoted, and the true purity of the selected samples lies between them.

Section 4.6.1 presents the modulation in simulation, where the composition of every selected sample is known. Section 4.6.2 examines the kinematics of the selected splittings. Section 4.6.3 compares the simulation with the collision data at equal thresholds.

4.6.1 Modulation in simulation

The mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of the jets selected in the simulated calibration sample is shown in Figure 4.13, as a function of the threshold on the $g \to q\bar{q}$ score along with the obtained purity of the selection. At the lowest measured threshold, the modulation of the selected jets is -0.122 ± 0.001, which is already stronger than the measurement on the inclusive sample of Section 4.3, -0.083, just by applying the matched score threshold. As the threshold increases, the modulation becomes steadily more negative and reaches -0.266 ± 0.016 at 0.60 while the purity goes from 0.18 to between 0.48 and 0.61, depending on the treatment of the jets without channel label. Thus, the modulation becomes more negative as the selected samples are enriched in the $g \to q\bar{q}$ channel, as expected from the stronger modulation of this channel found in Section 4.4.

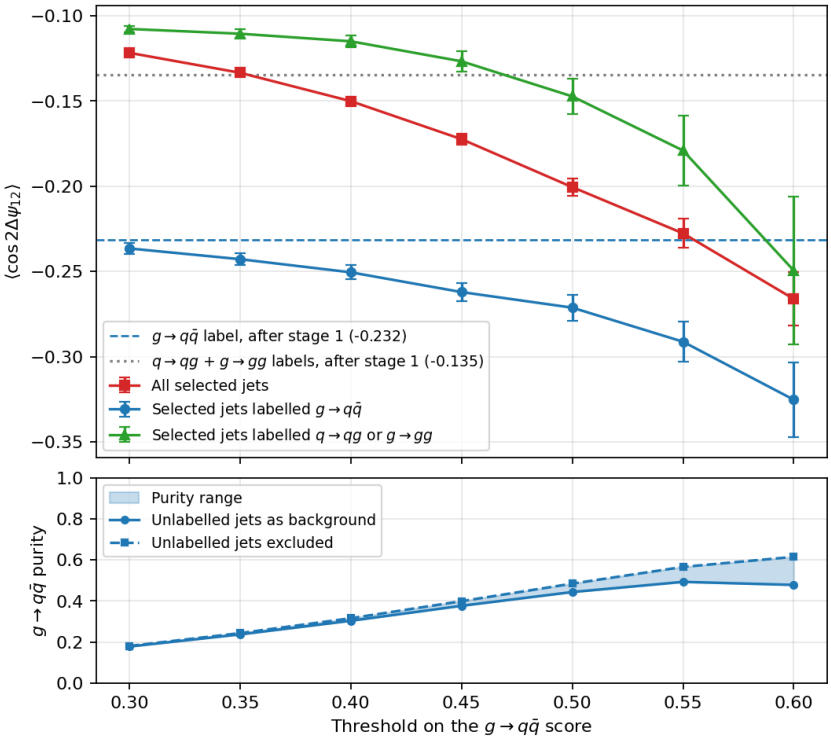


Figure 4.13 – Top: mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of all jets selected by the two-stage selection in the simulated calibration sample, and of the selected jets labelled $g \to q\bar{q}$ and labelled $q \to qg$ or $g \to gg$, as a function of the threshold on the $g \to q\bar{q}$ score. All jets have a matched score above 0.8. The horizontal lines show the mean values of the two classes with this requirement alone, without any threshold on the second score. Bottom: purity of the selected samples, with the jets without channel label counted as background and excluded, respectively.

The two purities coincide up to a threshold of about 0.45, which shows that the selected samples contain almost no jets without channel label at low thresholds. The selection as a whole therefore strongly disfavours these jets, which make up 31% of the inclusive sample. At higher thresholds, however, the purities become notably separated, and at a threshold of 0.60 a fifth of the selected jets carry no label. The tightest requirements thus retain proportionally more jets without a generator-level counterpart, which causes the purity counting them as background to saturate at about 0.5.

The behaviour of the selection is further clarified by following the jets of the signal and background classes separately. Among the jets passing the requirement on the matched score, those labelled $g \to q\bar{q}$ have a mean value of -0.232 ± 0.003, and those labelled $q \to qg$ or $g \to gg$ a mean value of -0.135 ± 0.001. Within the selection, the jets labelled $g \to q\bar{q}$ become steadily more negative as the threshold increases, from -0.237 ± 0.003 at 0.30 to -0.325 ± 0.022 at 0.60. However, the background classes also become more negative as the threshold increases, from -0.108 ± 0.002 at 0.30 to -0.249 ± 0.043 at 0.60, and the decrease accelerates at the highest thresholds. At the lowest thresholds their mean value lies above the reference of -0.135, which shows that the first requirement on the second score removes background configurations with a stronger than average modulation. The curve crosses the reference at a threshold of about 0.47, and at 0.60 the background approaches the signal, although these channels carry no or a slightly positive spin correlation. Therefore, the selection enriches the sample in the $g \to q\bar{q}$ channel and also favours, at its tightest requirements, configurations in which the modulation is stronger whatever their channel.

Nevertheless, the difference between the two classes is still notable. Between thresholds of 0.30 and 0.50, the jets labelled $g \to q\bar{q}$ lie between 0.12 and 0.14 below the background selected at the same threshold. The difference decreases only at the highest thresholds, to 0.11 at 0.55 and 0.08 ± 0.05 at 0.60, where both classes are dominated by the effect of the selection. If the selection shifted the modulation of all channels by the same amount, this shift would cancel in the difference between them, which would then remain constant even as both classes become more negative. This is what is observed up to a threshold of 0.50. The stability of the difference therefore indicates that, beyond the common effect of the selection, the $g \to q\bar{q}$ channel carries a modulation of its own, about 0.13 stronger than that of the background. At the highest thresholds the difference decreases, which suggests that the effect of the selection is no longer the same for all channels.

4.6.2 Kinematics of the selected splittings

The selection acts on the kinematics of each splitting as well as on their flavour. The splitting function of equation (2.5) favours symmetric $g \to q\bar{q}$ splittings, in contrast with those of $g \to gg$ and $q \to qg$, which are enhanced when one of the two products is soft. The distribution of the momentum fraction z_2 of the second splitting is shown in the left panel of Figure 4.14. As one can notice, both classes rise towards $z_2 = 1/2$, since the splitting of largest k_T is selected in each sequence and k_T grows with z. However, their shapes are different. With jets below $z_2 \approx 0.3$ the channels $q \to qg$ and $g \to gg$ are relatively more frequent, while above that those labelled $g \to q\bar{q}$ are more frequent. The selection amplifies this effect, and with it, at a threshold of 0.60, almost no selected jet has $z_2 < 0.25$, and the distribution rises much more steeply up to $z_2 = 1/2$.

The dependence of the modulation on z_2 is shown in the right plot. For the background, the modulation is most negative for asymmetric splittings, -0.19 at $z_2 \approx 0.12$, and weakens to about -0.12 for symmetric ones. These channels should carry no or a positive spin correlation, which reveals a component of the modulation that is not of spin origin and is strongest for asymmetric splittings. For the jets labelled $g \to q\bar{q}$ the modulation is instead nearly independent of z_2, at about -0.23 over most of the range, whereas at leading order in the collinear limit the spin correlation of this channel grows strongly towards symmetric splittings [18]. The difference between the two classes, which removes the component common to both, grows from about 0.08 for asymmetric splittings to about 0.11 for symmetric ones. It thus follows the expected direction, but varies much less than predicted. This discrepancy is discussed in Section 4.8.

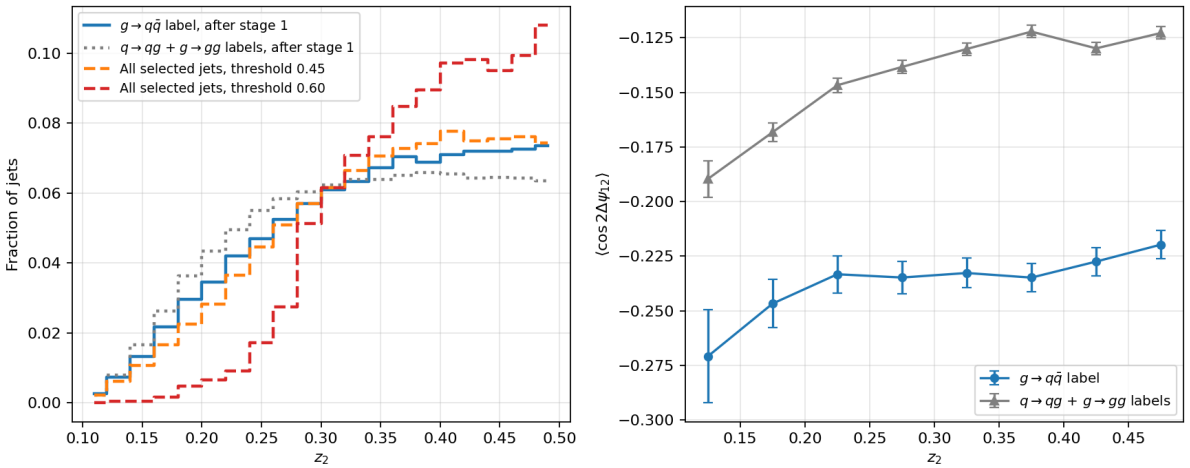


Figure 4.14 – Left: distribution of the momentum fraction z_2 of the second splitting for the jets labelled $g \to q\bar{q}$ and for those labelled $q \to qg$ or $g \to gg$ after the first stage, and for all jets selected at thresholds of 0.45 and 0.60, normalised to unit area. Right: mean value $\langle \cos 2\Delta\psi_{12} \rangle$ as a function of z_2 for the jets labelled $g \to q\bar{q}$ and for those labelled $q \to qg$ or $g \to gg$ after the first stage. All results are obtained on the simulated calibration sample.

These results explain part of the behaviour of Section 4.6.1. The first requirements on the second score remove mainly background splittings with low z_2, which carry the strongest modulation, so that the selected background initially lies above its reference. The shift towards symmetric splittings cannot, however, explain the strengthening of the modulation at the highest thresholds, since symmetric splittings show a weaker modulation in the background and an essentially unchanged one in the $g \to q\bar{q}$ class. Weighting the modulation of each class by the distribution of z_2 of the jets selected at 0.60 would give about -0.23 for the $g \to q\bar{q}$ class and -0.13 for the background, far from the observed values of -0.325 ± 0.022 and -0.249 ± 0.043. This strengthening must therefore arise from other properties of the splittings to which the classifier is sensitive, such as the momentum fraction of the first splitting, the opening angles or the shapes of the branches, whose identification is left for future work.

4.6.3 Comparison with collision data

The same selection procedure is applied to the collision data with the same networks trained on simulation and the same requirements on both scores. The mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of the selected jets is compared with that of the simulated calibration sample in Figure 4.15, however, the collision data is much smaller. At a threshold of 0.30 it contains $2\,607$ jets, about 110 times fewer than the $286\,380$ simulated ones, and only 162, 50 and 16 jets remain at 0.50, 0.55 and 0.60. Therefore, the statistical uncertainty of the data dominates the comparison at every threshold. Nevertheless, the fractions of jets at low thresholds are very similar between the two samples, differing by at most about 15%, which shows that the score distributions of data and simulation are close.

The quantitative comparison is restricted to thresholds up to 0.50 since, in this range, the data sample contains more than 150 jets, and its statistical uncertainty is at most about half of the difference of 0.097 between the two reference values. With thresholds above this value the uncertainty becomes comparable to this difference, and the data can no longer distinguish a sample dominated by $g \to q\bar{q}$ splittings from one dominated by background.

Then, at thresholds below 0.5, the mean values agree within less than one standard deviation, from -0.128 ± 0.014 in data and -0.122 ± 0.001 in simulation at 0.30 to -0.229 ± 0.054 and -0.201 ± 0.005

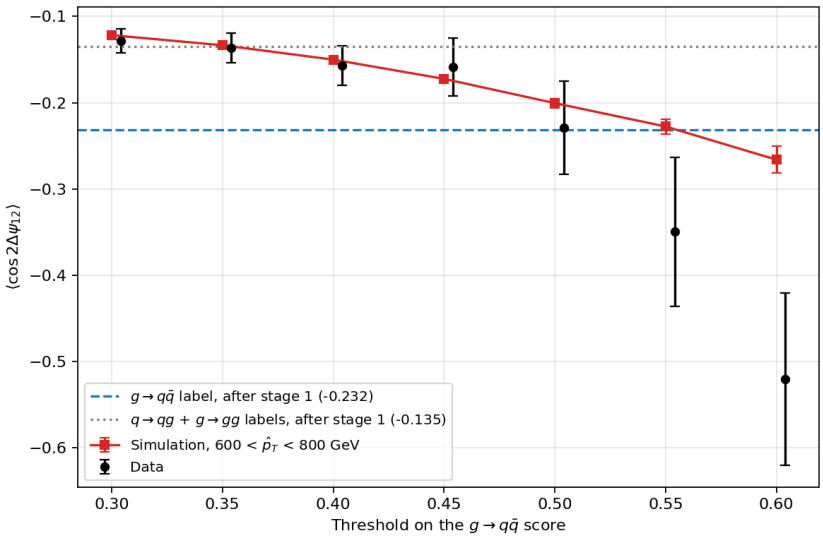


Figure 4.15 – Mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of the jets selected by the two-stage selection as a function of the threshold on the $g \to q\bar{q}$ score, in the collision data and in the simulated calibration sample. All jets have a matched score above 0.8. The horizontal lines show the mean values in simulation of the jets labelled $g \to q\bar{q}$ and of those labelled $q \to qg$ or $g \to gg$ after this requirement alone. The data points are shifted slightly to the right for readability, and the error bars show the statistical uncertainties.

at 0.50. Thus, the modulation in data strengthens with the threshold, just like in simulation. Since the selected samples are grouped together, the jets selected at 0.50 are also part of the sample at 0.30, and the uncertainty of the change between the two thresholds is approximately $\sqrt{\sigma^2_{0.50} - \sigma^2_{0.30}}$, smaller than the combination of the two uncertainties. Between 0.30 and 0.50 the mean value in data decreases by 0.101 ± 0.052, compared with 0.079 in simulation. Therefore, the strengthening is seen in data with a significance of about two standard deviations.

However, above the threshold of 0.50, the data lies below the simulation, by 1.4 standard deviations at 0.55 and by 2.5 at 0.60, where the mean value reaches -0.52 ± 0.10. These two points rest on very few jets and are strongly correlated. The 16 jets selected at 0.60 are also part of the 50 selected at 0.55, and the remaining 34 jets, with scores between 0.55 and 0.60, have a mean value of about -0.27, in agreement with the simulation. The deviation at both thresholds therefore comes mostly from the 16 jets of highest score. With so few jets the Gaussian approximation of the uncertainty is itself unreliable, and the central value at 0.60 even lies beyond $-1/2$, the most negative value allowed by a modulation of the form of equation (2.21). This illustrates the size of the statistical fluctuations at these thresholds, and a deviation of this kind in the last of several correlated points is not evidence of a disagreement with simulation.

Within the precision of the data, it is possible to say that the simulation describes both the response of the selection and the modulation of the selected jets. However this agreement does not establish the presence of spin correlation in data, since part of the strengthening of the modulation in simulation is common to all channels and is induced by the selection itself, as shown in Section 4.6.1, and no simulated sample without spin correlations is available to isolate the contribution of spin origin. These points are discussed in Section 4.8.

4.7 Three-point energy correlator

The three-point energy correlator of Section 3.4.4 gives a second probe of the spin correlations, built directly from jet constituents and independent of the declustering. Its distribution in the azimuthal angle ψ is computed over all triplets of constituents inside the window of equation (3.9), each with the weight of equation (3.12), and normalised to unit area over $(-\pi, \pi]$. Its modulation is summarised by the weighted mean

$$\langle \cos 2\psi \rangle = \frac{\sum_{\text{jets}} \sum_{abc} w_{abc} \cos 2\psi_{abc}}{\sum_{\text{jets}} \sum_{abc} w_{abc}}, \tag{4.3}$$

which is related to the coefficients of a modulation of the form of equation (2.21) by $a_2/a_0 = 2\langle \cos 2\psi \rangle$, as for $\Delta\psi_{12}$. The constituents of a jet share elements with one another and are therefore not independent, thus, statistical uncertainties are calculated by treating the jets as the independent units. Since the simulated calibration dataset is much larger than the collision dataset, the distributions of all jets are obtained from the first 5×10^5 events which makes the statistical uncertainty much smaller than that of the data while the latter uses all selected events. For the labelled classes in the simulated calibration dataset, all events were used.

The distribution of all jets is shown in the left panel of Figure 4.16. This distribution reveals a clear modulation, symmetric under $\psi \to -\psi$ as expected from parity invariance, with maxima at $\psi = 0$ and $\pm\pi$ and minima at $\pm\pi/2$. The mean values obtained are $+0.020 \pm 0.001$ in data and $+0.019 \pm 0.001$ in simulation, which is equivalent to $a_2/a_0 \approx +0.04$, and the ratio of data to simulation is compatible with unity within about 2% in every bin. The shape is closer to a single $\cos 2\psi$ harmonic, with, however, a minima somewhat deeper than this harmonic would produce, as it is possible to see when comparing to the red dashed line. With this, the obtained distribution has positive sign and thus it is opposite to that of the inclusive $\Delta\psi_{12}$ distribution of Section 4.3, and about four times smaller in magnitude.

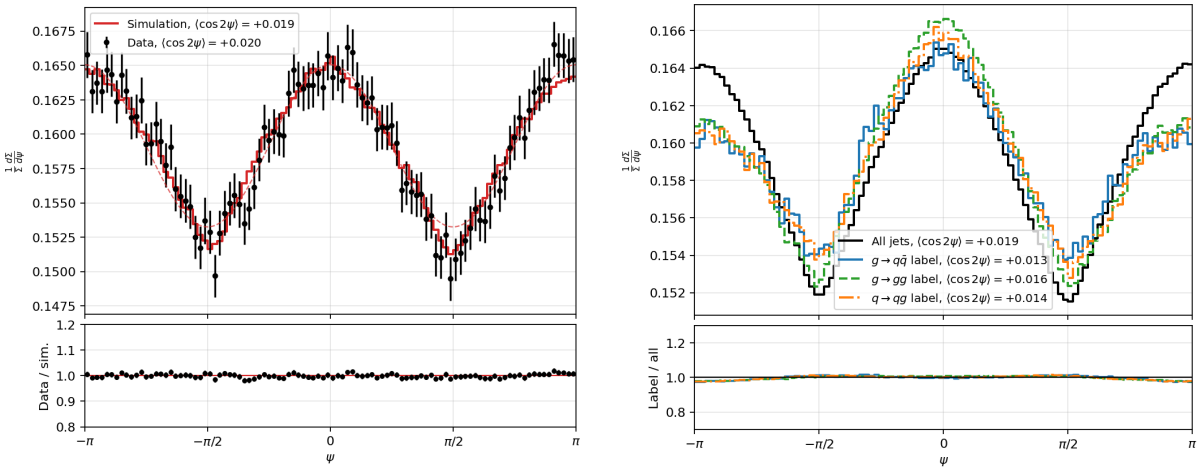


Figure 4.16 – Distribution of the azimuthal angle ψ of the three-point energy correlator, normalised to unit area. The left panel compares all jets in the collision data and in the simulated calibration sample, with the ratio of data to simulation in the lower panel. The right panel shows the simulated jets labelled $g \to q\bar{q}$, $g \to gg$ and $q \to qg$ together with all simulated jets, with the ratio of each class to all jets in the lower panel. The dashed line in the left panel is the pure $\cos 2\psi$ modulation with the mean value of the simulation.

The right panel of Figure 4.16 shows the separation by channel label, and demonstrates almost no dependence on the channel. All channels demonstrate a positive modulation of close size. The mean

value is $+0.0125 \pm 0.0006$ for the jets labelled $g \to q\bar{q}$, $+0.0136 \pm 0.0004$ for $q \to qg$ and $+0.0161 \pm 0.0002$ for $g \to gg$, and the distributions of the three classes differ from that of all jets by at most about 3% in every bin. The jets of the $g \to q\bar{q}$ channel have the weakest positive modulation and they differ from the other classes by less than 0.004. In comparison with the $\Delta\psi_{12}$ modulation, the same label has a much more negative modulation, as shown in Section 4.4. This main difference between the observables happens because of how they are built. The channel label corresponds to the second splitting of the Lund declustering, while the three-point energy correlator sums over every triplet inside a fixed angular window, most of which do not resolve that splitting. Therefore, a jet with a labelled channel of $g \to q\bar{q}$ mainly contributes to the correlator through triplets unrelated to its $g \to q\bar{q}$ splitting. The mean value of all jets, $+0.019 \pm 0.001$, is larger than that of every labelled class because it is dominated by the unmatched jets and by those without a label. These make up almost 90% of the jets and have mean values of $+0.0200$ and $+0.0196$.

The labelled channels also show a feature that is much weaker in the sample with all jets. Their maximum at $\psi = \pm\pi$ is about 3% lower than that at $\psi = 0$, which corresponds to a term in $\cos\psi$. Exchanging the two closest constituents a and b reverses the normal $\hat{n}_1$ of equation (3.10) and shifts ψ by π. Therefore, the distinction between $\psi = 0$ and $\psi = \pm\pi$ depends on which of the two is taken as a, here the one stored first. A term of this kind does not have the period π of the spin correlation and does not contribute to $\langle\cos 2\psi\rangle$.

The three-point energy correlator of the jets selected by the flavour selection procedure is shown in Figure 4.17 as a function of the threshold. After passing the first selection stage and the requirement of having a matched score above 0.8, the jets labelled $g \to q\bar{q}$ and the background have mean values of $+0.010$ and $+0.012$, which differ by only 0.002, compared with about 0.10 for $\Delta\psi_{12}$. Therefore, using the networks to enrich the samples with the $g \to q\bar{q}$ label can barely change the correlator. Nevertheless, in simulation the mean value decreases steadily with the threshold, from about $+0.014$ at 0.30 to about $+0.001$ at 0.60, and it lies outside the range spanned by the two references at both ends. Its size, about 0.013, is however an order of magnitude smaller than the corresponding change of $\Delta\psi_{12}$ seen in Section 4.6.1. The data follow the simulation. Up to a threshold of 0.50, the range retained in Section 4.6.3, they differ from it by at most about 1.3 standard deviations, from $+0.017 \pm 0.004$ against $+0.014$ at 0.30 to -0.007 ± 0.021 against $+0.010$ at 0.50. Above this threshold the uncertainty of the data exceeds the whole variation of the simulation.

Thus, the three-point energy correlator shows a modulation in which the simulation follows the collision data, although with a positive sign, nearly insensitive to the channel labelling of the Lund declustering, and only weakly affected by a flavour selection built on the declustering. It does not reproduce the strong negative modulation associated with $g \to q\bar{q}$ splittings in the $\Delta\psi_{12}$ observable. These results are compared with the theoretical expectations of [18] in Section 4.8.

4.8 Discussion

The previous sections presented the computation of the two spin-sensitive observables in collision data and simulation, before and after flavour selection. This section interprets these results, compares them with the theoretical expectations of [18], obtained at second order in the collinear limit for electron-positron collisions, and with the CMS measurement [2], which observed the spin correlations in $g \to q\bar{q}$ splittings with an analysis similar to the present one.

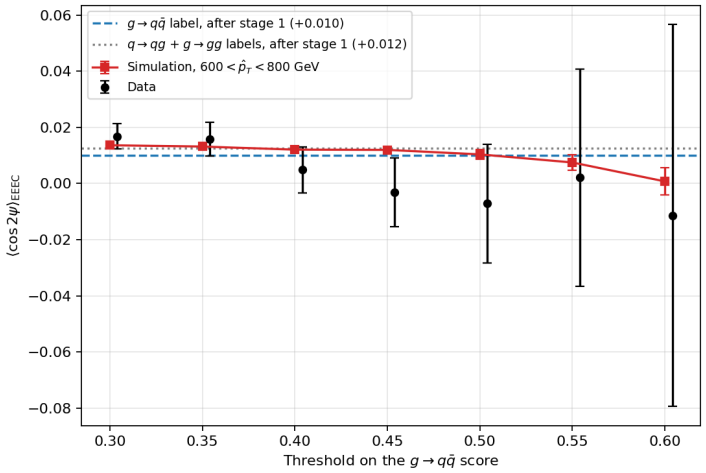


Figure 4.17 – Mean value $\langle \cos 2\psi \rangle$ of the three-point energy correlator of the jets selected by the two-stage selection as a function of the threshold on the $g \to q\bar{q}$ score, in the collision data and in the simulated calibration sample. All jets have a matched score above 0.8. The horizontal lines show the mean values in simulation of the jets labelled $g \to q\bar{q}$ and of those labelled $q \to qg$ or $g \to gg$ after this requirement alone. The data points are shifted slightly to the right for readability, and the error bars show the statistical uncertainties.

The discussion is organised in four parts. Section 4.8.1 examines whether the observed modulations have the properties expected from spin correlations. Section 4.8.2 discusses the effect of the flavour selection on the modulation, and Section 4.8.3 the comparison with the CMS measurement. Section 4.8.4 summarises the limitations of the analysis that follow from these comparisons.

4.8.1 Comparison with theoretical expectations

In simulation, the jets labelled as $g \to q\bar{q}$ revealed the most negative modulation, $\langle \cos 2\Delta\psi_{12} \rangle = -0.167 \pm 0.002$, but with half the magnitude of the collinear prediction of, -0.31. A weaker modulation is, however, expected for several reasons. The IFN label is a flavour proxy and does not coincide splitting by splitting with the vertex of the parton shower. The observable was built from reconstructed particles, after hadronisation and detector effects. The prediction also refers to quark-initiated jets in electron-positron collisions. This work is done in proton-proton collisions and a large fraction of the jets are initiated by gluons, for which the intermediate gluon is produced in a $g \to gg$ splitting with a different degree of polarisation, so no prediction specific to proton–proton collisions is available for $\Delta\psi_{12}$.

A larger deviation appears on the classes for which little or no spin correlation is expected. The jets, with labels $g \to gg$ and $q \to qg$, have mean values of -0.077 and -0.091, where the predictions are $+0.025$ and zero, and the unmatched jets have -0.059. For all jets together, the inclusive value of -0.083 in simulation and -0.078 in data compares with a prediction of about -0.013. Thus, a negative component of the modulation is present on each channel. Two of its properties are visible in the results. It depends strongly on the kinematics of the second splitting and is largest for asymmetric splittings, as shown in Section 4.6.2. It is also accompanied by higher harmonics in the inclusive distribution of Section 4.3, whereas a collinear spin correlation at this order contributes only a $\cos 2\Delta\psi_{12}$ term. Both properties suggest an origin in the kinematics of the declustering rather than in the spin of the intermediate gluon. The simulation reproduces this component in data, so it is described by the generator and the detector simulation, but its origin is not identified in this work. Its dominance is consistent with the CMS measurement, in which switching off the spin correlations in Pythia 8 changes the inclusive distribution of $\Delta\psi_{12}$ by only about 0.5% [2].

Since this component is common to all classes, the difference between the $g \to q\bar{q}$ class and the background is the quantity closest to the spin correlation of the $g \to q\bar{q}$ channel. Before any selection it amounts to 0.086, from -0.167 and -0.081, and after the first stage of the selection to 0.097, from -0.232 and -0.135. Both values are less than a third of the difference of about 0.33 between the predictions for the $g \to q\bar{q}$ and $g \to gg$ channels. Also, the dependence of this difference on z_2 is much weaker than expected. At leading order, the azimuthal dependence of the $g \to q\bar{q}$ splitting of a linearly polarised gluon is proportional to $1 - 2z(1-z) - 2z(1-z)\cos 2\phi$, so that its analysing power, $2z(1-z)/[1-2z(1-z)]$, grows from 0.22 at $z_2 = 0.1$ to 1 at $z_2 = 1/2$. The difference computed instead grows from about 0.08 below $z_2 = 0.2$ to about 0.10 above $z_2 = 0.3$. This may be either because the non-spin component does not cancel completely in the difference, or because the spin correlation is strongly diluted at reconstruction level by the smearing of z_2, by the cases where the reconstructed splitting does not correspond to the generator-level one, and by those in which the label does not correspond to the channel of the shower. The two effects cannot be separated without a reference in which the spin correlations are absent, as discussed in Section 4.8.3.

The three-point energy correlator departs further from the expectations. For quark-initiated jets in electron-positron collisions, In [18] predicts mean values $\langle \cos 2\psi \rangle$ of about -0.18 for the $g \to q\bar{q}$ channel, $+0.013$ for $g \to gg$ and -0.004 for all jets together. The result obtained here in this work is positive, $+0.0186 \pm 0.0001$ in simulation and $+0.0200 \pm 0.0012$ in data, and it depends little on the channel label. The jets labelled $g \to q\bar{q}$, with $+0.0125 \pm 0.0006$, differ from those labelled $g \to gg$ by only 0.004. The weak dependence on the label may come from the construction of the correlator, which sums over all triplets inside a fixed window and is thus not tied to the second splitting of the declustering, as discussed in Section 4.7. The positive sign is not explained here, but it may reflect the larger fraction of gluon-initiated jets at a hadron collider, whose $g \to gg$ splittings contribute positively, the transposition of the angular window from electron-positron collisions to the (η, ϕ) plane, or a non-spin component of the same kind as the one found for $\Delta\psi_{12}$. However, the present work does not discriminate between these possibilities.

4.8.2 Effect of the flavour selection

The flavour selection developed in this work strengthens the modulation of the selected jets from -0.122 at a threshold of 0.30 to -0.266 at 0.60, while the purity rises from about 0.18 to between 0.48 and 0.62. If the selection only changed the proportions of the channels, the jets of each class would keep the modulation they have after the first stage, -0.232 for the $g \to q\bar{q}$ class and -0.135 for the background, and the mean value of the labelled jets would lie between these two values. So, at a threshold of 0.60, where the observed purity was at most 0.62, such a mixture would give a modulation of -0.19. However, at this threshold, the labelled jets selected have instead a mean value of about -0.30, more negative than the reference of the $g \to q\bar{q}$ class itself.

Thus, the modulation of each class changes with the threshold. The jets labelled $g \to q\bar{q}$ become more negative, from -0.237 to -0.325, and the jets labelled $q \to qg$ or $g \to gg$ even more so, from -0.108 to -0.249, even though their label does not change. There are two possible explanations, and they are not exclusive. The first one is that the classifier, which does not use $\Delta\psi_{12}$ as an input, selects configurations in which the non-spin component of the modulation identified in Section 4.8.1 is stronger, through features correlated with it. Part of this effect is explained by the momentum fraction of the second splitting, as shown in Section 4.6.2. The second one is that the shift reflects the limitations of the IFN label. The label does not coincide splitting by splitting with the vertex of the parton shower, and the classifier, trained on the substructure of the branches, may recognise real $g \to q\bar{q}$ splittings better than the label does.

At high thresholds, the jets with the label of $g \to q\bar{q}$ would then mostly be those in which the label is correct, and the jets labelled as background would increasingly be $g \to q\bar{q}$ splittings that the label assigns to another channel. Thus, both would carry a stronger spin correlation, and the true purity would be higher than the purity defined by the IFN label. The mean value of the jets labelled $g \to q\bar{q}$ at the highest threshold, -0.325 ± 0.022, close to the prediction of -0.31, is consistent with this interpretation. As argued in Section 4.8.1, however, the negative modulation of the background before any selection cannot be attributed to such splittings.

The difference between the two classes stays close to 0.13 from a threshold of 0.30 up to 0.50, and decreases to 0.076 ± 0.049 at 0.60. According to the first explanation, the common shift cancels in this difference, and, according to the second one, the decrease at the highest thresholds would reflect the growing fraction of genuine $g \to q\bar{q}$ splittings among the jets labelled as background, although it is not significant within the statistical uncertainties. Unfortunately, the two explanations cannot be separated with the samples available in this work. They would differ only on a sample where no spin correlations are present, in which the first would still produce a shift and the second would not, and in the dependence of the additional modulation on z_2, which should grow towards symmetric splittings if it is of spin origin. In either case, the mean value of the selected sample combines the enrichment in $g \to q\bar{q}$ splittings with effects that cannot be separated from it in data. For this reason, the result is presented as a function of the threshold, and not instead as a function of the purity. A linear extrapolation of the modulation to a purity of one would assume that the modulations of the signal and of the background do not depend on the threshold, and the simulation shows that this assumption does not hold.

The comparison with the collision data is subject to the same limitation. Between thresholds of 0.30 and 0.50 the modulation in data strengthens by 0.101 ± 0.052, compared with 0.079 in simulation. Therefore, the collision data follows the simulation, but this agreement holds whichever of the two explanations applies, and it does not distinguish between them.

The energy correlator also behaves in a similar way. Its mean value decreases with the threshold by about 0.013 for all selected jets and by a similar amount for those labelled $g \to q\bar{q}$, from $+0.0093$ to -0.0017, although the two labelled references after the first stage differ by only 0.0025. Therefore, the selection changes the correlator mainly through properties of the selected jets other than their label.

4.8.3 Comparison with the CMS measurement

The CMS measurement used 34.7 fb^{-1} of proton-proton collisions recorded in 2022 at $\sqrt{s} = 13.6$ TeV [2]. The distribution of $\Delta\psi_{12}$ in the selected jets was compared with the predictions of Pythia 8 and Herwig 7, each with and without the angular correlations induced by the gluon spin. The hypothesis without correlations is disfavoured with an observed significance of 6.8 standard deviations, for an expected 8.1, while the data differ from the hypothesis with correlations by 1.9 standard deviations and favour a slightly weaker modulation than predicted. Its strategy is very close to the one followed here. The composition of the simulated samples is very similar, as shown in Section 4.4. The purity reached by CMS, 38% for a score above 0.6 in its four-class network, is also of the same order as the purities obtained here, from 0.38-0.40 at a threshold of 0.45 to 0.44-0.49 at 0.50, although the two analyses define the purity with different matching criteria.

Nevertheless the two analyses differ in several aspects. CMS reconstructs jets using the PUPPI algorithm for pileup mitigation, while the jets in this work are corrected with charged-hadron subtraction, and it applies the requirements on the constituents and on the second subjet discussed in Section 3.6.4. CMS'

classifier has four classes and is trained on the Herwig 7 sample without spin correlations, so that it cannot exploit the modulation itself. Here the selection is performed in two stages and trained on a Pythia 8 sample. However, the four-class network studied as an alternative in Section 4.5 performs comparably to the two-stage selection, so that the choice of architecture is not expected to be the main difference between the two results.

The decisive difference lies in the predictions against which the data are compared. In CMS, the predictions with and without spin correlations are processed through the same selection, so that every effect of the selection that does not originate from the spin is present in both and cancels in their comparison. These predictions also show how small the spin-induced part of the modulation is. In Pythia 8, switching off the correlations changes the inclusive distribution by only about 0.5%, and the distribution of the selected $g \to q\bar{q}$ category by at most 12% [2]. This is consistent with the large non-spin component found in Section 4.8.1, which the selection does not remove. The same comparison would also separate the two explanations of Section 4.8.2, since a shift caused by the selection of the non-spin component would persist without spin correlations, whereas a shift caused by genuine $g \to q\bar{q}$ splittings assigned to other channels would disappear.

In Pythia 8 the correlations are included as an average modulation of the angle between the production and splitting planes of the gluon [2], and the nominal sample used here therefore corresponds to a prediction with correlations. The agreement between data and simulation found in Section 4.6.3 is thus consistent with the CMS observation that the data agree with Pythia 8 with correlations. No sample without spin correlations is available among the CMS Open Data simulations of the period studied here, however, so this agreement cannot be turned into a statement on the presence of spin correlations in the data. A numerical comparison of the modulation itself is not possible either, since CMS presents the distribution of $\Delta\psi_{12}$, at detector and at particle level, but not its mean value $\langle \cos 2\Delta\psi_{12} \rangle$. The present analysis thus reproduces the strategy of the CMS measurement with public data from an earlier period, but cannot by itself reach its conclusion. What it adds is a decomposition of the modulation by channel class in simulation, which shows how the flavour selection acts on each class and why the modulation of the selected samples cannot be interpreted as that of the $g \to q\bar{q}$ channel alone.

4.8.4 Limitations

The main limitations of this analysis come from the preceding subsections. The most important is the absence of a simulation without spin correlations. The simulation used here includes the approximate treatment of the correlations of Pythia 8, and the CMS measurement shows that the spin-induced part of the modulation is small compared with the non-spin component. Without a reference using the same selection, but in which the correlations are turned off, the two components cannot be separated, and neither can the two explanations of the effect of the selection discussed in Section 4.8.2.

This analysis also relies on a single generator. Pythia 8 with the CP5 tune is used for the training of the classifier, for the calibration of the selection, for the determination of the purity and for the comparison with data. The dependence of the composition and of the purity on the description of the parton shower and of the hadronisation is therefore not tested, as discussed in Section 3.6.2. This dependence is likely to be relevant, since Pythia 8 and Herwig 7 treat the spin correlations differently and predict different strengths of the modulation [2].

The computed purity is defined through the IFN label, which is a flavour proxy assigned through an angular matching. About 31% of the simulated jets carry no label, which is why the purity is quoted in an interval,

and the true fractions of $g \to q\bar{q}$ splittings in the selected samples may differ from the labelled one, as discussed in Section 4.8.2.

The collision data correspond to a single period of data taking, with about 4.6×10^4 jets with a valid $\Delta\psi_{12}$. Their statistical precision limits the comparison with simulation to thresholds up to 0.50, and it is insufficient to resolve differences of the size expected between predictions with and without spin correlations. All results are also obtained at detector level, without correction for the resolution and acceptance of the detector, so their comparison with theoretical predictions is only qualitative. Finally, only statistical uncertainties are given. The systematic sources of Section 3.6 are not propagated, and the analysis choices that differ from the CMS measurement, discussed in Section 3.6.4, are not varied.

Within these limitations, the analysis shows that the full chain, from the released data to a flavour-enriched measurement of the modulation, can be carried out with public data and tools. It also shows that the simulation describes the data well in the observables central to this work, namely the plane (z_1, z_2), the modulation as a function of the threshold and the energy correlator. The extensions needed to turn this agreement into a statement on the presence of spin correlations are discussed in Chapter 5.

5 CONCLUSIONS

5.1 Summary of the results

This work studied the collinear spin correlations in QCD jets with CMS Open Data datasets, using two observables sensitive to the polarisation of an intermediate gluon. The first computed observable is the azimuthal angle $\Delta\psi_{12}$ between the planes of two successive splittings of the Lund declustering. The second is the azimuthal modulation of the three-point energy correlator. Both of them were measured in jets with $p_{\mathrm{T}} > 700$ GeV in the 2016 collision data and compared with the corresponding Pythia 8 simulation, before and after a flavour selection designed to enrich the samples in $g \to q\bar{q}$ splittings. The results are summarised below in relation to the objectives of Section 1.2.

A fully complete and reproducible analysis was developed over the released data. It extracts the jets and their constituents information inside the CMS software environment, and performs the declustering, the generator-level labelling of the splittings, the computation of the energy correlator and the flavour selection outside it, with public tools. The simulation describes the collision data kinematics, and the primary and secondary Lund jet planes show the structure expected from the emission pattern of QCD and from the boundaries set by the grooming condition and by the jet radius. The momentum fractions distributions of the two selected splittings agree between data and simulation as well, with mean values of 0.270 and 0.283 in data against 0.268 and 0.283 in simulation.

The observed $\Delta\psi_{12}$ shows a clear $\cos(2\Delta\psi_{12})$ modulation even before any flavour selection, with $\langle \cos 2\Delta\psi_{12} \rangle = -0.078 \pm 0.003$ in data and -0.083 in simulation. In simulation, the jets labelled as $g \to q\bar{q}$ have a modulation about twice as strong as the background, -0.167 against -0.081, with a negative sign as predicted for this channel. However, every channel carries a negative modulation, including those in which little or no spin correlation is expected. This non-spin component is largest for asymmetric splittings and dominates the inclusive modulation, in line with the small difference between the Pythia 8 predictions with and without spin correlations reported by CMS. The three-point energy correlator also shows a modulation that the simulation describes well in data, $+0.0200 \pm 0.0012$ against $+0.0186 \pm 0.0001$, but it has a positive sign and is almost independent of the channel, in contradiction with the theory prediction. In the configuration used here, it is therefore not a channel-selective probe of the spin correlations.

A two-stage flavour selection was developed, in which a first network separates matched from unmatched splittings and a second network identifies the $g \to q\bar{q}$ channel among the matched ones. This selection raises the purity of the selected samples from 0.03 before any selection to between 0.48 and 0.62 at the highest threshold. It performs best among the methods studied at low efficiency, while a four-class network developed as an alternative test, similar to the one used by CMS, is better at high efficiency. The purity counted against all jets saturates at about 0.5, because the tightest requirements retain proportionally more jets without a generator-level counterpart. This sets the practical limit of the selection with the features used.

Using this selection, the modulation of the selected jets strengthens steadily with the threshold, from -0.122 at 0.30 to -0.266 at 0.60 in simulation. The data follows the simulation within one standard deviation up to a threshold of 0.50, beyond which their statistical precision is insufficient. However, a modulation of

a pure $g \to q\bar{q}$ sample could not be established. The modulation of each labelled class changes with the threshold, so the selected samples do not behave as mixtures of classes with fixed modulations, and a linear extrapolation to a purity of one is not justified. The most reasonable quantity is the difference between the $g \to q\bar{q}$ class and the background, which stays close to 0.13 up to a threshold of 0.50, and which is less than half of the difference expected at leading order. The shift present in both classes can be explained by a selection of configurations with a stronger non-spin component, or by genuine $g \to q\bar{q}$ splittings that the label assigns to other channels. Despite of these hypothesis, the samples available do not allow them to be studied or tested.

The sources of systematic uncertainties were identified, with those related to the purity limiting the interpretation most. These are the dependence of the purity on the generator and on the definition of the channel label. However, the most important limitation of this work is the absence of a simulated dataset without spin correlations processed using the selection adopted here. A dataset of that kind allowed CMS to establish the presence of the correlations, and it is not available among the released simulations of the period studied. Nevertheless, the present analysis reproduces the strategy of the CMS measurement with public data, and its results are consistent with the CMS observation that the data agree with Pythia 8 with spin correlations, although, it cannot establish by itself their presence. Its main contribution is the decomposition of the modulation by channel class, which shows that the modulation of a flavour-selected sample cannot be interpreted as that of the $g \to q\bar{q}$ channel alone. This applies to any measurement of this kind, and it supports comparing the data with predictions without spin correlations rather than interpreting the selected modulation directly.

5.2 Future work

The most important extension would be a prediction without spin correlations, processed with the same selection as in data. In Pythia 8 the correlations can be switched off in the parton shower, and Herwig 7 treats them explicitly through the spin-density matrix, so that samples with and without correlations can be generated with both. A full simulation of the CMS detector is computationally demanding, but a first study can be carried out at particle level, where the flavour selection, the labelling and the observables can be applied without change. Such samples would separate the spin and non-spin components of the modulation, and would distinguish between the two explanations of the effect of the selection discussed in Section 4.8.2.

The flavour selection procedure could also be improved in two directions. Although $\Delta\psi_{12}$ is not an input of the classifier, the shift that the selection induces in every class shows that its score is correlated with $\Delta\psi_{12}$ through other features. Training another classifier explicitly decorrelated from $\Delta\psi_{12}$, using, for instance, a penalty on the distance correlation between its score and $\Delta\psi_{12}$ [39] or with an adversarial network [40], would limit this shift, possibly at the cost of some discriminating power. Another and a simpler solution would be training the network on a sample without spin correlations, as done by CMS. This would also ensure that any residual correlation would not originate from spin. The dependence of the purity on the definition of the channel label should also be studied, by comparing the matching of the two branches of the second splitting used here with alternatives such as the matching of subjets 2 and 3 used by CMS. A second generator would finally test how much the composition and the purity depend on the description of the parton shower and of the hadronisation.

On the collision data side, the statistical precision could be increased by using other datasets of other periods of the 2016 data taking available in CMS Open Data. This would permit a comparison with

simulation at the highest thresholds. The systematic uncertainties identified in Section 3.6 should be propagated to the results, and the analysis choices that differ from the CMS measurement, in particular the thresholds on the transverse momentum of the constituents and of the second subjet, should be varied. An unfolding of the measured distributions to particle level would finally allow a direct comparison with theoretical predictions and with the CMS result.

The three-point energy correlator requires a different treatment in order to become sensitive to the $g \to q\bar{q}$ channel. Its angular window was transposed from electron-positron collisions, and its optimisation for jets at a hadron collider, together with predictions for proton-proton collisions, would clarify the origin of the positive modulation observed here. Restricting the triplets to the constituents of the branches selected by the Lund declustering would also tie the correlator to the labelled splitting, and allow the flavour selection to act on it in the same way as on $\Delta\psi_{12}$.

Concluding, this analysis chain is not restricted to proton-proton collisions. CMS also released heavy-ion data, in which the same observables could be used to study whether the spin correlations of the parton shower are modified by the quark-gluon plasma. However the simulated heavy-ion datasets, that are suitable for this analysis, do not include jet quenching and their maximum jet radius is at maximum of $R = 0.5$. Nevertheless, they can provide a reference without medium effects, but not a prediction of how the medium modifies the correlations, which would require dedicated samples generated with a model of jet quenching. The reference provided by the proton-proton measurement, together with the understanding of the selection obtained in this work, is a necessary starting point for such a study.

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A SUPPLEMENTARY NUMERICAL RESULTS

This appendix collects the numerical values behind the figures of Chapter 4 that are not tabulated in the main text. The tables follow the order in which the results are presented there, and all uncertainties are statistical.

Table A.1 – Number of jets with a valid $\Delta\psi_{12}$, mean values of the momentum fractions z_1 and z_2 of the two selected splittings, and correlation coefficient between them, for all jets in the collision data and in the simulated calibration sample, and for the simulated jets in each class of the channel label of the second splitting. The fraction is given relative to all simulated jets. The statistical uncertainties are below 0.001 for the mean values and below 0.005 for the correlation coefficients.

Sample	Jets	Fraction	$\langle z_1 \rangle$	$\langle z_2 \rangle$	$\rho(z_1, z_2)$
Data, all jets	45 755		0.270	0.283	−0.032
Simulation, all jets	8 495 207	100.0%	0.268	0.283	−0.031
no label	2 647 865	31.2%	0.267	0.285	−0.026
unmatched	4 397 983	51.8%	0.264	0.275	−0.028
$g \to q\bar{q}$	179 810	2.1%	0.275	0.315	−0.064
$q \to qg$	358 498	4.2%	0.294	0.294	−0.066
$g \to gg$	857 140	10.1%	0.273	0.301	−0.086
rest	53 911	0.6%	0.299	0.305	−0.048

Table A.2 – Performance of the first stage of the flavour selection as a function of the threshold on the matched score, evaluated on the test set of the training sample. Among the labelled splittings of this set, a fraction 0.234 is matched. The purity is the fraction of matched splittings among those selected, the efficiency is the fraction of matched splittings that are selected, and the last column gives the fraction of unmatched splittings that are selected. The threshold of 0.8 is the one adopted in this work.

Threshold	Selected fraction	Purity	Efficiency	Unmatched selected
0.30	0.449	0.427	0.820	0.336
0.40	0.323	0.516	0.714	0.204
0.50	0.241	0.599	0.618	0.126
0.60	0.184	0.672	0.530	0.079
0.70	0.143	0.733	0.449	0.050
0.75	0.123	0.764	0.403	0.038
0.80	**0.102**	**0.797**	**0.349**	**0.027**
0.85	0.080	0.830	0.284	0.018
0.90	0.054	0.865	0.198	0.009
0.95	0.012	0.917	0.046	0.001

Table A.3 – Mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of the jets selected by the two-stage selection as a function of the threshold on the $g \to q\bar{q}$ score, in the simulated calibration sample and in the collision data. All jets have a matched score above 0.8. The purity is given as the interval between the values obtained by counting the jets without a channel label as background and excluding them. In simulation, the mean value is given for all selected jets, for those labelled $g \to q\bar{q}$, and for those labelled $q \to qg$ or $g \to gg$. After the requirement on the matched score alone, these two labelled classes have mean values of -0.2315 ± 0.0027 and -0.1346 ± 0.0011. The uncertainties are statistical.

	Simulation					Data	
			$\langle \cos 2\Delta\psi_{12} \rangle$				
Threshold	Jets	Purity	All selected	$g \to q\bar{q}$	$q \to qg$, $g \to gg$	Jets	$\langle \cos 2\Delta\psi_{12} \rangle$
0.30	286 380	0.179–0.181	-0.1218 ± 0.0013	-0.2365 ± 0.0030	-0.1078 ± 0.0018	2 607	-0.1281 ± 0.0138
0.35	165 321	0.238–0.244	-0.1335 ± 0.0017	-0.2428 ± 0.0034	-0.1106 ± 0.0025	1 613	-0.1366 ± 0.0174
0.40	90 503	0.304–0.316	-0.1501 ± 0.0023	-0.2505 ± 0.0041	-0.1150 ± 0.0036	924	-0.1568 ± 0.0231
0.45	42 982	0.377–0.399	-0.1724 ± 0.0034	-0.2621 ± 0.0053	-0.1268 ± 0.0059	423	-0.1585 ± 0.0336
0.50	18 220	0.444–0.485	-0.2006 ± 0.0052	-0.2712 ± 0.0075	-0.1473 ± 0.0105	162	-0.2289 ± 0.0539
0.55	6 548	0.493–0.566	-0.2276 ± 0.0085	-0.2912 ± 0.0118	-0.1791 ± 0.0204	50	-0.3495 ± 0.0864
0.60	1 852	0.478–0.615	-0.2661 ± 0.0156	-0.3252 ± 0.0221	-0.2495 ± 0.0433	16	-0.5202 ± 0.0997

Table A.4 – Mean value $\langle \cos 2\Delta\psi_{12} \rangle$ as a function of the momentum fraction z_2 of the second splitting, for the simulated jets labelled $g \to q\bar{q}$ and for those labelled $q \to qg$ or $g \to gg$, after the requirement of a matched score above 0.8. The last column gives the difference between the two classes. The values correspond to the right panel of Figure 4.14, and the uncertainties are statistical.

z_2	$g \to q\bar{q}$	$q \to qg$ or $g \to gg$	Difference
0.10–0.15	-0.2709 ± 0.0213	-0.1896 ± 0.0084	-0.0813 ± 0.0229
0.15–0.20	-0.2467 ± 0.0110	-0.1682 ± 0.0042	-0.0785 ± 0.0118
0.20–0.25	-0.2333 ± 0.0085	-0.1467 ± 0.0033	-0.0866 ± 0.0091
0.25–0.30	-0.2348 ± 0.0074	-0.1383 ± 0.0030	-0.0965 ± 0.0080
0.30–0.35	-0.2327 ± 0.0068	-0.1301 ± 0.0029	-0.1026 ± 0.0074
0.35–0.40	-0.2348 ± 0.0065	-0.1222 ± 0.0028	-0.1126 ± 0.0071
0.40–0.45	-0.2275 ± 0.0064	-0.1298 ± 0.0028	-0.0977 ± 0.0070
0.45–0.50	-0.2197 ± 0.0064	-0.1228 ± 0.0029	-0.0969 ± 0.0070

Table A.5 – Mean value $\langle \cos 2\psi \rangle$ of the three-point energy correlator for all jets in the collision data and in the simulated calibration sample, and for the simulated jets in each class of the channel label of the second splitting. All jets of the ntuples are included, also those without a valid $\Delta\psi_{12}$. The last two rows give the labelled classes used as references in Figure 4.17, after the requirement of a matched score above 0.8 alone. The coefficient of the modulation is $a_2/a_0 = 2\langle \cos 2\psi \rangle$. The uncertainties are statistical and treat the jets as independent.

Sample	Jets	Triplets	$\langle \cos 2\psi \rangle$
Data, all jets	60 566	1.54×10^8	$+0.0200 \pm 0.0012$
Simulation, all jets	11 476 817	2.58×10^{10}	$+0.0186 \pm 0.0001$
no label	5 629 475	5.21×10^9	$+0.0196 \pm 0.0002$
unmatched	4 397 983	1.35×10^{10}	$+0.0200 \pm 0.0001$
$g \to q\bar{q}$	179 810	7.18×10^8	$+0.0125 \pm 0.0006$
$q \to qg$	358 498	1.73×10^9	$+0.0136 \pm 0.0004$
$g \to gg$	857 140	4.36×10^9	$+0.0161 \pm 0.0002$
rest	53 911	2.79×10^8	$+0.0130 \pm 0.0010$
Matched score > 0.8, $g \to q\bar{q}$	64 104	3.24×10^8	$+0.0100 \pm 0.0009$
Matched score > 0.8, $q \to qg$ or $g \to gg$	388 308	2.70×10^9	$+0.0125 \pm 0.0003$

Table A.6 – Mean value $\langle \cos 2\psi \rangle$ of the three-point energy correlator of the jets selected by the two-stage selection, as a function of the threshold on the $g \to q\bar{q}$ score, in the collision data and in the simulated calibration sample. All jets have a matched score above 0.8. The last two columns refer to the selected simulated jets labelled $g \to q\bar{q}$. The uncertainties are statistical and treat the jets as independent.

	Data		Simulation, all selected		Simulation, $g \to q\bar{q}$ label	
Threshold	Jets	$\langle \cos 2\psi \rangle$	Jets	$\langle \cos 2\psi \rangle$	Jets	$\langle \cos 2\psi \rangle$
0.30	$2\,607$	$+0.0168 \pm 0.0046$	$286\,380$	$+0.0136 \pm 0.0004$	$51\,186$	$+0.0093 \pm 0.0010$
0.35	$1\,613$	$+0.0158 \pm 0.0061$	$165\,321$	$+0.0132 \pm 0.0006$	$39\,305$	$+0.0093 \pm 0.0011$
0.40	924	$+0.0049 \pm 0.0082$	$90\,503$	$+0.0121 \pm 0.0008$	$27\,512$	$+0.0095 \pm 0.0014$
0.45	423	-0.0032 ± 0.0122	$42\,982$	$+0.0120 \pm 0.0011$	$16\,197$	$+0.0082 \pm 0.0018$
0.50	162	-0.0071 ± 0.0211	$18\,220$	$+0.0104 \pm 0.0017$	$8\,084$	$+0.0044 \pm 0.0026$
0.55	50	$+0.0021 \pm 0.0387$	$6\,548$	$+0.0075 \pm 0.0027$	$3\,229$	$+0.0035 \pm 0.0040$
0.60	16	-0.0114 ± 0.0679	$1\,852$	$+0.0008 \pm 0.0048$	886	-0.0017 ± 0.0073